Sources, Return and Apparatus Certificates

Technical Supplement to HAFF-D

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Technical documentation for HAFF-D main paper (HAFF_D_manuscript.pdf). This supplement contains the paper's complete technical appendices, not a new paper or an additional scientific result. The assumptions, open joins and scope of each certificate are retained. Appendix letters and all pre-existing result, equation, figure, table and citation numbers agree with the main paper's complete source. References to main sections and results link to the companion PDF. The main paper's certificate guide maps these proofs to its claims.

A Finite-return estimates and spatial clock work

This appendix gives the fixed-window estimates used in Section 9. Its common Hamiltonian domain, bounded interaction and source norms are part of the assumptions. The spatial clock-work construction is a separate model.

A.1 A finite prefix is a sufficient response object

The source need not have finite second moment over all time to determine a finite episode. This matters for actual material radiation, whose tails are not inferred from a finite smooth-source certificate.

Proposition A.1 (Finite-prefix response and its source tolerance)**.** *Fix $u_* < U$ and matched data (a_*, b_*, γ_*) at u_* in the traceless single-chiral JT sector on a common patch with $1 + uv \neq 0$. For $F \in L^1([u_*, U])$, define*

$$a(u) = a_* + \kappa \int_{u_*}^{u} zF(z)dz, \quad b(u) = b_* - \kappa \int_{u_*}^{u} F(z)dz, \quad \gamma(u) = \gamma_* + \kappa \int_{u_*}^{u} z^2 F(z)dz.$$

Then $\Phi = (a + bu + \gamma v - auv)/(1 + uv)$ solves the JT equations on this window with those data, and $\mathcal{C}(u) = a(u)^2 + b(u)\gamma(u)$. No future-source assumption enters either reading. If two sources have $\int_{u_*}^{U}(1 + z^2)|\delta F|dz \leq D$, then

$$\begin{aligned} |\delta a|, |\delta b|, |\delta\gamma| &\leq \kappa D, \\ |\delta\Phi| &\leq \frac{\kappa D(1 + |uv| + |u| + |v|)}{|1 + uv|}, \\ |\delta\mathcal{C}| &\leq \kappa D(2|a| + |b| + |\gamma|) + 2\kappa^2 D^2. \end{aligned} \tag{32}$$

Proof. The coefficient derivatives are $a' = \kappa uF$, $b' = -\kappa F$ and $\gamma' = \kappa u^2 F$. They cancel from the first derivatives of the numerator, giving the field equations in their weak sense. Direct substitution in $\Phi^2 - L^2(\nabla\Phi)^2$ gives $a^2 + b\gamma$. The three integrands are bounded in absolute value by $1 + z^2$. Expanding the numerator and invariant gives the estimates. □

For the complete quantum material, Q-RAIF C bounds smooth interacting stress tests on one positive Hamiltonian with all old references retained [14]. Its cross and creation contributions are essential. The bound is a test-functional certificate, whereas Eq. (32) uses a weighted L^1 certificate. They are not interchangeable. With a smooth preparation cutoff, smearing a finite-resolution dilaton reading gives smooth retarded kernels. Each such kernel can be used as a source test with its own Q-RAIF C form constant. Recovering a sharp prefix additionally requires a bound on unresolved source structure. This gives a direct finite-window interface without claiming an unproved all-time tail estimate or discarding controller and support sources.

A.2 The same quantum reader, support and source

A reflecting boundary alone does not supply its support. The material branch uses a positive dilaton-dependent mass and varies the worldline. It contains the finite reader, history clock, blank and old purification in its mass. Classical scalar fields couple to an angle θ; storage y has positive kinetic and spring energies. Their emitted powers are

$P_\pm = g_\pm^2 \dot\theta^2$, derived by characteristics, and their reaction is included in the mechanical equation. The two exterior invariants evolve with the proper worldline and obey the mass junction. Appendix J gives this complete mean-action construction and its admissible positive domain.

The quantum core is

$$H_c = H_{\rm hist} + E_{\rm off} I + \lambda(\Phi - \phi_0) Z_P + \frac{\chi_R}{2} P_R (y-d)^2.$$

Its first active clock edge writes an actual Y_P outcome; the next swaps the probe with a paid blank. The record spring reacts while acquisition is in progress. Variation gives $F_y = \chi_R(y-d)\langle P_R\rangle$ and $F_\Phi = \lambda\langle Z_P\rangle$, with $\dot E_c = F_y \dot y + F_\Phi \dot\Phi$. Consequently $M = m_{\rm rest} + U + e + E_c$ obeys $\dot M = (U' + F_\Phi)\dot\Phi - P_+ - P_-$. These are quantum expectations in a retained pure-state calculation, not a sampled record fed to an independent source generator.

Proposition 9.1 proves a nonzero source return through its shortest-path expansion. Either $\lambda = 0$ or $\chi_R = 0$ removes the proper-power difference between its two initial geometries. Both interactions present produce a nonzero coefficient at order t^{2r+7}, where r is the acquisition edge. The finite 33-state example retains its old probe reference and all work balances. This is a mean-worldline realization; quantum support fluctuations are not removed by taking the quantum expectation of its force.

The reader must also remain accurate when the response is allowed to move, rather than being supplied beforehand. The following estimate controls this coupled problem.

Proposition A.2 (Joint tolerance of an acquired-record source return)**.** *Write the mean-action equations as $\dot z = \mathcal{F}(z,\Psi)$ and $i\dot\Psi = H_c(z)\Psi$, retaining the old reference in Ψ. Allow $H_c(z) = H_* + V_c(z)$, where H_* is a common semibounded self-adjoint internal Hamiltonian and V_c and its first two coordinate derivatives are bounded on the stated domain. For an unbounded core, require strong initial data in $D(H_*)$ and propagators preserving this common domain along the stated coordinate paths. This includes the finite core above and regular Gaussian internal data with bounded controller coupling; it does not include an unbounded coordinate-force operator merely because its free field is Gaussian. Include the complete mass M among the classical variables, with its actual initial value and work equation*

$$\dot M = (U' + \langle \partial_\Phi V_c\rangle)\dot\Phi - P_+ - P_-.$$

Let exact and approximate solutions, with the same core and reference space, lie in a compact subset of the positive domain of Proposition J.4, with $|y| \le Y$ and $|\dot\theta| \le V$. Use the maximum norm for classical variables nondimensionalized in fixed material and clock units. Suppose the approximate equations have classical residual at most $r_z(t)$, including their mass-work residual. Let $r_H(t)$ bound the Hermitian Hamiltonian error applied to the approximate quantum vector; the propagators and this residual must exist on their specified domains. Choose constants on this domain such that

$$|\mathcal{F}(z,\Psi) - \mathcal{F}(\widehat z,\widehat\Psi)| \le L_z|z-\widehat z| + L_\Psi \|\Psi - \widehat\Psi\|.$$

Here the same reference is compared, not just its reduced probe state. For the displayed sensor and spring the Hamiltonian Lipschitz constant is $L_H = |\lambda| + \chi_R(Y + |d|)$. For any $\sigma > 0$ put $L = \max\{L_z + \sigma L_H, L_\Psi/\sigma\}$ and $E = |z-\widehat z| + \sigma\|\Psi - \widehat\Psi\|$. Then

$$E(t) \le e^{Lt}E(0) + \int_0^t e^{L(t-s)}[r_z(s) + \sigma r_H(s)]ds. \tag{33}$$

Any terminal record probability differs by at most $2E/\sigma$, and the total proper emitted power differs by at most $2\gamma VE$, where $\gamma = g_+^2 + g_-^2$. These estimates cover the actual quantum-force reaction in the classical response, including its contribution to mass and clock conversion.

Proof. For normalized states, $|\langle A\rangle_\Psi - \langle A\rangle_{\widehat{\Psi}}| \le 2\|A\|\|\Psi - \widehat{\Psi}\|$. In particular the P_R and Z_P expectations have that bound. The force differences are bounded by

$$|\delta F_y| \le \chi_R|\delta y| + 2\chi_R(Y + |d|)\|\delta\Psi\|, \quad |\delta F_\Phi| \le 2|\lambda|\|\delta\Psi\|,$$

The mechanical work $-F_y\dot{y}$ cancels its contribution $F_y\dot{y}$ to the internal energy, leaving the stated mass equation. Thus mass is compared through its initial value and integrated work, not by bounding $\langle H_*\rangle$ with the state-vector error. The mass, square roots $\beta_\pm$ and null-clock denominators are smooth with bounded derivatives on the stated compact domain; these formulas give finite L_z, L_Ψ for the actual coupled equations. All clock and material energy remains in the initial mass. The quadratic spring and the probe term give the displayed L_H. Duhamel comparison using the exact unitary propagator gives

$$\|\delta\Psi(t)\| \le \|\delta\Psi(0)\| + \int_0^t [L_H|\delta z(s)| + r_H(s)]ds.$$

It contains no spurious growth from the norm of a common clock Hamiltonian. Add the classical integral bound with weight σ and apply Gronwall. Finally $P_+ + P_- = \gamma\dot{\theta}^2$ gives the power estimate, and a record effect has norm at most one. □

For the strong-domain solutions just specified, the Schrödinger equation gives

$$\frac{d}{dt}\langle H_* + V_c(z)\rangle = \sum_j \langle \partial_j V_c(z)\rangle \dot{z}_j.$$

The common H_* contributes no explicit coordinate work. In its interaction picture only bounded perturbation differences enter Duhamel's estimate, so the same proof applies without bounding $\langle H_*\rangle$ by a vector-norm error. Extending to all energy-form initial data would additionally require an energy-controlled approximation preserving this work identity. Spatial field observables still require the separate material form bounds of Q-RAIF C. This extension treats the core as internal material of the specified thin worldline; it does not equate spatial field stress with a point source. Smooth tests of the line stress $Mu^\mu u^\nu \delta_\Gamma$ are Lipschitz on the same compact tube, so their errors are bounded by a constant times $\int E(s)ds$. The constants and admissible tube may change when the initial clock mass grows; no uniform-in-clock-size geometric limit follows.

The work equation is necessary even when the retained quantum states are close. Q-RAIF C gives a positive clock-driven oscillator whose actual total energy is mJ, whereas its scheduled product-clock target has energy $mJ + b^2/(2\omega)$ for every m, despite vanishing joint state error [14]. A nearby rotated clock can match that scalar energy, but does not thereby match local stress. For a scheduled comparison with internal Hamiltonian jumps $h_{b-1} \to h_b$, subtract the switching work $W_b = \langle h_b - h_{b-1}\rangle_{t_b}$ from its clock ledger. Together with the continuous coordinate work this gives the mass equation used above. Initial mass mismatch and any missing work belong in $E(0)$ and r_z; they cannot be dropped by declaring the clock state error small. This ledger is not the energy expectation of the unmodified product-clock target.

A source contrast remains nonzero whenever the two power-error bounds sum to less than its ideal contrast at a chosen time. Thus finite implementation errors can be admitted while preserving the causal sensor-removal and feedback-removal tests. The domain condition is verified by a strict ideal margin and an error tube smaller than that margin; it cannot be certified from a reduced-state fidelity alone. These are the fixed-window mean-action bounds in Table 1.

Foundation III uses this action for a frame learned while the original quantum content is held [56]. In a finite positive parameter family, the acquired record changes the apparatus's own source; its geometric response enters an X-basis reading of the old probe into the already spent cell. The first-order geometry signal is generated after record acquisition, with no new probe or blank port. The matched intervention keeps the complete initial state, mass and geometric constraints fixed and removes only the first record's mechanical use. A sensor-edge diagnostic separates the geometric path from direct probe and cursor effects. This is one finite semiclassical mean-action scalar/JT return. The completed-window geometry-edge contrast, averaged over the stated plant-sign prior, is about -4.98×10^{-9} in the example; its small size is reported separately from the task advantage. It is distinct from the spatial realization below and from the sustainable sharp-exhaust conjecture.

A.3 A complete spatial clock-work source

The work correction can be a spatial tensor rather than a scalar debit. Consider the local quadratic action

$$S_{\chi u} = -\frac{1}{2}\int \sqrt{-g}\left[I(\nabla\chi)^2 + (\nabla u)^2 + I\mu^2\chi^2 + (\omega u + \lambda\chi/\omega)^2\right] d^2x, \tag{34}$$

where $I, \mu, \omega > 0$. In canonical variables $(\sqrt{I}\chi, u)$ its mass matrix is

$$\begin{pmatrix} \mu^2 + \lambda^2/(I\omega^2) & \lambda/\sqrt{I} \\ \lambda/\sqrt{I} & \omega^2 \end{pmatrix}.$$

Its determinant is $\mu^2\omega^2 > 0$. Constant orthogonal diagonalization gives two positive-mass Klein–Gordon fields with their complete spatial mode towers. Choose the interacting static vacuum and a finite-energy Weyl displacement with initial data $\chi = u = \dot{u} = 0$, $\dot{\chi} = \Omega f$, $f \in C_c^\infty$. Its added clock energy in flat space is $I\Omega^2\|f\|_2^2/2$. One fixed local renormalization prescription is used for the interacting vacuum and all displaced states.

Put $L_a = \partial_t^2 - \partial_x^2 + a^2$. In flat space let

$$L_\mu c = 0, \qquad L_\omega u_0 = -\lambda c, \qquad L_\mu \eta_0 = -\lambda u_0 - \lambda^2 c/\omega^2, \tag{35}$$

where c has the prepared clock data and the other two fields have zero data. Define

$$T^{(a)}_{\alpha\beta}[q] = \partial_\alpha q \partial_\beta q - \tfrac{1}{2} g_{\alpha\beta}[(\nabla q)^2 + a^2 q^2], \tag{36}$$

$$B_{\alpha\beta}[c, \eta] = \partial_\alpha c \partial_\beta \eta + \partial_\alpha \eta \partial_\beta c - g_{\alpha\beta}(\nabla c \cdot \nabla \eta + \mu^2 c\eta). \tag{37}$$

The work-corrected comparison source is

$$T_{\mathrm{cmp}} = IT^{(\mu)}[c] + B[c, \eta_0] + T^{(\omega)}[u_0] - g(\lambda c u_0 + \lambda^2 c^2/(2\omega^2)) + T_{\mathrm{vac},I}. \tag{38}$$

The large positive clock source and the same interacting-vacuum source are present on both sides of the comparison.

Proposition A.3 (Full-mode stress matching)**.** *For each fixed finite window and compact smooth tensor test W, the actual coherent quantum state of* (34) *satisfies*

$$\left| \int \sqrt{-g}\, W^{\alpha\beta}(\langle T_{\alpha\beta}\rangle - T_{\mathrm{cmp},\alpha\beta}) \right| \le C_W/I. \tag{39}$$

The constants are independent of I above a finite threshold. In flat space the comparison obeys the exact local Ward identity. The source difference also tends to zero in every fixed C^k norm on a compact window. The same statements hold on a fixed static metric using its finite-window Klein–Gordon energy constants.

Proof. The actual means obey $L_\mu \chi_I = -\lambda u_I/I - \lambda^2 \chi_I/(I\omega^2)$ and $L_\omega u_I = -\lambda \chi_I$. For $\eta_I = I(\chi_I - c)$ these imply the exact identities

$$L_\mu \eta_I = -\lambda u_I - \lambda^2 \chi_I/\omega^2, \qquad u_I - u_0 = -\lambda G_\omega \eta_I/I,$$

where G_ω is the zero-data retarded operator. Use $\|q\|_{s,a}^2 = \|\dot q\|_{H^s}^2 + \|q_x\|_{H^s}^2 + a^2\|q\|_{H^s}^2$. The energy inequality is $\|G_a F(t)\|_{s,a} \le \int_0^t \|F(r)\|_{H^s} dr$. Set $C = |\Omega|\|f\|_{H^s}$, $a_T = |\lambda|T/\mu$, $b_T = |\lambda|T/\omega$, $d_T = \lambda^2 T/(\omega^2\mu)$ and $h_T = a_T b_T + d_T$. For $I \ge 2h_T$ it gives

$$\sup \|\eta_I\|_{s,\mu} \le 2h_T C, \qquad \sup \|u_I - u_0\|_{s,\omega} \le 2a_T h_T C/I, \tag{40}$$

$$\sup \|\eta_I - \eta_0\|_{s,\mu} \le 2h_T^2 C/I. \tag{41}$$

These are full spatial energy estimates, without a mode cutoff. The exact quantum covariance equals the interacting-vacuum covariance; only the mean is displaced. Metric variation therefore gives

$$\langle T\rangle = IT^{(\mu)}[c] + B[c,\eta_I] + I^{-1}T^{(\mu)}[\eta_I] + T^{(\omega)}[u_I] - g(\lambda\chi_I u_I + \lambda^2\chi_I^2/(2\omega^2)) + T_{\mathrm{vac},I}.$$

Polarization of each quadratic tensor and the displayed energy estimates prove (39). Smooth initial data, higher Sobolev estimates and the wave equations give the C^k result. Finally $\nabla^\alpha B_{\alpha\beta} = (\lambda u_0 + \lambda^2 c/\omega^2)\partial_\beta c$; the material and contact terms have the opposite divergence. □

The correction has a measurable tensor component. Relative to the common clock and vacuum terms in (38), the flat-space preparation gives

$$\delta T_{tt} = O(t^4), \qquad \delta T_{xx} = -\lambda^2\Omega^2 f^2 t^2/\omega^2 + O(t^4). \tag{42}$$

The clock pays the order-t^2 energy transfer; pressure does not cancel. This subtraction defines a comparison, not a second experiment with the coupling removed and a different vacuum. For metric-only matter on a fixed AdS patch, the matched sources drive the linear JT dilaton equation with the same initial constraints and boundary data. Its finite-window response inherits the source estimate. A positive dilaton domain may require initial homogeneous data growing with I; that resource and a detector's gain cannot be removed from the account. This is the quadratic clock/material entry of Table 1; its receiver and thermal interfaces are separate.

At finite spatial bandwidth, these source and readback steps can be placed in one mean-action evolution. Proposition J.7 includes an elastic bar, a physical synchronization field, two retained quantum spins and the full oscillator Hilbert space of the chosen modes. The two preparations have the same initial gravitational constraints. Their later pressure

contrast produces a dilaton contrast, which changes the retained sensor and the next material force. The synchronization field is varied, not supplied as a prescribed clock. This establishes the finite spatial return listed in Table 1.

The retained quartet admits a full-mode bulk step toward that construction. Proposition J.8 evolves its physical clock, material director and exact renormalized quantum source together with constrained JT data. The normal-order conversion and causal-state identification are nonlinear. The result supplies an interior patch for specified global quantum data; moving endpoints, the other gate windows and the finite-horizon renewal error must still be joined to it.

B Complete export and gravitational transport

B.1 Joint entropy accounts with a reused memory

The memory can also be reused without being decorrelated after each correction. Let correction i act on VA and a fresh thermal port B_i, initially independent of the whole state of VA and all earlier ports, and put $s_i = S(V_i) - S(V_i')$ and $x_i = S(B_i') - S(B_i)$. Suppose each intervening noise step is a CPTP map on V, tensored with the identity on A and the previously exported ports; updates involving A belong to the charged correction. Entropy conservation gives

$$x_i = s_i - [S(A_i') - S(A_i)] + I(V_i' : A_i') - I(V_i : A_i) + I(V_i' A_i' : B_i').$$

Data processing between corrections and telescoping therefore imply

$$\sum_{i=1}^{n} x_i \geq \sum_{i=1}^{n} s_i - [S(A_n') - S(A_1)] - I(V_1 : A_1). \tag{43}$$

For a fixed finite A, the last two terms are bounded. Reused memory purity and initial correlations cannot absorb a positive entropy-removal rate indefinitely. Here x_i refers to each exported port's marginal entropy. Summing those marginals can overcount correlated records. There is a stronger joint-port statement under the same fresh-input and local-noise assumptions:

$$S(B_{1:n}') - \sum_{i=1}^{n} S(\tau_{B_i}) \geq \sum_{i=1}^{n} s_i - [S(A_n') - S(A_1)] - I(V_1 : A_1). \tag{44}$$

To see this, retain all previously exported ports in the mathematical state. A noise step on V, with the identity on those ports, changes its total entropy by $\Delta S(V) - \Delta I(V : AB_{<i}) \geq \Delta S(V)$, by data processing. The corrections preserve total entropy, and each new port adds $S(\tau_{B_i})$. At the end, subadditivity between VA and all exported ports gives Eq. (44) after the V entropies telescope. Thus finite reused memory cannot hide the maintained rate in correlations between outputs.

The same account applies when the noise also acts on the retained controller. Put $W = VA$ and let each correction be a unitary on WB_i with a fresh, independent input τ_{B_i}, uncorrelated with W and the earlier ports. Noise between corrections acts on W alone. With $s_i^W = S(W_i) - S(W_i')$, the exact identity is

$$S(B_{1:n}') - \sum_{i} S(\tau_{B_i}) = \sum_{i} s_i^W + I(W_n' : B_{1:n}') + \sum_{i=1}^{n-1} \ell_i \;\geq\; \sum_{i} s_i^W, \tag{45}$$

where $\ell_i = I(W_i' : B_{1:i}') - I(W_{i+1} : B_{1:i}') \geq 0$ is the information lost to the intervening noise. Indeed, that noise changes the entropy of the joint state by $\Delta S(W) + \ell_i$; the correction preserves it. Summing these changes and expanding the final entropy as $S(W_n') + S(B_{1:n}') - I(W_n' : B_{1:n}')$ gives the identity. Returning an old port to an operation would change these hypotheses. The joint VA rate, rather than a sum of system marginal entropies, is what an apparatus with noisy internal memory must export.

B.2 The complete output deficit

Proposition B.1 (Output deficit and direct renewal)**.** *Use the complete outgoing carrier B_τ, its physical translation generator H_τ, and the thermal input of Proposition 10.2. Let the actual charged entropy removal satisfy $s_\tau/\tau \to x > 0$, set $T_* = T + 6\hbar x/(\pi c_{\mathrm{ex}} k_B)$, and write $\tau_* = e^{-H_\tau/(k_B T_*)}/Z_\tau$. Assume the extensive Gibbs limits used above and that input, charge, interaction, resource and boundary remainders are $o(\tau)$. If the full entropy account gives $L_\tau = S(\rho_{B_\tau}) - S_T - s_\tau \geq -o(\tau)$, then*

$$Q_\tau - \tau \left(k_B T x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}} \right) = k_B T_* \left[D(\rho_{B_\tau} \| \tau_*) + L_\tau \right] + o(\tau). \tag{46}$$

Thus asymptotic equality requires, and under these hypotheses is equivalent to, both deficit and entropy-slack rates tending to zero.

For Q-RAIF C's original quartet preparation followed by direct per-species ring renewal [14], let d_{ring} bound its reset error against the actual old outside marginal, η its residual zero-mode amplitude, and e_H the full-joint clock comparison error over the specified horizon. Suppose each round has disjoint emission modes, these bounds hold uniformly, and no later converter changes the exported modes. With $t = d_{\mathrm{ring}} + 2\arcsin\eta + 2e_H < 1/2$, the output of M rounds obeys

$$D(\rho_{B_\tau} \| \tau_*) \geq M \left[h(\epsilon) - 2h(t) \right], \qquad h(p) = -p \ln p - (1-p)\ln(1-p), \tag{47}$$

when the zero-chemical-potential free-rail reference factorizes across the original species. If $M/\tau \to \nu$ and $h(\epsilon) > 2h(t)$, the asymptotic heat excess above the floor is at least $k_B T_* \nu [h(\epsilon) - 2h(t)]$ per unit time. Temporal independence of thermal packets is not required.

Proof. The finite-carrier Gibbs identity is $D(\rho\|\tau_*) = (E_\rho - E_*)/(k_B T_*) - S(\rho) + S_*$. Here $S_* - S_T = x\tau + o(\tau)$ and $E_* - E_T = \tau[k_B T x + 3\hbar x^2/(\pi c_{\mathrm{ex}})] + o(\tau)$, which gives (46).

The current purifier's occupation pair is 00 or 11, with probabilities $1-\epsilon, \epsilon$. Operations on the receiver leave its joint marginal with previous outputs unchanged. The reset certificate compares that pair with the same actual past marginal; it need not be independent of the current readout record. The exact linear scattering column emitting an old zero mode d_j has squared norm $1 - \eta^2$. CAR therefore give the extracted mode $b_j = \sqrt{1-\eta^2}\, d_j + \eta e_j$, where e_j is a canonical orthogonal mode in the actual body and incoming field. A two-mode rotation of angle $\arcsin\eta$ compares b_j with d_j, retaining every old reference. The two purifier species cost $2\arcsin\eta$; replacement of the comparison joint state and its outside marginal costs at most $2e_H$.

Measure their even occupations in the disjoint emitted modes, giving $X_{1:M}, Y_{1:M}$. For each round the joint law of $(X_j, X_{<j})$ is within variation t of $\mathrm{Bern}(\epsilon) \otimes P_{X_{<j}}$, and $\Pr(X_j \neq Y_j) \leq t$. Conditional binary entropy continuity and the binary Fano bound give $H(X_j|X_{<j}) \geq h(\epsilon) - h(t)$ and $H(X_j|Y_j) \leq h(t)$. Hence $I(X_{1:M} : Y_{1:M}) \geq M[h(\epsilon) - 2h(t)]$.

The reference measurement law is $q_X q_Y$, even though each sequence may be thermally correlated. Data processing and $D(P_{XY}\|q_X q_Y) = I(X:Y) + D(P_X\|q_X) + D(P_Y\|q_Y)$ prove (47). □

All entropies in this proposition use the finite-carrier account and its extensive limit, not a trace entropy of a Type III interval. A fixed-charge canonical reference needs its sector-weight correction; it has the same rate conclusion only when that correction is sublinear. A conditional account must use the same charged removal. For a complete usable system $W = ZM$ with a conserved classical record Z, write $s^W = S(W) - S(W')$. A unitary interaction with a port B gives

$$S(B'|Z) - S(B|Z) = s^W + I(M' : B'|Z) - I(M : B|Z).$$

When the port is initially independent of W, its thermal entropy replaces $S(B|Z)$ and the last term vanishes. Old purifications remain in the physical state; they are usable side information only if the apparatus has that access. Fresh-port sequences with noise acting only on M also accumulate nonnegative information losses to earlier outputs. Reading an old output or updating Z needs a joint account of that operation.

For a classical record X retained at input and output, put $L_c = S(B_{\mathrm{out}}|X) - S(B_{\mathrm{in}}|X) - s_\tau$. When the input marginal entropy is S_T, the same Gibbs identity is

$$\begin{aligned} Q_\tau - \tau f(x) &= k_B T_* \left[D(\rho_{B_{\mathrm{out}}}\|\tau_*) + L_c + I(X : B_{\mathrm{out}}) - I(X : B_{\mathrm{in}})\right] + o(\tau), \\ f(x) &= k_B T x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}. \end{aligned} \tag{48}$$

Indeed $L_\tau = L_c + I_{\mathrm{out}} - I_{\mathrm{in}}$, while $D(\rho_{XB}\|\rho_X \otimes \tau_*) = D(\rho_B\|\tau_*) + I(X : B)$. These are two forms of one account; thermal input marginals alone do not remove old correlations. The direct protocol's deficit lower bound tends to $\nu h(\epsilon) > 0$ as its certified error vanishes. A physical converter that redistributes those correlated records, with its own work and retained memory charged, is therefore necessary for that route to approach the sharp floor.

The incoming state and conserved charges matter. For a nonthermal stream with energy current e_{in} and dimensionless entropy current σ_{in}, put $E_{\mathrm{th}}(\sigma) = 3\hbar\sigma^2/(\pi c_{\mathrm{ex}})$. The same variational argument gives

$$\dot{Q} + e_{\mathrm{in}} - E_{\mathrm{th}}(\sigma_{\mathrm{in}}) \geq k_B T_\sigma x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}, \qquad T_\sigma = \frac{6\hbar\sigma_{\mathrm{in}}}{\pi c_{\mathrm{ex}} k_B}. \tag{49}$$

The nonequilibrium energy above the entropy-matched thermal stream can pay for export. A pure coherent input therefore cannot be charged as though it were a zero-energy thermal reservoir. At nonzero chemical potential the corresponding Gibbs comparison must hold all retained charges fixed. For example, with a level-one $U(1)$ current normalized as in Ref. [12, Eq. (84)], the unitary CFT Gibbs currents in units $\hbar = k_B = 1$ are $e = \pi c_{\mathrm{ex}} T^2/12 + \pi\mu^2/2$, $j = \pi\mu$ and $\sigma = \pi c_{\mathrm{ex}} T/6$ [12, Eq. (93)]. Equal incoming and outgoing charge currents cancel the chemical energy in their difference; an apparatus that absorbs charge can instead consume it. Equation (29) does not cover that consumption without a charge ledger.

For independent exhausts at a common temperature, with central charges c_j and nonnegative entropy currents x_j, the same argument and Cauchy–Schwarz give

$$\dot{Q}_{\mathrm{tot}} \geq k_B T \dot{s} + \frac{3\hbar\dot{s}^2}{\pi \sum_j c_j}, \qquad \sum_j x_j \geq \dot{s}. \tag{50}$$

The capacity term is sharp at the channel level: equality requires thermal outputs and $x_j = c_j \dot{s} / \sum_k c_k$. Additional channels are an additional physical resource. Intermittent export through the same capacity cannot lower the long-window Gibbs floor: apply it to the complete output with its total entropy. If a local-current description is valid, $\overline{x^2} \geq \bar{x}^2$ gives the same conclusion.

Heat alone does not bound every geometric observable. Even a positive E_0 can give zero or negative $\delta\mathcal{C}$ in Eq. (70); its response depends on other moments and on the background. A positive weighted integral $\int K(u)F(u)du \geq K_{\min} \int F du$ requires both a lower bound on K over the exhaust and $F \geq 0$. Quantum excess stress need not have that sign. Proposition 10.3 instead reads E_0 through the late dilaton coefficient slope, without either assumption.

B.3 A horizon reading of the maintenance price

For a matched running single-chiral JT source, let $\mathcal{C} = R^2 > 0$ and $\xi = (bu^2 + 2au - \gamma)/(2R) > 0$ on the regular future outer patch. The marginal horizon has $\Phi_h = R$ and $S_h = S_0 + 2\pi R/\kappa$. In its rapidity clock $d\eta = du/\xi$,

$$\frac{dS_h}{d\eta} = 2\pi F \xi^2.$$

When the charged heat is exactly $F\xi^2$ and the physical exhaust input is thermal at $T_H = 1/(2\pi)$ in this same clock, the long-window channel bound gives

$$\overline{\dot{S}_h} \geq \dot{s} + \frac{6\dot{s}^2}{c_{\mathrm{ex}}}.$$

Appendix H proves the horizon balance and its domain conditions. The entropy here is classical Wald entropy; a generalized-entropy statement needs its matter contribution.

Steady positive flux cannot be extrapolated in one fixed u chart: its boundary dilaton numerator eventually vanishes. Nor does a new clock prepare a reservoir. For an actual source field thermal in η, the vacuum-subtracted physical flux contains $q_{\mathrm{ref}} = -c(\delta' + \delta^2/2)/(24\pi)$, where $\delta = R'/R$. Its self-consistent equations are

$$R' = R\delta, \qquad \delta' = \frac{\kappa q - R\delta}{\kappa c/(24\pi)} - \frac{\delta^2}{2}.$$

The positive growing family of Proposition H.5 has $R = R_0 + \kappa q \eta + O(1)$ and finite accumulated reference correction. Hence its asymptotic horizon rate is $2\pi q = s + 6s^2/c$. A partially transmitting local thermal contact produces a nonthermal output with a strictly larger energy price at the same entropy current; changes of angle add their own pump work. The detailed state, source and clock calculations are in Appendix H. Their global source-clock solution does not grant infinite life to a finite material energy reserve.

C Sustained renewal and a sharp informative return

The complete target below uses the dephased quartet and pointer debits of Proposition 11.1. It requires sustained thermal renewal (target B) and the joint sharpness/return conditions (target C), beyond the finite mean-action return (target A).

Conjecture C.1 (Pointer-aligned selection with a sharp complete exhaust)**.** *Fix $\nu > 0$, $c_{\mathrm{ex}} > 0$ and thermal input at $T_H = 1/(2\pi)$ in the running rapidity clock, in natural units. For this dephasing family there are local models with one reused body during each N-cycle run, for arbitrarily large N, restoring both preparations to the declared error and realizing the following requirements jointly.*

Renewal acts on the current resettable A/R source, with joint control of both pairs explicitly permitted. Independently held unknown content H and its inaccessible old reference R_{old} remain preserved by the specified content-holding task throughout the loop. No reset of H or operation on R_{old} is granted by source renewal. Compatibility of that task with sustained renewal and the response controller is part of the conjecture.

The selected coupling supplies a complete chiral output carrying the charged heat and entropy. Every retained usable register, sensor and program update enters $s_{1:N}$, with $s_{1:N}/N \to \bar{s} > 0$ for the actual adaptive run. Initial usable purity and correlation capacity are $o(N)$; there is no positive rate of uncharged purity or nonequilibrium resource consumption. The excess above the complete-output floor

$$\limsup_{N\to\infty} \left[\frac{2\pi Q_{1:N} - s_{1:N}}{N} - \frac{6\nu\bar{s}^2}{c_{\mathrm{ex}}} \right] \tag{51}$$

can be made arbitrarily small along the family. This is sharpness for the whole adaptive output, including correlations and every atypical label.

The physical comparison has a declared block and decision clock. Let L_N be the sorting block length, $W_N \geq L_N$ the number of cycles used for one reading, and $d_N \geq 0$ the causal transport, sorting and sensing delay, all in cycle units. The program is held through each frame of $B_N = W_N + d_N$ cycles. Its outgoing observation interval $I_j = [\eta_j^{\mathrm{out}}, \eta_j^{\mathrm{out}} + W_N/\nu]$ is physically assigned to that program, and its record is available only by the frame's end, before the next decision. Require $K_N = \lfloor N/B_N \rfloor - 1 \to \infty$, so each tested decision and its following program are inside the run; when block sorting is used, $L_N \to \infty$. For the block converter, start at a sorting boundary and choose $W_N = m_N L_N$, $m_N \in \mathbb{N}$. Pad the output latency to $r_{\mathrm{out}} L_N$ and the subsequent readout latency to $r_{\mathrm{read}} L_N$, choosing $r_i = \max\{1, \lceil \delta_i/L_N \rceil\}$ for the physical latencies δ_i in cycle units, $i \in \{\mathrm{out}, \mathrm{read}\}$; these may depend on N. The window I_j contains m_N complete output blocks after the r_{out} blocks; the final r_{read} blocks allow its record to arrive before the next decision. Thus $d_N = (r_{\mathrm{out}} + r_{\mathrm{read}})L_N$ aligns all frames and windows. Buffers, padded waiting, operation during delay and the final flush enter the complete entropy and energy account.

For fixed $Q_{\mathrm{ref}} > 0$ the normalized reading is

$$\overline{Y}_j = \frac{S_h(\eta_j^{\mathrm{out}} + W_N/\nu) - S_h(\eta_j^{\mathrm{out}})}{2\pi Q_{\mathrm{ref}} W_N}. \tag{52}$$

Whole-frame X and Z program interventions, made at the same pre-program state, history and geometric data and averaged over the declared actual-run pre-frame histories, give $\mathbb{E}(\overline{Y}_X - \overline{Y}_Z) \geq \delta_Y > 0$ uniformly as implementation and sharpness errors vanish. This is a per-cycle-normalized window contrast, not a one-cycle switch inside a pooled block.

The response changes the next program toward Z when the measured maintenance response is larger. To isolate this edge, let $P_j^{\mathrm{on/off}}(Z \mid \mathcal{F}_j)$ be the next-program probabilities with the physical response input present or blocked, respectively. The paired intervention keeps the upstream history and executed commands fixed up to that decision, including

all non-response controller inputs. With $\mathcal{F}_j$ drawn from the actual run before this tested input, require

$$G_N = \frac{1}{K_N}\sum_{j=1}^{K_N}\mathbb{E}\big[P_j^{\mathrm{on}}(Z \mid \mathcal{F}_j) - P_j^{\mathrm{off}}(Z \mid \mathcal{F}_j)\big] \geq g_0 > 0. \tag{53}$$

The denominator includes every declared decision opportunity, not only selected informative or exploration windows. The response-channel replacement used in the test and its fixed reference are specified as part of the sensor. Its finite resolution, range, stress and entropy are charged. Neither a counterfactual fixed-program contrast nor a conditional gain on rare events substitutes for this operational test. For binary X/Z, the same opportunity/history average obeys $\langle P^{\mathrm{off}}(X)\rangle_N = \langle P^{\mathrm{on}}(X)\rangle_N + G_N \geq \langle P^{\mathrm{on}}(X)\rangle_N + g_0$. These are blocked-input probabilities on matched actual-run histories, not frequencies of a separately evolved off-history; if the on average tends to zero, the blocked average remains at least g_0. The actual adaptive output still satisfies Eq. (51)*.*

Each model supplies a conserved complete stress tensor, the heat transport and a timelike worldline in a positive dilaton domain. Its actual clock has $0 < z_{\min} \leq d\tau/d\eta \leq z_{\max} < \infty$ uniformly along the family, with physical inlet temperature $T_\tau = 1/(2\pi z)$. Thermal preparation, reference stress and matching errors give a sublinear accumulated remainder in the horizon account. Drives are recovered, stored or routed away from the heat reading, or enter as explicit nonthermal sources in a matched background comparison. Their program-dependent residual remains in the contrast. Support work, clock preparation, thermal banks, buffer stages, ground offsets and final stored energy belong to the same ledger. Size, installed energy, control capacity and the dilaton domain may grow with N; each run keeps one body and a controller whose initial entropy and usable purity and correlation capacity are $o(N)$. Temperature in the rapidity frame, external exhaust capacity and cycle frequency stay fixed.

The conjecture keeps sustained renewal, sharp complete export and an informative causal return in one dynamics. At fixed $\nu, c_{\mathrm{ex}}, T_H, \bar{s}$, the quadratic excess over bare Landauer remains. Proposition B.1 excludes direct purifier dumping from the sharp target. Arbitrary-unitary reservoir constructions approach Landauer's bound [13, Section 6], but do not keep this exhaust capacity fixed.

C.1 What the fixed-source converter does not supply

This converter supports specified joint-source renewal, not correction of unknown content against an inaccessible purifier. For example, if R were instead forbidden to all operations, the noisy Z pair would have negativity $(1-2q)\sqrt{\epsilon(1-\epsilon)}$, below its target $\sqrt{\epsilon(1-\epsilon)}$. A channel on A and independent thermal ports cannot increase that negativity: partial transpose on R commutes with the channel, and trace norm contracts on Hermitian operators. It therefore cannot restore the entangled target. Preserving the distinct held content H/R_{old} alongside this converter requires the stated separate content-holding interface.

In a long fixed-program calibration window, suppose its full debit is $\bar{s}_p$ and converter and horizon remainders are sublinear. For $F(s) = s + 6\nu s^2/c_{\mathrm{ex}}$ the matched mean reading at the floor is

$$\overline{Y}_p = \frac{F(\bar{s}_p)}{2\pi Q_{\mathrm{ref}}} + o(1), \qquad \Delta\overline{Y} = \frac{F(\bar{s}_X) - F(\bar{s}_Z)}{2\pi Q_{\mathrm{ref}}} + o(1). \tag{54}$$

For the quartet alone, $\nu = 1, c_{\rm ex} = 4$ give $F(h_Z) = .0993021514$, $F(h_X) = .2576276961$ and $\Delta\overline{Y} = .02519829/Q_{\rm ref}$. Here $\overline{Y}_p$ is the normalized window reading (52), under the stated fixed-program calibration assumptions, not a consequence of the adaptive sharpness clause. A single changed preparation within a pooled L-cycle block changes its mean debit by $O(L^{-1})$ and its normalized heat by $O(L^{-1})$; it does not produce this whole-window contrast.

Separate fixed-program sharpness does not prove the whole adaptive target. If time fractions a_p are occupied by long individually sharp phases with debits $\bar{s}_p$, their combined heat has the excess

$$\sum_p a_p F(\bar{s}_p) - F\left(\sum_p a_p \bar{s}_p\right) = \frac{6\nu}{c_{\rm ex}} \operatorname{Var}_a(\bar{s}_p). \tag{55}$$

This is positive for nonzero fractions and unequal debits. Joint block sorting can remove that energy gap at the channel level, but the present proof supplies no local program contrast or causal return after that sorting. Correlations with retained program history also enter the slack in Eq. (48). These are joint obstacles to Eq. (51). Sparse exploration with mean X-window share $a_N \to 0$ removes this phase-variance term but settles neither window gain. For independent noisy window tests with a fixed nonzero information per trial, vanishing estimation error requires the expected test count $K_N a_N \to \infty$. If window accuracy improves with N, the requirement concerns the accumulated physical detection information instead; counting windows alone is insufficient. Let b_N be the frequency of response-sensitive opportunities under the same opportunity/history measure as G_N. If on/off kernels coincide elsewhere, $|G_N| \le b_N$. When updates occur only in X frames, $b_N \le a_N$ with a_N evaluated over the same tested opportunities and actual-run histories. For equal-length windows, a_N is the expected X share of observed cycles, not the fraction of the full run covered by X observations. Hence $a_N \to 0$ violates Eq. (53) despite fixed conditional contrast. Exploration and update duty are not identical in general: continuous monitoring during Z windows could keep b_N nonvanishing. Its compatibility with charged sensing, whole-output sharpness and a causal return is unproved, not excluded.

Under isotropic depolarization the two noisy quartet spectra and full debits instead coincide. Their separately entropy-optimal matched heat outputs have zero asymptotic program contrast; unsorted source contrast can become recovered work. This excludes that particular sharp-heat mechanism, not different full control debits or every causal realization using depolarization. A fixed initial record also obeys $I(X : Y_{1:N}|Z) \le H(X|Z)$ for charged baseline Z; sublinear initial record entropy cannot supply a positive new-information rate indefinitely.

The exact marginal-horizon balance gives $\overline{Y}_j = Q_{I_j}/(W_N Q_{\rm ref})$ when the entire matched source is charged heat. Drive, other and reference-source integrals add $[E_{{\rm drive},I_j} + E_{{\rm other},I_j} + E_{{\rm ref},I_j}]/(W_N Q_{\rm ref})$ in the same normalization. A common background cancels in program comparisons only with a justified common clock and transport; program-dependent residuals remain. Table 1 separates the available finite readouts from this joint inlet, renewal, sensor and outlet task. In particular, Q-RAIF C's massless sorter and massive-outside full-angle gate are different layouts [14].

The sorting-based route fails without source entropy concentration or controlled energy tails. Any joint realization must also avoid positive asymptotic output deficit or slack, loss of program contrast after drive and sensing charges, and failure of causal routing, temperature matching or the operating worldline.

Fresh, independently discarded quadrature drives have a separate accuracy cost [21]: at error ϵ_{drv} their input energy is at least $\omega_h[\pi/(4\sqrt{\epsilon_{\mathrm{drv}}})+O(1)]$, where $\omega_h^{-1} = \int |f|^2\omega^{-1}d\omega$. Reused drives retain correlations and correlated failures; recovery is not established by fresh-drive accuracy. At $\epsilon_{\mathrm{drv}} = .01$ the finite bound is $6.74238\omega_h$. Choose the source parameter $\epsilon = .01$ independently of ϵ_{drv}. With $q = .05$, $\nu = 1$ and $c_{\mathrm{ex}} = 4$, the complete quartet debits are $(h_Z, h_X) = (.01370657, .19851524)$. Their heat floors $F(h_p)/(2\pi)$ are $(.00222632, .04100272)$. With the additional benchmark choice $\omega_h = 1$, the drive/heat ratios are about $(3028, 164)$; unrecovered fractions $(.0003302, .0060813)$ already equal one corresponding heat unit. At the main example's $\epsilon = .1$, the Z floor is .01580443 and that ratio is about 427, while the X values are unchanged. These quartet-only benchmarks exclude additional charged controller debits and do not identify the drive with the quartet's full renewal device. Gate spectrum, duration and debit set the ratio; no universal $\omega_h \geq \nu$ or universal drive/heat multiplier follows. Outgoing coherent drive energy uses the nonthermal ledger in Eq. (49). Recoverable energy, stored energy and exported heat are separate entries; the same outgoing energy is never charged twice.

The exact moving-clock conversion is

$$z = \frac{Ly}{R}, \qquad y = \sqrt{R_b^2 + L^2\dot{\Phi}_b^2} - L\dot{\Phi}_b, \quad R_b^2 = \Phi_b^2 - R^2, \tag{56}$$

with proper-time dot. The tracking-clock conversion is derived in Appendix H.1, Eq. (102). Thus $z = L\rho$ is a stationary limit. The physical input needs $zT_\tau = T_H$, and support power is $(L^{-1} - K)\dot{\Phi}_b/\kappa$ in Eq. (96), including its actual sign and source. Prescribing the trajectory or transforming temperature constructs neither support nor inlet. When the monitored source carries heat plus a sublinear reference remainder, its source–clock sector has $R = R_0 + \kappa Q_{1:N} + o(N)$ at a positive bounded mean heat; an exterior operating domain must therefore reach $\Phi_b > R = \Omega(N)$. With bounded z, an isolated reserve paying all per-cycle losses also requires initial energy $\Omega(N)$; replenished charged work is an alternative. The family must report its upper as well as lower size, installed-energy, clock-domain and compiler scalings. A stronger resource target would be $O(N)$ installed energy and dilaton range, with polynomial control capacity; this is an optional benchmark, not a hypothesis of Conjecture C.1. Faster growth would miss that benchmark without refuting the stated conjecture. Fixed-gap and fixed-cutoff examples alone give no uniform growing-run certificate.

This loop would connect selection to its source and backreaction. Its finite pointer algebra still needs a map to the modular-stable geometric algebra, and Conjecture 4.2 still asks for the origin of the gravitational action.

D Common-action implementations and their unresolved joins

Q-RAIF C supplies a local common action with dynamical support, memory and clock fields and a specified positive finite canonical ordering [14]. This permits a concrete attack on the missing source: freeze the texture only as a controlled limit, quantize the retained material and controls together, and compare the resulting local gates with the actual acquired-history Hamiltonian. Direct substitution fails already because dynamical support and clock introduce kinetic cross terms and forces absent from a fixed-profile gate. Moreover a spatial canonical cutoff preserves its positive form but does not by itself

provide a covariant renormalized Ward identity. Classical on-shell conservation therefore cannot certify the complete quantum source. The coupled tolerance above closes the trajectory/reader comparison in the retained mean-action branch; it cannot remove these two operator and continuum obstacles.

For the real-quartet carrier, Proposition J.8 supplies that Ward identity and the actual full-mode bulk state. Its projective and Weyl conversion is varied with the clock and director; adding a complete vacuum effective action to the retained canonical stress would double count the quantum response. This bulk construction does not yet transport the entire reset protocol through dynamical walls.

The finite receiver-renewal family of Q-RAIF C holds clock mean energy and norm fixed while its space, time and paid Gibbs bank grow. It is a real finite update result, with old references retained. It renews a receiver rather than the entire interacting material. In the 31-site full-program diagnostic, the receiver occupation .157712 exceeds the target .13 by .027712; this is not a full-program state error. The independently reported completion probability is .857524. An additional complete-ring construction renews the same body to fixed per-cycle tolerance through four thermal outlet species, retaining old correlations and restoring the original mixed receiver and its current purifier by local neutral pumps [14]. At the same incoming history, changing the maintained logical axis changes the stress emitted by that reset, with a strictly positive finite-error bound; no separate program-tagged emitter is added. Its prescribed drives are paid, and its energy-constrained storage stays bounded at fixed parameters. Its finite smooth schedule can also be driven by a work-clock field whose source includes the old incoming correlations [14, Section on a spatial work clock]. That result has static endpoints and a supplied frame. The bulk covariant construction and this fixed-frame schedule have not yet been joined with those endpoints and the sharp excess-heat limit required here. The finite-program diagnostic remains evidence against combining successful components without a common state and dynamics. The conjectured common theory would be refuted by an unavoidable extra extensive source or renewal price, by a persistent mismatch of its selected modular data, or by a response gain that vanishes under the stated fixed resources. These are different failure modes, not a general impossibility claim.

E Modular definitions and candidate checks

E.1 Preliminaries

Standard references are Takesaki [40], Bratteli–Robinson [41] and Haag [42].

Definition E.1. *A **von Neumann algebra** $\mathcal{M}$ on $\mathcal{H}$ is a $*$-subalgebra of $\mathcal{B}(\mathcal{H})$ containing the identity and closed in the weak operator topology; equivalently $\mathcal{M} = \mathcal{M}''$, where $\mathcal{M}'$ is the commutant. It is a **Type III_1 factor** if $\mathcal{M} \cap \mathcal{M}' = \mathbb{C}\mathbf{1}$, it admits no semifinite normal trace, and its Connes spectrum is $\mathbb{R}_{\geq 0}$.*

Under standard conditions, local algebras of quantum field theories are Type III_1 factors [42, 43].

A vector $|\Omega\rangle$ is **cyclic** for $\mathcal{M}$ if $\mathcal{M}|\Omega\rangle$ is dense, and **separating** if $M|\Omega\rangle = 0$ implies $M = 0$. For cyclic and separating $|\Omega\rangle$ the antilinear map $S_\Omega : M|\Omega\rangle \mapsto M^*|\Omega\rangle$ is closable, and its closure has the polar decomposition $S_\Omega = J_\Omega \Delta_\Omega^{1/2}$ with modular conjugation J_Ω and modular operator Δ_Ω.

Theorem E.2 (Tomita–Takesaki [40]). *With $\mathcal{M}$, $|\Omega\rangle$, J_Ω, Δ_Ω as above, $J_\Omega \mathcal{M} J_\Omega = \mathcal{M}'$, and the modular group $\sigma_t^\Omega(M) := \Delta_\Omega^{it} M \Delta_\Omega^{-it}$ satisfies $\sigma_t^\Omega(\mathcal{M}) = \mathcal{M}$ for all real t.*

A state ω is KMS at parameter β in the modular-time convention used here if for all $A, B \in \mathcal{M}$ there is a function $F_{A,B}$ analytic in $0 < \operatorname{Im} z < \beta$ and continuous on the closure, with $F_{A,B}(t) = \omega(\alpha_t(A)B)$ and $F_{A,B}(t + i\beta) = \omega(B\alpha_t(A))$. This ordering gives $\beta = 1$ for $\sigma_t^\omega(A) = \rho^{it} A \rho^{-it}$ in a density-matrix representation; the usual future physical Hamiltonian time has the opposite modular orientation.

Theorem E.3 (Takesaki [40]). *A faithful normal state ω on $\mathcal{M}$ is KMS at $\beta = 1$ for σ_t^ω, and σ_t^ω is the unique one-parameter automorphism group with this property.*

The modular flow is thus the unique time evolution for which the given state looks thermal. In the Rindler wedge it is the Lorentz boost, and the KMS condition at $\beta = 2\pi$ gives the Unruh temperature $T_U = a/(2\pi)$.

Definition E.4 ([44]). *An inclusion $\mathcal{N} \subset \mathcal{M}$ with common cyclic and separating vector $|\Omega\rangle$ is a **half-sided modular inclusion** (HSMI) if $\sigma_t^{\mathcal{M}}(\mathcal{N}) \subset \mathcal{N}$ for all $t \leq 0$.*

Theorem E.5 (Wiesbrock [44]). *For an HSMI in the stated $t \leq 0$ convention, the closure of $P = (\log \Delta_{\mathcal{N}} - \log \Delta_{\mathcal{M}})/(2\pi)$ is positive. The canonical unitary group is $U(a) = e^{iaP}$, fixes $|\Omega\rangle$, and obeys $\mathcal{N} = U(1)\mathcal{M}U(1)^*$ and $\Delta_{\mathcal{M}}^{it} U(a) \Delta_{\mathcal{M}}^{-it} = U(e^{-2\pi t} a)$ [15, Thm. 2.1].*

The translation from $\mathcal{M}$ to $\mathcal{N}$ is therefore encoded algebraically, without background geometry. This is the standard tool for extracting translation structure from modular data.

E.2 Example 1: Rindler wedge

Consider a free massless scalar field in $(1 + 3)$-dimensional Minkowski spacetime with Fock space $\mathcal{H}_U = \mathcal{F}(L^2(\mathbb{R}^3))$ and vacuum $|\Omega\rangle$. Let $\mathcal{W}_R = \{x > |t|\}$ be the right Rindler wedge and $\mathcal{R}(\mathcal{W}_R) = \{e^{i\phi(f)} : \operatorname{supp} f \subset \mathcal{W}_R\}''$. For this check we take $\mathcal{M}_U = \mathcal{R}(\mathcal{W}_R)$, which makes the ambient modular action well defined without performing a nontrivial selection.

The Reeh–Schlieder theorem [42] makes $|\Omega\rangle$ cyclic and separating for $\mathcal{R}(\mathcal{W}_R)$, so (U1) holds. By the Bisognano–Wichmann theorem [45, 46], $\Delta_\Omega = e^{-2\pi K}$ with K the global boost generator in the x-direction. The modular action is $\sigma_t^\Omega(X) = e^{-2\pi i t K} X e^{2\pi i t K}$; its spacetime action preserves the wedge, giving $\sigma_t^\Omega(\mathcal{R}(\mathcal{W}_R)) = \mathcal{R}(\mathcal{W}_R)$, which is (U2). No decoherence semigroup is specified, so (U3) is not verified. Haag duality, $\mathcal{R}(\mathcal{W}_R)' = \mathcal{R}(\mathcal{W}_L)$, does not imply that every proper extension of the wedge algebra contains the opposite wedge algebra, nor any maximality statement.

On the $t = 0$ Cauchy slice the full modular generator is

$$K_{\mathrm{mod}} = -\ln \Delta_\Omega = 2\pi K, \qquad K = \int_{\mathbb{R}^3} x\, T_{00}(0, x, y, z)\, d^3x. \tag{57}$$

Formally this is $2\pi(K_R - K_L)$, with $K_R = \int_{x>0} x T_{00}\, d^3x$ and $K_L = \int_{x<0} (-x) T_{00}\, d^3x$. The right-wedge expression $2\pi K_R$ alone is not $-\log \Delta_\Omega$: in a factorizing regulator it is the right reduced-state modular Hamiltonian, up to its normalization constant. In the Type III theory the separate one-sided charges require an appropriate quadratic-form or regulator interpretation. For a local unitary excitation in the right wedge, the

opposite-wedge expectation is unchanged, so the change of the full boost charge equals the right-sided change when those differences exist. This does not identify the two operators. The example reproduces the Bisognano–Wichmann geometric modular action; it does not show that (U1)–(U3) select the wedge algebra.

E.3 Example 2: Qubit chain with dephasing

Consider n qubits with Hamiltonian $\hat{H} = -J\sum_i \sigma_z^{(i)}\sigma_z^{(i+1)} - h\sum_i \sigma_x^{(i)}$, $J > 0$, coupled to an environment through

$$\mathcal{T}_t(\rho) = e^{t\mathcal{L}}(\rho), \qquad \mathcal{L}(\rho) = -i[\hat{H}, \rho] + \gamma \sum_i \left(\sigma_z^{(i)} \rho\, \sigma_z^{(i)} - \rho\right), \tag{58}$$

with Heisenberg dual $\mathcal{E}_t := \mathcal{T}_t^*$ and global state $\omega = Z^{-1}e^{-\beta\hat{H}}$. For an exact check we take $h = 0$; for $h \neq 0$ the diagonal algebra is stable only approximately in a strong-dephasing regime.

For $h = 0$ the diagonal observables are fixed by the Heisenberg semigroup, and the candidate is the maximal abelian algebra

$$\mathcal{A}_{\mathrm{geo},\mathbf{c}} = \{f(\sigma_z^{(1)}, \ldots, \sigma_z^{(n)})\}'' = \mathcal{A}_{\mathrm{diag}}. \tag{59}$$

The thermal state is positive definite at finite β, so $\omega(|s\rangle\langle s|) = \langle s|\rho_{\mathrm{th}}|s\rangle > 0$ for every computational basis state, which gives (U1); diagonal elements of a positive definite matrix are positive in any basis. The ambient modular flow is $\sigma_t^{\omega,U}(X) = \rho_{\mathrm{th}}^{it} X \rho_{\mathrm{th}}^{-it}$; for $h = 0$, ρ_{th} is diagonal and commutes with $\mathcal{A}_{\mathrm{diag}}$, giving (U2). For $h \neq 0$ the conjugation generally leaves $\mathcal{A}_{\mathrm{diag}}$. Under the dephasing dual, off-diagonal matrix units decay as $e^{-2\gamma\, d(s,s')\, t}\, |s\rangle\langle s'|$ with Hamming distance d; for $h = 0$ every diagonal observable is fixed, so (U3) holds and $\mathcal{A}_{\mathrm{diag}} \subset \mathrm{Fix}(\mathcal{E})$. This does not prove maximal setwise invariance, since the full matrix algebra is always mapped into itself.

The modular flow acts trivially on the abelian pointer algebra, and the entropy of a block A within $\mathcal{A}_{\mathrm{diag}}$ is the Shannon entropy of the marginal distribution $p(s_A)$. The example checks one modular- and semigroup-stable candidate, which under exact fixation is the pointer algebra; it neither verifies the Type III regime nor selects the candidate uniquely by (U1)–(U3). Selection of another type is available here: for $h = 0$ the fixed algebra is the commutant of the $\sigma_z^{(i)}$, and the window theorem of Paper B shows that every sufficiently long observation window selects it as the unique maximal stable algebra [20].

F Source identification, carriers and profiles

F.1 Continuing calibration and identification of the source

Source control need not interrupt stored quantum content. With an independently prepared vacuum mode and full output collection, Q-RAIF B's corrected parity instrument has branches

$$\mathcal{I}_\pm(\rho) = \tfrac{1}{2}(1 \pm e^{-2k(1-\alpha a)^2})\rho.$$

It yields the same calibration statistic without preparing a new body state. The coherent controlled-parity identity $Re^{-imZ_L Q_f}R = I_L \otimes e^{-imQ_f}$ goes further: it protects the body's marginal for an arbitrary mode state by transferring the interaction to the field alone [21].

For a centered independent input the coupling-added stress is still $m^2 F_f$. The meter can be returned coherently; retaining a calibration outcome is a separate read and memory operation. Partial collection loses coherence during the inquiry itself. Brain Anchor prices that loss, bounds continuing retuning under drift and gives an exact posterior criterion for when an extra inquiry pays for its source and apparatus price [57]. No geometric response or entropy export is needed for these preservation identities.

The inquiry and maintenance choices can also share one running body. Brain Anchor tracks both the interface posterior b_x and the coherence-weighted mass c_x in a finite hidden-state model. Equal b_x and even equal reduced body states can require different next commands, because past loss has different correlations with the interface. Q-RAIF B realizes collection by finite resonant contacts, including successive contacts with a finite packet. In the matched field realization, the source at the next encounter is still $m_j^2 F_f$, irrespective of the accessible correction, at the interaction output before collection. A collector can redistribute that energy, so this is not its full downstream stress. For a history-dependent policy the coefficient is computed from that policy's hidden-path weights, not from a new independent gain draw at every step. The four-encounter joint example gives $\sum_j \mathbb{E} m_j^2 = .588965$, against .640000 with permitted maintenance but no retained calibration reads. These are source accounts; the supplied objective's prices are not an entropy-export measurement. Using these sources in the JT response also requires their profiles and emission times and any collector contribution. The continuous collection route below specifies a free-field remainder, but does not supply the wall, thermal exhaust or entire sensing apparatus of the response model.

F.1.1 Continuous collection changes the outgoing profile

Q-RAIF B also gives an excitation-preserving, positive-frequency collection model [21]. A retained oscillator of frequency Ω couples to a continuum $\omega \in [\Omega - B, \Omega + B]$, with real even form factor $g(\omega - \Omega)$. A prescribed finite source drive prepares the matched packet. The one-particle propagator S then determines both the collected fraction and the outgoing wave. Its finite waveform certificate allows clock, arrival and switching errors; it is not a white-noise limit. The field interaction is band-smeared, so its spatial tails are part of the model. Parameters are fixed within each encounter.

Let $z_{\mathrm{out}}(\omega)$ denote the asymptotic free outgoing amplitude for a unit source drive, at the same null-time origin as the incoming packet, and let the body-conditioned amplitudes be $\pm m\beta z_{\mathrm{out}}$. With the current normalization of Eq. (10), put

$$\chi(u) = -i \int_{\Omega - B}^{\Omega + B} \sqrt{\frac{\omega}{4\pi}} z_{\mathrm{out}}(\omega) e^{-i\omega u} d\omega, \quad F_{\mathrm{out}}(u) = 4m^2\beta^2 [\mathrm{Re}\, \chi(u)]^2. \tag{60}$$

For independent vacuum preparation, this is the outgoing free-field stress for every logical input. Tracing the body gives a mixture of opposite coherent amplitudes, whose normal-ordered squared currents agree. Accessible parity correction does not change this marginal stress. For a capture pulse ending at T, the rotating-frame amplitude is $z_{\mathrm{out}}(\omega) = e^{i(\omega - \Omega)T} [P_{\mathrm{field}} S(0, w)](\omega)$, with the prescribed source carrier phase included. Removing the final free propagation phase fixes the common time origin; it must not be counted again when placing the source in the JT model. The input occupation is $k_{\mathrm{eff}} m^2 = \beta^2 m^2 \|w\|^2$. For real envelopes and an even band, the source, receiver and free remainder energies in units $\hbar = 1$ are, respectively,

$$\Omega k_{\mathrm{eff}} m^2, \qquad \Omega \xi k_{\mathrm{eff}} m^2, \qquad E_0^{\mathrm{out}} = \Omega(1 - \xi) k_{\mathrm{eff}} m^2. \tag{61}$$

Capture moves energy into the receiver. A later vacuum swap exports it to a separate exhaust. The net prescribed capture-switch work is zero for this symmetric packet family, while source preparation supplies the first energy in Eq. (61). The receiver, exhaust, controls and supports do not disappear from a gravitational account.

Frequency regularity matters for using this source in the JT moment formulas. A hard flat band generally gives $F_{\mathrm{out}}(u)$ tails of order u^{-2}, so finite energy alone does not ensure a finite second moment. Instead choose $g(\delta) = \sqrt{J_0}[1 - (\delta/B)^2]$ inside the band and zero outside. For a finite source envelope and finite collection time, $z_{\mathrm{out}}(\omega)$ is g times a smooth spectral function. Thus $\sqrt{\omega} z_{\mathrm{out}} \in H_0^1$ and Plancherel gives

$$E_2^{\mathrm{out}} = \int u^2 F_{\mathrm{out}}(u) du = m^2 \beta^2 \int \left| \partial_\omega (\sqrt{\omega} z_{\mathrm{out}}) \right|^2 d\omega < \infty. \tag{62}$$

E_1^{out} is finite by Cauchy–Schwarz. This specifies a legitimate free-field source for the response dictionary, rather than reusing the uncaptured packet unchanged. It does not determine the total apparatus stress or assert compact temporal support. The JT background, boundary access and matter normalization remain the declared inputs of the response model.

In the worked collection setting $\Omega = 20$, $B = 5$, $T = 16$ and $J_0 = 1/(2\pi)$, a fixed 8721-bit waveform table and stated clock tolerances certify $1 - \xi < 4.43 \times 10^{-7}$ and a relative preparation-strength uncertainty below .001. This is enough to keep the collection-induced loss below .001 in the preceding 4096-per-setting calibration example. The bound does not include parity-gate or detector errors. Repeated encounters require independent vacuum fields and exhausts, or a separately proved return/reset bound on one reused field. A fixed-capacity thermal cycle and the source–sensor return of Section 10 are not supplied by this finite-bank construction.

With several Gaussian field species, however, logical calibration alone may not identify the actual source. Let the species have the same switch profile and independent centered inputs, with quadrature covariance kW in their species coordinates. Put $G = (\mathbf{g}_1, \mathbf{g}_2)$ and $\mathbf{v}(a) = \mathbf{g}_1 - a\mathbf{g}_2$. The logical coherence and coupling-added stress are respectively

$$r_W(a) = e^{-2k\mathbf{v}(a)^T W \mathbf{v}(a)}, \qquad T_{\mathrm{add}}(u) = \|\mathbf{v}(a)\|^2 F_f(u). \tag{63}$$

The first measures $D_W = G^T W G$; the second needs $C = G^T G$. The incoming stress of the reference preparation is an additional source.

Proposition F.1 (Protection records do not always identify stress)**.** *Take* $W = \mathrm{diag}(1, 4)$ *and the two coupling families*

$$\mathbf{g}_1 = (1, 0), \quad \mathbf{g}_2 = (1, 1), \qquad \widetilde{\mathbf{g}}_1 = (0, 1/2), \quad \widetilde{\mathbf{g}}_2 = (2, 1/2).$$

Every command gives the same logical channel, with weighted residual $1 - 2a + 5a^2$*. Their stress coefficients are* $1 - 2a + 2a^2$ *and* $1/4 - a/2 + 17a^2/4$*. Thus no finite adaptive experiment using only these fresh-port logical channels distinguishes them. The noise-minimizing command is* $a = 1/5$ *for both, but their source-minimizing commands are* $1/2$ *and* $1/17$*.*

Proof. Insert each vector pair in Eq. (63). Equality of conditional channels at every command gives equality of every adaptive logical record law by induction, also with retained ancillas. Minimizing each positive quadratic gives the stated commands. □

At the noise optimum their added stress coefficients are 17/25 and 8/25. This is an ambiguity of the specified logical access, not an inability to observe stress through another instrument. It persists with the same incoming preparation and does not come from a changed background.

Proposition F.2 (Switched references identify the physical source)**.** *In the same two-species family, suppose the apparatus can independently prepare known centered Gaussian references with covariances kW_1, kW_2, where $W_1 = \mathrm{diag}(1,4)$ and $W_2 = \mathrm{diag}(4,1)$. At each reference measure the logical coherence means at $0, +h, -h$, $h > 0$. They determine the physical source Gram matrix exactly:*

$$D_{j,11} = -\frac{\log r_j(0)}{2k}, \qquad D_{j,12} = \frac{\log r_j(h) - \log r_j(-h)}{8kh},$$
$$D_{j,22} = -\frac{\log r_j(h) + \log r_j(-h) - 2\log r_j(0)}{4kh^2}, \qquad C = \frac{D_1 + D_2}{5}. \tag{64}$$

More generally, known reference weights with $I = \sum_j t_j W_j$ suffice, with $C = \sum_j t_j D_j$.

For independent ± 1 trials, N at each of the six settings, suppose all true means are at least $r_{\min} > 0$. Let $\eta = \sqrt{2\log(12/\zeta)/N} \le r_{\min}/2$, clip sample means to $[r_{\min}/2, 1]$ and use the same formulas. With probability at least $1-\zeta$, uniformly on $|a| \le A$,

$$|\widehat{V}(a) - \|\mathbf{v}(a)\|^2| \le \Delta_C := \frac{2\eta}{5kr_{\min}}\left(1 + \frac{A}{h} + \frac{2A^2}{h^2}\right). \tag{65}$$

Minimizing $\widehat{V}$ on any supplied finite command grid incurs actual excess source coefficient at most $2\Delta_C$ over its best point.

Proof. The log characteristic function is the quadratic form $-2kD_j$. Its three coefficients give Eq. (64); $W_1 + W_2 = 5I$ gives C and proves the general sufficient condition. The union Hoeffding bound covers all six means. Each log error is at most $2\eta/r_{\min}$, so the errors of C_{11}, C_{12}, C_{22} are at most $2\eta/(5kr_{\min})$, $\eta/(5khr_{\min})$ and $4\eta/(5kh^2 r_{\min})$. Insert them in $V = C_{11} - 2aC_{12} + a^2 C_{22}$ to prove the uniform bound. Two applications of that bound compare the chosen grid point with the best one. □

These are $6N$ new preparations, couplings and reads, not six free exact means. The reference covariances, their switches and their incoming energies have to be calibrated and charged. Separate logical test preparations are permitted here; applying these references to held content would also require the protection operations of Q-RAIF B and their collection account. Equation (64) identifies the coupling-added source under the declared profile and species architecture, not the full stress of an unknown apparatus. There is another route when the apparatus has an oriented field readout rather than only logical records.

Proposition F.3 (Signed field calibration with held content)**.** *For d independent input modes put $Q_s = b_s + b_s^\dagger$ and $P_s = i(b_s^\dagger - b_s)$. Assume centered independent Gaussian modes, initially independent of the body and its retained reference, a known common k, and $\mathrm{Var}(P_s)/(4k) \le \sigma_{\max}^2$. Controlled total parity conjugates $Z_L\sqrt{k}\sum_s v_s(a)Q_s$ to $I_L\sqrt{k}\sum_s v_s(a)Q_s$. After this coherent echo the signed readings $Y_s = -P_s/(2\sqrt{k})$ have mean $v_s(a)$ while the logical state remains unchanged. Two commands, $a = 0, h$ with $h \ne 0$, identify*

$$\mathbf{g}_1 = \mathbb{E}\mathbf{Y}(0), \qquad \mathbf{g}_2 = \frac{\mathbb{E}\mathbf{Y}(0) - \mathbb{E}\mathbf{Y}(h)}{h}.$$

This does not require the Gaussian variances as labels.

Suppose $|v_s(0)|, |v_s(h)| \le B$ and use a finite readout that clips Y_s to $[-Y, Y]$, $Y > B$, then rounds to the nearest point of a prescribed uniform grid of spacing $q > 0$ in that interval, adding the two endpoints if needed. This alphabet has at most $\lceil 2Y/q \rceil + 2$ symbols and absolute rounding error at most $q/2$. For N fresh queries at each command, define

$$\eta = Y\sqrt{\frac{2\log(4d/\zeta)}{N}} + \frac{q}{2} + 2\sigma_{\max}\varphi\left(\frac{Y-B}{\sigma_{\max}}\right), \qquad \varphi(z) = \frac{e^{-z^2/2}}{\sqrt{2\pi}}.$$

With probability at least $1-\zeta$, the estimated residual on $|a| \le A$ has vector error at most $e_v = \sqrt{d}\,\eta(1 + 2A/|h|)$. If its true norm is bounded by B_v, then

$$|\widehat{V}(a) - \|\mathbf{v}(a)\|^2| \le 2B_v e_v + e_v^2. \tag{66}$$

The readout uses a finite alphabet, with at most $\lceil \log_2(\lceil 2Y/q \rceil + 2) \rceil$ bits per species per query.

Proof. Total parity reverses every Q_s, so the coherent parity calculation of Q-RAIF B [21] gives a field-only displacement. The canonical relation $[Q_s, P_s] = 2i$ shifts P_s by $-2\sqrt{k}v_s(a)$ and leaves its variance unchanged. For a Gaussian with $|\mu| \le B$ and standard deviation at most $\sigma_{\max}$, the clipping bias is at most $2\sigma_{\max}\varphi((Y-B)/\sigma_{\max})$, by integrating the two tails. Hoeffding for the bounded clipped readings and a union over the $2d$ means give the first term of η; rounding adds at most $q/2$. Thus each coordinate error of $\mathbf{g}_1$ is at most η and of $\mathbf{g}_2$ at most $2\eta/|h|$. The Euclidean bound follows. Expanding $\|\mathbf{v} + \delta\mathbf{v}\|^2$ proves Eq. (66). □

The sign reference, species-resolved quadrature detector and complete mode control are its additional permissions. Two commands here mean $2N$ new field queries, each with d digitized readings; mode preparation, coupling, capture and memory remain charged. Unknown incoming means would confound the shift. Together with the matched JT equations either identified source predicts a response from actual records rather than from a supplied coupling label. Boundary sensing still checks the physical response itself. Neither route replaces the wall, clock and thermal-export construction in a closed loop.

F.2 The command carrier matters

How strongly the gravitational field records the command statistics depends on how the command is physically carried.

Proposition F.4 (Phase-carried commands)**.** *Keep the coupling switched on at every opportunity and carry the command in the drive phase: a correction uses the input $|\sqrt{N}\rangle$ and no correction uses $|i\sqrt{N}\rangle$. Then (i) the no-correction branch flips the bit with probability $1 - s_N$ and the correction branch with probability s_N; (ii) both inputs carry the same field energy $N\bar{\omega}$, and the branch sources are $4N[\mathrm{Re}\,\psi]^2 + F_g$ and $4N[\mathrm{Im}\,\psi]^2 + F_g$; (iii) for the worked packet both branch sources have the same E_0, E_1, E_2, so $\mathcal{C}_\infty$ does not depend on the command rate, while the dilaton coefficients $A(u)$, $B(u)$ at finite u do.*

Proof. With $Q = b + b^\dagger$, the bit channel multiplies coherences in the X basis by $\langle\beta|e^{-2i\lambda Q}|\beta\rangle = e^{-2\lambda^2}e^{-4i\lambda\,\mathrm{Re}\,\beta}$. For $\beta = i\sqrt{N}$ this factor is $e^{-2\lambda^2}$, real and positive, which gives (i). The stress of a coherent input β is $(\psi\beta + \psi^*\beta^*)^2$, which is $4N[\mathrm{Re}\,\psi]^2$ for

$\beta = \sqrt{N}$ and $4N[\operatorname{Im}\psi]^2$ for $\beta = i\sqrt{N}$, and the coupling adds F_g in both branches. The equal moments of the two quadrature profiles give (iii) through Eq. (21); the difference of A between branches is proportional to $\operatorname{Re}[a + i(u - u_0)]^{-2}$, which is nonzero for generic u. □

The hiding has a price. With phase-carried commands the uncorrected steps also err, so the world mean obeys $m \mapsto c(2s_N - 1)[(1 - h)m + h\nu]$ on the calibrated branch. At $N = 16$ and a known interface the stationary loss rises from 0.1896 with gated couplings to 0.2334, and the switching work per opportunity rises from $0.0039\bar{\omega}$ to $0.0386\bar{\omega}$. The phase must also be set on the incoming packet before the encounter, which for full-line packets has the same causal limitation as on-demand emission. At $N = 16$ and $\kappa = 10^{-4}$ the coefficient A at the packet centre differs between the two branch sources by 3.4×10^{-4}, and the difference decays as u^{-2}.

So the gravitational trace of a selection is not universal. With gated couplings it sits in the rate of an energy-carrying event; with phase-carried commands it sits in the shape of the source and disappears from the asymptotic invariant of this example. A general law connecting selection to gravitational response must therefore name the carrier of the selection.

F.3 A worked packet

Take the normalized mode $f(\omega) = 2a\sqrt{\omega}\, e^{-a\omega} e^{i\omega u_0}$, for which $\bar{\omega} = 1/a$ and

$$\psi(u) = -\frac{ia}{\sqrt{\pi}\,[a + i(u - u_0)]^2}. \tag{67}$$

The two quadrature profiles $4[\operatorname{Re}\psi]^2$ and $4[\operatorname{Im}\psi]^2$ have different shapes but the same normalized moments $(E_0, E_1, E_2) = (1, u_0, u_0^2 + a^2)/a$. They have power-law tails: $[\operatorname{Im}\psi]^2 \sim a^2/(\pi u^4)$ and $[\operatorname{Re}\psi]^2 = O(u^{-6})$. The absolute third moment of F_g diverges logarithmically; the first three moments are finite and are exactly those used in Eq. (21). The switch determined by $g' = 4\lambda \operatorname{Im}\psi$ and $g(\pm\infty) = 0$ is

$$g(u) = -\frac{4\lambda a(u - u_0)}{\sqrt{\pi}\,[a^2 + (u - u_0)^2]}. \tag{68}$$

Both g and F_g have full-line support. The formulas and table below are exact for this scattering idealization with asymptotic coefficient data. A command decided at a finite time cannot gate its entire switch; finite-time feedback uses the following compact approximation.

Proposition F.5 (Causal compact approximation)**.** *Choose a fixed even $\eta \in C_c^\infty(\mathbb{R})$ with $0 \le \eta \le 1$, $\eta = 1$ on $[-1, 1]$ and $\eta = 0$ outside $[-2, 2]$, and set $g_T(u) = \eta((u - u_0)/T)g(u)$. For $T \ge a$, write $F_T = (g_T')^2/4$. The unrescaled switching stresses obey*

$$\begin{aligned} \left| \int (F_T - F_g)du \right| &= O(\lambda^2 a^2 T^{-3}), \\ \int (u - u_0)(F_T - F_g)du &= 0, \\ \left| \int (u - u_0)^2 (F_T - F_g)du \right| &= O(\lambda^2 a^2 T^{-1}). \end{aligned} \tag{69}$$

The one-particle smearing vectors converge in norm, with difference $O(|\lambda|a/T)$. Hence the corresponding coherent-drive gate channels converge. A gate chosen before $u_0 - 2T$ can be applied causally. Its duration and any delay between the read and the gate are part of its task schedule.

Proof. With $w = u - u_0$ and $C = 4|\lambda|a/\sqrt{\pi}$, the tails satisfy $|g| \le C/|w|$ and $|g'| \le C/w^2$ for $|w| \ge T \ge a$. The derivative of the cutoff obeys the same w^{-2} bound on its annulus, up to a constant depending only on η. Integrating w^{-4} and w^{-2} gives the first and third bounds; parity gives the second. For $h = g_T - g$, $\|h\|_2 = O(|\lambda|aT^{-1/2})$ and $\|h'\|_2 = O(|\lambda|aT^{-3/2})$. Plancherel and Cauchy–Schwarz bound $\int_0^\infty \omega|\hat{h}|^2 d\omega/(4\pi)$ by $\|h\|_2\|h'\|_2/2$, giving the one-particle estimate. The coherent-state characteristic function is continuous in this norm. □

At finite T, the drive mode, its overlap with the incoming beam and the gate error must be recomputed from g_T before assigning values to a finite-cutoff table. If the drive is matched to the new mode, its quadrature profiles and first three stress moments converge as well. The real and imaginary profiles are Hilbert-transform pairs. For $r = (g_T - g)'$, the identity $u\mathcal{H}r = \mathcal{H}(ur) + \pi^{-1}\int r$ in the convention $\mathcal{H}r(u) = \pi^{-1}\,\mathrm{pv}\int r(v)/(u - v)dv$ has zero last term. The L^2 isometry of $\mathcal{H}$ therefore gives $\|u\mathcal{H}r\|_2 = \|ur\|_2 \to 0$. Together with convergence of the mode normalization, this proves the weighted profile convergence. Equation (13) holds exactly for each compact implementation with its actual gate parameters. The same discrete feedback law applies when the target is held between its read and gate; noise during a new delay must instead be included in the transition law.

With $A_{\mathrm{tot}} = N + p\lambda^2$ the asymptotic invariant is $\mathcal{C}_\infty = H_0^2 + 2\kappa H_0 u_0 A_{\mathrm{tot}}/a - \kappa^2 A_{\mathrm{tot}}^2$. Relative to the same beam without couplings,

$$\delta\mathcal{C}(p) = p\lambda^2\Big[\frac{2\kappa H_0 u_0}{a} - \kappa^2\big(2N + p\lambda^2\big)\Big], \tag{70}$$

which has an explicit quadratic term in p. Table 2 uses $a = 1$, $u_0 = 3$, $H_0 = 1$, $\kappa = 10^{-4}$, the task parameters $\nu = 0.8$, $c = 0.9$, $h = 0.4$ of Ref. [21], and the uncalibrated ($q = \frac{1}{2}$) and calibrated (32 passive blocks, $q = 0.0404888$) coupling rates.

N	gate error	p, $q = \frac{1}{2}$	p, q_{32}	$\delta\mathcal{C}$, $q = \frac{1}{2}$	$\delta\mathcal{C}$, q_{32}	beam share	quad./lin.
1	0.3544	0.15525	0.11413	5.746×10^{-5}	4.224×10^{-5}	7.0×10^{-2}	-1.2×10^{-6}
4	0.1327	0.15153	0.10698	1.402×10^{-5}	9.897×10^{-6}	4.1×10^{-3}	-2.7×10^{-7}
16	0.0371	0.15033	0.10468	3.476×10^{-6}	2.420×10^{-6}	2.5×10^{-4}	-6.7×10^{-8}
64	0.0095	0.15002	0.10408	8.657×10^{-7}	6.006×10^{-7}	1.6×10^{-5}	-1.7×10^{-8}

Table 2: JT invariant response to the coupling-added source of the worked packet. "Beam share" is $\delta\mathcal{C}(p)$ at q_{32} divided by the beam's own change $\mathcal{C}_\infty(N) - \mathcal{C}_\infty(0)$; "quad./lin." is the ratio of the p^2 term of Eq. (70) to the term linear in p.

At $N = 16$ the ratio of the two responses is 0.696342074, against the coupling-rate ratio 0.696342054. For $\kappa = 0.01$ and 0.05 ($\kappa N = 0.16$ and 0.8) the response ratios are 0.6963442 and 0.6963560; the quadratic term stays small because $p\lambda^2$ is small, not because κN is. The calibration-dependent part is a small fraction of the beam's own response, and that fraction varies roughly as N^{-2}: it is 7% at $N = 1$, where the gate fails 35% of the time, and 1.6×10^{-5} at $N = 64$. Its size is therefore a property of the chosen

drive and background, not a general feature of selection, and a field-energy ratio such as $p\lambda^2/N$ is not a general invariant response ratio. In this example the bracket of Eq. (70) is positive, so better calibration lowers $\delta\mathcal{C}$; for other backgrounds the bracket can vanish or change sign.

The mean-source solution is not the mean of branch solutions. For a single command $I \sim \text{Bernoulli}(p)$, $\mathbb{E}\,\mathcal{C}_\infty(N + \lambda^2 I) - \mathcal{C}_\infty(N + p\lambda^2) = -\kappa^2\lambda^4 p(1-p)$, which counts only the classical command randomness and not field fluctuations within a branch. With M independent copies of the whole experiment, each with its own hidden instrument, calibration records and drives, and with $G = G_1/M$, the classical source-fraction variance of smooth stress readings falls as $1/M$. A common hidden label shared by all copies does not give this independence, and a coherent drive still has nonzero energy variance. The exact mean-source solution does not show that a single quantum world has a definite classical geometry.

F.4 What remains input

The construction takes as given the JT action and its anti-de Sitter background, the matter field and its central-charge matching, the boundary data and the identification of the downstream null coordinate with the preparation coordinate, the semiclassical use of mean stress, and the actuator's field, switch, clock and fresh ports. The two-dimensional sector theorem permits either one or several cones; it neither derives this matter–gravity identification nor forbids a dynamical route to it [23]. The source is computed outside the apparatus, for the free outgoing field; the stress of the switch and its support is not included, and a covariant detector stress tensor shows that it cannot simply be dropped in the coupling region [59]. Within these inputs the response is exact.

G Finite boundary sensing and returned records

G.1 A conditional readback route

The matter action in Section F.4 does not directly couple to Φ. A bulk detector of the matter metric alone therefore does not acquire a local dilaton readout. Boundary motion offers a different route if a physical reflecting boundary with a known dilaton condition is available.

Proposition G.1 (A dilaton-blind record law)**.** *Compare two dilaton solutions with the same matter metric, the same accessible matter input states and the same initial controller state. Suppose every instrument and conditional evolution used by the controller depends on this metric and its earlier records, but not on the dilaton or a boundary that senses it. Then every finite adaptive record sequence has the same distribution in the two solutions.*

Proof. The first instrument has the same input and conditional outcome law. Given any earlier record sequence, the next instrument and input evolution are again the same. Induction gives equality of the joint record laws. Private random bits can be included in the controller state and the induction. □

This statement does not exclude inference from previously recorded sources. It excludes a new dilaton signal from an otherwise identical metric-only experiment. A reflecting boundary evades its hypothesis because its position depends on Φ.

Proposition G.2 (Reconstruction from three boundary echoes)**.** *In a source-free interval let a, b, γ be constant, $\mathcal{C} = a^2 + b\gamma > 0$, and suppose the apparatus can time reflections at the supplied boundary $\Phi = \Phi_b > \sqrt{\mathcal{C}}$. Use one calibrated null coordinate frame. Three reflection pairs (u_i, v_i), with distinct u_i and $d_i = (a + \Phi_b)u_i - \gamma \neq 0$, determine a, b, γ uniquely:*

$$\begin{pmatrix} 1 - u_1 v_1 & u_1 & v_1 \\ 1 - u_2 v_2 & u_2 & v_2 \\ 1 - u_3 v_3 & u_3 & v_3 \end{pmatrix} \begin{pmatrix} a \\ b \\ \gamma \end{pmatrix} = \Phi_b \begin{pmatrix} 1 + u_1 v_1 \\ 1 + u_2 v_2 \\ 1 + u_3 v_3 \end{pmatrix}. \tag{71}$$

Writing this matrix as M, its determinant is

$$\det M = \frac{2\Phi_b(\mathcal{C} - \Phi_b^2) \prod_{i<j}(u_j - u_i)}{\prod_i d_i} \neq 0.$$

If the u_i are exact and the measured returns are $v_i + \delta v_i$, an invertible measured matrix $\widetilde{M}$ yields the exact coefficient error

$$\delta(a, b, \gamma)^T = \widetilde{M}^{-1}(\delta v_i d_i)_{i=1}^3, \qquad \|\delta(a, b, \gamma)\|_2 \leq \|\widetilde{M}^{-1}\|_2 \|(\delta v_i d_i)_i\|_2. \tag{72}$$

Proof. The boundary is $v = V(u) = (a + bu - \Phi_b)/[(a + \Phi_b)u - \gamma]$, with $V' = (\Phi_b^2 - \mathcal{C})/[(a+\Phi_b)u-\gamma]^2 > 0$, so it is timelike. Its equation gives Eq. (71). Multiplying row i of M by d_i leaves evaluations of three quadratic polynomials; their coefficient determinant is $2\Phi_b(\mathcal{C} - \Phi_b^2)$, giving the displayed Vandermonde factor. In the noisy equation the difference between the measured right side and $\widetilde{M}(a, b, \gamma)^T$ is exactly $(\delta v_i d_i)_i$. □

Null signals can carry these timestamps back to a receiver before the next command. They reconstruct the coefficients, hence $\mathcal{C}$, in a declared source-free interval. The inverse becomes ill-conditioned near $\Phi_b = \sqrt{\mathcal{C}}$ or coincident sample times. Exact reconstruction is a test-probe limit, not three cost-free finite-energy measurements. A usable protocol must supply the reflecting boundary, its support, calibrated clocks, timing resolution, causal waiting time and probe stress; the last changes the background beyond that limit. Six timestamps stored with b bits each occupy $6b$ logical bits, not a determined physical energy or complete memory cost. This gives an explicit sensing interface to implement, rather than granting the controller direct access to Φ.

G.2 Probe backreaction and a returned program

A finite probe contributes stress, including the reflected component. The inverse can retain both chiral components instead of treating the probe as free. Work in the mean-source JT equations of Section F.4, with the same trace-anomaly subtraction. Suppose the complete additional bulk source in the sensing interval is calibrated as $T_{uu} = f(u)$, $T_{vv} = h(v)$, $T_{uv} = 0$. Define

$$\begin{aligned} A_f(u) &= -\kappa \int_{u_*}^{u} (u - z) f(z)\, dz, & B_f(u) &= -\kappa \int_{u_*}^{u} z(u - z) f(z)\, dz, \\ A_h(v) &= -\kappa \int_{v_*}^{v} (v - z) h(z)\, dz, & B_h(v) &= -\kappa \int_{v_*}^{v} z(v - z) h(z)\, dz, \\ K(u, v) &= A_f(u) + vB_f(u) + A_h(v) + uB_h(v). \end{aligned} \tag{73}$$

The matching convention fixes the lower limits and the homogeneous coefficient data. These are calibrated sources, not a derived reflection law for the wall. If reflection or

support stresses introduce other components or unknown source terms, the following inverse must be changed accordingly.

Proposition G.3 (Source-compensated echo inverse)**.** *For a known K as in Eq.* (73)*, the solution is*

$$\Phi(u,v) = \frac{a + bu + \gamma v - auv + K(u,v)}{1 + uv}. \tag{74}$$

Three points on the supplied boundary $\Phi = \Phi_b$ determine the coefficients $\boldsymbol{\theta} = (a, b, \gamma)^T$ whenever their matrix M in Eq. (71) *is invertible:*

$$M\boldsymbol{\theta} = \big(\Phi_b(1 + u_i v_i) - K(u_i, v_i)\big)_{i=1}^{3}. \tag{75}$$

The source-free determinant formula need not hold for these altered echoes; invertibility persists under sufficiently small perturbations of an invertible source-free matrix.

Suppose the measured u_i, v_i have errors at most δ_u, δ_v, the calibrated K has pointwise error at most η_K, and the measured matrix $\widetilde{M}$ is invertible. On a domain joining true and measured points, bound the derivatives of

$$R(u,v) = (\Phi_b - a) + (\Phi_b + a)uv - bu - \gamma v - K(u,v)$$

by $|R_u| \le L_u$ and $|R_v| \le L_v$. Then

$$e := \|\widehat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|_2 \le \sqrt{3}\|\widetilde{M}^{-1}\|_2(L_u\delta_u + L_v\delta_v + \eta_K), \tag{76}$$

$$|\widehat{\mathcal{C}} - \mathcal{C}| \le \sqrt{4a^2 + b^2 + \gamma^2}\, e + e^2, \qquad \widehat{\mathcal{C}} = \widehat{a}^2 + \widehat{b}\widehat{\gamma}. \tag{77}$$

If the next projection angle is $\vartheta_{\text{next}} = F(\widehat{\mathcal{C}})$ for a supplied Lipschitz program F, its error relative to $F(\mathcal{C})$ is at most $\mathrm{Lip}(F)$ *times Eq.* (77)*.*

Proof. The f contribution is the retarded solution already obtained in Section 8; interchange u, v for the h contribution. Their sum satisfies all three JT equations and gives Eq. (74). Evaluation on the boundary yields Eq. (75). The true residual $R(u_i, v_i)$ is zero. The residual at each measured point, including the calibration error, is bounded by $L_u\delta_u + L_v\delta_v + \eta_K$. Multiplication by $\widetilde{M}^{-1}$ gives Eq. (76). Expanding $\widehat{a}^2 + \widehat{b}\widehat{\gamma}$ leaves a linear term with gradient $(2a, \gamma, b)$ and the quadratic term $(\delta a)^2 + \delta b\delta\gamma$, whose magnitude is at most e^2. The program bound follows from its Lipschitz property. □

An apparatus needs a certified parameter range to bound L_u, L_v and the conditioning of the inverse before use. Neither the assumed calibration of K nor this error bound supplies a timing POVM or the wall's support energy.

Here is a causal one-step return in that declared geometry. Take the selected projection of Section 8.4, a fresh vacuum field and the independent bit state $(I + 0.8Z)/2$. Use a smooth compact switch

$$g(u) = A\exp\left[-\frac{1}{1 - ((u-3)/0.25)^2}\right] \quad (|u-3| < 0.25), \qquad g(u) = 0 \quad \text{otherwise,} \tag{78}$$

with A fixed by $\int (g')^2/4\, du = 0.25$. Unlike the full-line packet of Section 8, this source finishes before the readout begins. Set $H_0 = L = 1$, $\kappa = 0.05$ and $\Phi_b = 3$. The source moments are $E_0 = 0.25$, $E_1 = 0.75$, $E_2 = 2.257177876$; a coupled population r gives

$$(a, b, \gamma) = (1 + \kappa r E_1, -\kappa r E_0, \kappa r E_2).$$

The projections P_0 and $P_{\pi/2}$ give $r = 0.9$ and 0.5 respectively, and hence $\mathcal{C} = 1.067496366$ and 1.037498878.

Supply a timelike receiver at $v = -\alpha/u$, $\alpha = 1/4$. Its proper clock is $\tau = 2L\sqrt{\alpha}\ln u/(1-\alpha)$, up to an additive constant. Emit at receiver coordinates $t_i = 4, 6, 9$, with $v_i = -\alpha/t_i$. The outward ray has fixed v_i and meets the boundary at

$$u_i = \frac{a - \Phi_b + \gamma v_i}{(a + \Phi_b)v_i - b} > t_i.$$

The reflected ray has fixed u_i; reception at the receiver therefore occurs at the same coordinate u_i, with $v = -\alpha/u_i > v_i$. For $r = 0.9$ the receive coordinates are $8.18983, 12.56501, 19.53471$; for $r = 0.5$ they are $8.10371, 12.30538, 18.81553$. Throughout these intervals the receiver obeys $\sqrt{\mathcal{C}} < \Phi < \Phi_b$ and $1 + uv > 0$, inside the usual cutoff and outside the stationary horizon. A new compact coupling can start after $u = 20$. Writing $\mathcal{C}_{\mathrm{mid}}$ for the midpoint of the two invariants, choose

$$F(\mathcal{C}) = \frac{\pi}{4} + 10(\mathcal{C} - \mathcal{C}_{\mathrm{mid}}). \tag{79}$$

The two returned angles are 0.935385602 and 0.635410725. If the inverse error stays below the corresponding separation margin, the measurement changes the next coupled projection and its represented algebra $\{P_{F(\mathcal{C})}\}'$. This is a response-dependent return map, not a proof that this program is optimal or that its interface formed through learning. The controller uses the measured invariant; computing one from assumed initial data is a different protocol.

The listed times use the source-free geometry. With finite probes, the return must instead use Eq. (75) and the altered boundary. As a bulk check, two calibrated positive impulsive fluxes of integrals 10^{-5} in u and 2×10^{-5} in v give a coefficient bias of about 5.7×10^{-6} if ignored; source compensation removes it to numerical precision. This is not a physical reflection or timing experiment. For a translation-only readout with the listed proper-delay difference $\Delta\tau = 0.05001338$, the sharp signed-delay bound in Q-RAIF B requires at least $11.04\hbar$ in this clock's energy units for binary error 0.1 [21]. A real echo can also carry redshift and amplitude information, so that restriction is not a universal bound on geometric sensing. It shows why one cannot assign both negligible probe stress and a finite resolution without constructing the readout. The boundary, receiver, timing instrument, full probe-induced stress and support costs remain explicit apparatus inputs to this conditional return; mean-source evolution does not bound stress fluctuations.

G.3 A quantum boundary record without echo timestamps

A current detector gives a second readback route. It measures fluctuations over a prescribed interval of proper time, rather than the arrival time of a reflected signal. The boundary condition still does essential work: the detector is carried by the supplied timelike curve $\Phi = \Phi_b$. A detector on the same fixed bulk worldline in both solutions would remain subject to Proposition G.1.

Use a transparent chiral current interface, with the vacuum defined in u and the current normalization of Section 8. Its input on the interval to be sensed has zero mean and covariance $\langle j(u)j(u')\rangle = -[4\pi(u-u'-i0)^2]^{-1}$. The apparatus supplies a proper-time clock, a degenerate detector bit initially in $|0\rangle$, and a real compact switch $\zeta(\tau)$. The finite interaction is

$$U_s = e^{-iX_sQ_g}, \qquad Q_g = \int g(u)j(u)\,du, \qquad g(u(\tau)) = \zeta(\tau). \tag{80}$$

Equivalently its integrated interaction is $\int \zeta(\tau) X_s \dot{u}(\tau) j(u(\tau))\, d\tau$. The fixed detector operator and c-number current commutator make the time-ordering correction a scalar phase. This is the same gapless-detector method used for exact field channels [8]; the finite switch defines a smeared current, not an unsmeared instantaneous Hamiltonian. We read Z_s after contact and retain the bit $R = 0, 1$, with $R = 1$ the flipped outcome. Clock calibration, preparation and recording permissions are supplied resources.

Proposition G.4 (Invariant-dependent finite detector law)**.** *On a source-free background with $0 < \mathcal{C} < \Phi_b^2$, a future boundary segment has*

$$\dot{u} = \frac{bu^2 + 2au - \gamma}{2L\sqrt{\Phi_b^2 - \mathcal{C}}} > 0, \qquad \lambda_b = \frac{\sqrt{\mathcal{C}}}{L\sqrt{\Phi_b^2 - \mathcal{C}}}. \tag{81}$$

In the test-background calculation, Eq. (80) *has the record law*

$$k_b(\mathcal{C}) = \frac{1}{4\pi} \int_0^\infty \omega \coth \frac{\pi\omega}{\lambda_b} |\widehat{\zeta}(\omega)|^2 \, d\omega, \qquad \widehat{\zeta}(\omega) = \int e^{i\omega\tau} \zeta(\tau)\, d\tau, \tag{82}$$

$$\Pr(R = 1 \mid \mathcal{C}) = \frac{1 - e^{-2k_b(\mathcal{C})}}{2}. \tag{83}$$

For every nonzero switch with finite variance, the probability is strictly increasing in $\mathcal{C}$. It depends on the invariant, not on the three coefficients separately. For two equally likely backgrounds the optimal decision from this one bit has error $[1 - |p_1 - p_0|]/2$, where $p_i = \Pr(R = 1 \mid \mathcal{C}_i)$.

Proof. Pulling back the metric to the level curve gives $d\tau/du = 2L\sqrt{\Phi_b^2 - \mathcal{C}}/(bu^2 + 2au - \gamma)$. The Riccati equation in Eq. (81) makes $u(\tau)$ a fractional-linear transform of $e^{\lambda_b \tau}$ on each regular segment. Möbius invariance of the current two-point function therefore gives

$$\langle \dot{u}(\tau) j(u(\tau)) \dot{u}(\tau') j(u(\tau')) \rangle = -\frac{\lambda_b^2}{16\pi \sinh^2[\lambda_b(\tau - \tau' - i0)/2]}.$$

Its symmetrized spectrum yields Eq. (82); the pullback vacuum is KMS at temperature $\lambda_b/(2\pi)$ in this proper clock. For a centred Gaussian field, $\langle e^{2iQ_g} \rangle = e^{-2\langle Q_g^2 \rangle}$, hence the detector's Z_s expectation is e^{-2k_b}. The factors λ_b and $\coth(\pi\omega/\lambda_b)$ strictly increase with $\mathcal{C}$ for $\omega > 0$. A nonzero compact switch has Fourier weight on a set of positive measure, so the integral strictly increases. The binary decision formula is the total-variation formula for two Bernoulli laws. □

This response uses the familiar Hawking–Unruh covariance. It does not add a direct matter–dilaton coupling, and the KMS temperature here is the boundary proper-time temperature, not an automatically thermal exhaust in the running rapidity clock. Without calibration of L, Φ_b and the proper-time scale, these records identify λ_b rather than $\mathcal{C}$. A clock-scale error can mimic an invariant change; this instrument alone cannot distinguish them.

G.4 The sensing interaction changes the background

The same detector interaction adds the chiral mean stress $f(u) = [g'(u)]^2/4$. To include it, choose a boundary-source completion: this current is transmitted into the JT region, the coupling-added bulk source is precisely $T_{uu} = f(u)$, and the other state-dependent chiral

source is zero. Keep the trace-anomaly subtraction of Section F.4. This is a specified boundary injection model, not the reflected-probe model of Section G.1. No unknown reflected component is fitted as a known function. Forces supporting the boundary, an ultraviolet realization of the contact and the clock's stress require a further apparatus model.

Proposition G.5 (Self-consistent sensing source in mean JT geometry)**.** *In this completion, put* $R_b = \sqrt{\Phi_b^2 - \mathcal{C}}$ *and let dots denote proper-time derivatives. Starting with the matched coefficient data, solve*

$$\dot{u} = \frac{bu^2 + 2au - \gamma}{2LR_b}, \qquad f(u(\tau)) = \frac{\dot{\zeta}^2}{4\dot{u}^2},$$
$$(\dot{a}, \dot{b}, \dot{\gamma}) = \frac{\kappa\dot{\zeta}^2}{4\dot{u}}(u, -1, u^2). \tag{84}$$

On a regular finite segment this gives the boundary trajectory and the complete coupling-added bulk source together. Its exact balances are

$$\dot{\mathcal{C}} = \frac{\kappa L}{2} R_b \dot{\zeta}^2, \qquad R_b(\tau) = R_b(0) - \frac{\kappa L}{4}\int_0^\tau \dot{\zeta}(s)^2\, ds, \tag{85}$$

$$E_s^{(\tau)} := \int f\dot{u}^2\, d\tau = \frac{1}{4}\int \dot{\zeta}^2\, d\tau = \frac{R_b(0) - R_b(\tau_f)}{\kappa L}, \quad E_s^{(u)} := \int f\, du = \frac{1}{4}\int \frac{\dot{\zeta}^2}{\dot{u}}\, d\tau. \tag{86}$$

For the actual, generally nonstationary map $u(\tau)$*, the detector variance is*

$$k[g] = \frac{1}{4\pi}\int_0^\infty \omega|\widehat{g}(\omega)|^2 d\omega = \frac{1}{8\pi}\int_{\mathbb{R}^2} \frac{[g(u) - g(v)]^2}{(u-v)^2}\, du\, dv. \tag{87}$$

Its record probability remains $(1 - e^{-2k[g]})/2$*; replacing this variance by the stationary KMS integral during contact would neglect backreaction.*

Proof. The retarded JT solution has running coefficients $a' = \kappa u f$, $b' = -\kappa f$ and $\gamma' = \kappa u^2 f$, with primes in u. In the derivative of the boundary level equation the source contributions cancel: its numerator derivative is b and its denominator derivative is $a+\Phi_b$. Thus its slope is still $(\Phi_b^2 - \mathcal{C})/[(a+\Phi_b)u - \gamma]^2$ with running coefficients. The induced metric gives the first equation in Eq. (84), while $g' = \dot{\zeta}/\dot{u}$ gives the second. Differentiating $a^2 + b\gamma$ and substituting $\dot{u}$ gives Eq. (85). The energy balances follow by change of variable. The source f follows from $U_s^\dagger j U_s = j + X_s g'/2$ and the initially independent, centred detector with $\langle X_s\rangle = 0$. The incoming current on the deterministic mean trajectory is Gaussian; its vacuum variance has the Fourier expression in Eq. (87). Parseval's identity and $\int(1 - \cos\omega x)/x^2\, dx = \pi|\omega|$ give the second. □

In this model the mean source fixes a classical trajectory before the detector outcome is conditioned on. It is not a theory of outcome-conditioned quantum geometry. The inequality $\kappa L\int \dot{\zeta}^2/4 < R_b(0)$ prevents closure of the boundary gap; one must also check $\dot{u} > 0$ and $1 + uv > 0$ over the chosen window. The two energies in Eq. (86) refer to different clocks. Neither is the total preparation, support, readout or erasure cost. For any sequence obeying this positive-source completion at unchanged Φ_b, L, κ, the cumulative proper-clock source charge satisfies

$$\sum_n E_{s,n}^{(\tau)} < \frac{\sqrt{\Phi_b^2 - \mathcal{C}_{\text{initial}}}}{\kappa L}. \tag{88}$$

This follows directly from $R_b > 0$ and Eq. (85). A nonzero charge per contact therefore rules out infinitely many such contacts at a fixed cutoff. It is an upper budget for this positive-source boundary family, not a universal lower bound on sensing energy or a prohibition of protocols with a different net source.

Proposition G.6 (Finite sensing survives its own added source)**.** *Take two source-free initial backgrounds with different invariants, with the same Φ_b, L and a common proper-time contact interval on which both unperturbed curves are regular. Fix a nonzero C^2 compact switch ζ_0. At fixed κ, the completion in Proposition G.5, with $\zeta = \epsilon\zeta_0$, has distinct record laws for all sufficiently small $\epsilon > 0$. The added energies and invariant changes are finite and of order ϵ^2.*

Proof. On the compact regular segment the trajectory equations are locally Lipschitz, and their source-dependent terms are $O(\kappa\epsilon^2)$. Continuous dependence preserves the domain and gives $u_\epsilon - u_0 = O(\kappa\epsilon^2)$ in C^1. The corresponding compact smears obey $\|g_\epsilon - \epsilon g_0\|_0 = O(\kappa\epsilon^3)$: the inverse clocks and their first derivatives have the same perturbative order, and the H^1 bound controls the vacuum $H^{1/2}$ seminorm. Thus $k[g_\epsilon] = \epsilon^2 k[g_0] + O(\kappa\epsilon^4)$ and $p_\epsilon = \epsilon^2 k[g_0] + O((1+\kappa)\epsilon^4)$. The leading variances differ strictly by Proposition G.4. Their record difference cannot vanish for all sufficiently small nonzero ϵ. Equation (86) gives the energy order. □

G.5 A finite record returns to a physical selection

The two compact-source branches of Eq. (78) can be sensed by the same contact, without giving their coefficient values to the program. Start the boundary clock at $u = 4$, keep $\kappa = 0.05$, and use the C^2 compact profile

$$\zeta(\tau) = \frac{4}{3}\left[1 - \left(\frac{\tau - 4}{4}\right)^2\right]^3 \quad (0 \le \tau \le 8), \qquad \zeta = 0 \quad \text{otherwise.} \tag{89}$$

It has finite variance and energy; smooth approximants give the same limiting quantities. Solving Eq. (84) and integrating Eq. (87) gives

Original population r	Initial $\mathcal{C}$	Final $\mathcal{C}$	$\Pr(R = 1)$	Final u
0.5	1.037498878	1.1206720	0.31748037	57.481
0.9	1.067496366	1.1505122	0.31771688	52.458

The contact charge is exactly $E_s^{(\tau)} = 512(4/3)^2/(385\cdot 8) = 0.29552670$. The corresponding null energies are 0.06577751 and 0.06774638. These are deterministic numerical evaluations with converged quadratures, not certified intervals. The single-bit total variation is about 2.3650×10^{-4}, so one contact is a weak discriminator, with equal-prior error about 0.49988175. It is a nonzero physical return, not a high-confidence reconstruction or an optimal sensing strategy. Many repeats would need a new account of field correlations, preparation and accumulated source; they cannot be declared independent by multiplying this record law. Even ignoring all other operations, Eq. (88) allows at most 190 contacts with this unchanged charge on either initial branch. This is a necessary budget, not a 190-contact construction.

Retain R and let it select the next coupled projection: $P_0 = (I + Z)/2$ if $R = 0$ and $P_{\pi/2} = (I + X)/2$ if $R = 1$. Use the independent bit state $(I + 0.8Z)/2$ and a later

compact switch ζ_{next}, scheduled by the same proper clock after $\tau = 8$. Its realised null smear is $g_{\mathrm{next}}(u(\tau)) = \zeta_{\mathrm{next}}(\tau)$. The program supplies that proper-time profile and selects its projection using only R; it need not know $u(\tau)$ or $\mathcal{C}$. The mean population coupled to the next field is

$$r_{\mathrm{next}} = 0.9 - 0.4\Pr(R = 1), \qquad T^{\mathrm{added}}_{uu,\mathrm{next}} = r_{\mathrm{next}}\frac{(g'_{\mathrm{next}})^2}{4}. \tag{90}$$

It is 0.77300785 and 0.77291325 in the two branches. Thus the earlier selection changes a source, that source changes the boundary record law, and the recorded bit changes both the next pointer algebra and the next mean source. The controller uses no numerical dilaton input. For example, use the profile in Eq. (89) rescaled to amplitude $1/4$ and the proper-clock interval $9 \le \tau \le 11$. Its added proper-clock source charge is $r_{\mathrm{next}}\, 512(1/4)^2/(385 \cdot 2)$. The source kernel in u is the one on each branch's actual mean trajectory; the population difference alone is not a comparison of two complete fixed-shape stresses. The same trajectory equations apply with $\dot\zeta^2$ replaced by $r_{\mathrm{next}}\dot\zeta^2_{\mathrm{next}}$, continuing the causal mean solution beyond readout.

There is a same-current subtlety in this return. On the union of the sensor and later null intervals, take the input to have the vacuum restriction. A preceding compact Weyl source in a disjoint earlier interval leaves this restriction unchanged, since the relevant smeared currents commute. Conditioning on the sensor bit applies $\cos Q_g$ or $\sin Q_g$ to that restriction. Both resulting states are invariant under simultaneous current parity, so every later current has zero conditional mean. The next bit is independently prepared, and the linear stress cross term vanishes in each record branch, proving Eq. (90) without resetting the field. Its conditional covariance generally changes. The later field is not a fresh vacuum port, and a vacuum or thermal gate-error formula cannot be carried over to it.

For a realised switch $\widetilde{g}$, define the vacuum seminorm $\|g\|_0 = \sqrt{k[g]}$. If $\|\widetilde{g} - g\|_0 \le e$, then

$$|\widetilde{p} - p| \le |k[\widetilde{g}] - k[g]| \le 2\sqrt{k[g]}e + e^2.$$

This prices the needed control of the actual smeared interaction; clock and worldline errors must be included in $\widetilde{g}$, not only in its amplitude. Two record laws remain distinct if the sum of their probability-error bounds is below their stated separation. Section G.6 adds the independent source requirement, and Section G.8 gives a finite thermal replacement. The clock, contact and continuous exhaust with their complete stress remain the joint apparatus task of Conjecture C.1. The finite run consumes a prepared detector bit and an independently prepared next bit. Resetting the record without retained side information must export its binary entropy; it is not supplied by coherent field return.

G.6 A joint certificate for records and gravitational sources

A detector can be accurate as an instrument and inaccurate as a source. The vacuum seminorm controls a half derivative of its switch, whereas the injected energy controls a full derivative. This distinction is already visible in a single contact, before questions of thermal reuse arise.

Proposition G.7 (Record accuracy does not control source energy)**.** *Let* $g_n(u) =$

$n^{-3/4}e^{-u^2/2}\cos(nu)$, $n \geq 1$, *and* $E[g] = \int (g')^2 du/4$. *Then*

$$k[g_n] = \frac{n^{-3/2}}{8}[n\sqrt{\pi}\ \mathrm{erf}(n) + 2e^{-n^2}],$$
$$E[g_n] = \frac{n^{-3/2}\sqrt{\pi}}{8}[n^2 + \tfrac{1}{2} + \tfrac{1}{2}e^{-n^2}]. \tag{91}$$

Its detector flip probability tends to zero while its source energy diverges. Every member has finite absolute stress moments through order two.

Proof. The Fourier transform is the sum of two shifted Gaussians. Integrating its squared modulus with weights ω and ω^2 gives the two formulas. Consequently $k[g_n] \sim \sqrt{\pi}/(8\sqrt{n})$ and $E[g_n] \sim \sqrt{\pi n}/8$. Gaussian decay gives the finite moments. □

This is a comparison on a fixed null-coordinate field. A member with too much energy can leave a chosen gravitational operating domain. A fixed bandwidth excludes this sequence, but that bandwidth is then part of the source certificate. The controlled weak-switch result above uses stronger regularity and is unaffected.

Proposition G.8 (Simultaneous record, source and response control)**.** *Write the realised switch as* $g + h$, *put* $w(u) = 1 + u^2$, *and assume* $g, h \in H^1$ *with weighted square-integrable derivatives. Define*

$$e_0 = \sqrt{k[h]}, \qquad A = \|\sqrt{w}\, g'\|_2, \qquad e_1 = \|\sqrt{w}\, h'\|_2, \qquad D_s = \tfrac{1}{2}Ae_1 + \tfrac{1}{4}e_1^2.$$

For the independent vacuum input of the current detector,

$$|\widetilde{p} - p| \leq 2\sqrt{k[g]}e_0 + e_0^2,$$
$$\int w\,|\widetilde{f} - f|du \leq D_s, \qquad f = (g')^2/4, \quad \widetilde{f} = (g' + h')^2/4. \tag{92}$$

The second line bounds each of the three stress-moment errors by D_s. *For matched retarded data, at each common* u *the running coefficients satisfy* $|\delta a|, |\delta b|, |\delta\gamma| \leq \kappa D_s$, *hence*

$$|\delta\mathcal{C}| \leq \kappa D_s(2|a| + |b| + |\gamma|) + 2\kappa^2 D_s^2,$$
$$|\delta\Phi(u,v)| \leq \frac{\kappa D_s(1 + |uv| + |u| + |v|)}{|1 + uv|}. \tag{93}$$

For a single contact, even when the detector has a retained reference, the half trace distance of the complete output from its ideal output is at most $\sqrt{1 - e^{-k[h]}} \leq e_0$.

Proof. Cauchy–Schwarz applied to $\widetilde{f} - f = g'h'/2 + (h')^2/4$ proves the weighted bound. The inequalities $1, |u|, u^2 \leq w$ cover both full and prefix moments; the retarded coefficient equations then give the stated errors. Expand $(a+\delta a)^2 + (b+\delta b)(\gamma+\delta\gamma)$ and the dilaton numerator for the response bounds. For the final claim, purify the initial detector and reference. The Weyl relation makes $U_g^\dagger U_{g+h}$ equal to $e^{-iX_sQ_h}$ up to one scalar phase, since $X_s^2 = I$. Its vacuum overlap is $e^{-k[h]/2}$ independently of the detector state. The pure-state trace-distance formula and partial-trace contraction give the bound. This claim concerns the declared vacuum input, not an arbitrary field left by earlier instruments. □

The coefficient bounds also control the clock on a regular window. If both compared fixed-cutoff solutions have $R_b \geq r_* > 0$, $P \geq p_* > 0$, and $R_b \leq r^*$, put $\Delta_C = \sup|\delta\mathcal{C}|$ and $\Delta_P = \sup \kappa D_s(1+|u|)^2$ on a null interval of length ℓ. Matching the initial proper time gives

$$\sup|\delta\tau(u)| \leq 2L\ell\left(\frac{\Delta_C}{2r_*p_*} + \frac{r^*\Delta_P}{p_*^2}\right).$$

An inverse-clock bound follows by dividing by a positive lower bound on $d\tau/du$. The interval and gap conditions must hold in both solutions. They cannot be inferred from fidelity alone.

A finite timing tolerance has its own source condition. For a translation error $h_\delta(u) = g(u-\delta) - g(u)$ and g with the indicated second derivative norms finite,

$$\begin{aligned}\sqrt{k[h_\delta]} &\leq |\delta|\sqrt{k[g']},\\ \|\sqrt{w}\,h_\delta'\|_2 &\leq |\delta|\|\sqrt{w}\,g''\|_2 + \tfrac{1}{2}\delta^2\|g''\|_2.\end{aligned} \tag{94}$$

The first inequality follows from $|e^{i\omega\delta}-1| \leq |\omega\delta|$. For the second, integrate $g''(u-s)$ between 0 and δ and use $\sqrt{1+u^2} \leq \sqrt{1+(u-s)^2} + |s|$. This is a null-clock translation certificate. A proper-clock error must be pulled through its actual worldline map; a finite quantum clock also needs control of its tails and correlations. Small arrival error alone does not supply a continuum stress bound.

G.7 Boundary energy, work and clock conversion

Choose the timelike Dirichlet boundary term, with the normal pointing toward increasing dilaton,

$$I_\partial = \frac{1}{\kappa}\int_{\partial M}\sqrt{-h}\,\Phi(K - L^{-1})\,dt. \tag{95}$$

Its lapse variation fixes the proper-time quasilocal energy

$$E_b = \frac{1}{\kappa}(\Phi_b/L - n^\mu\partial_\mu\Phi).$$

This is the finite-cutoff JT energy with a stated counterterm, rather than a total apparatus energy [60, Sec. 6]. For example, in Gaussian normal coordinates $ds^2 = dr^2 - N^2dt^2$, integrating the bulk curvature term by parts and adding Eq. (95) leaves the lapse boundary term $N(\partial_r\Phi - \Phi/L)/\kappa$. Its variation is $-E_b\delta N$. The dilaton boundary momentum is $(K - L^{-1})/\kappa$.

Proposition G.9 (Energy and work on a running level curve)**.** *In the traceless single-chiral source completion let $\Phi_b = \Phi_b(\tau)$ be a smooth prescribed boundary value. On a future timelike segment put*

$$R_b^2 = \Phi_b^2 - \mathcal{C} > 0, \quad S_b = \sqrt{R_b^2 + L^2\dot\Phi_b^2}, \quad y = S_b - L\dot\Phi_b > 0, \quad q_\tau = f\dot u^2.$$

Then

$$\begin{aligned}\dot u &= \frac{P}{2Ly}, & \dot{\mathcal{C}} &= 2\kappa Lyq_\tau, & \dot R_b &= \frac{\Phi_b\dot\Phi_b - \kappa Lyq_\tau}{R_b},\\ E_b &= \frac{\Phi_b - S_b}{\kappa L}, & K &= \frac{\Phi_b + L^2\ddot\Phi_b + \kappa L^2 q_\tau}{LS_b}, & \dot E_b &= q_\tau + \frac{L^{-1} - K}{\kappa}\dot\Phi_b.\end{aligned} \tag{96}$$

At a fixed cutoff these reduce to $E_b = (\Phi_b - R_b)/(\kappa L)$ and $\dot E_b = q_\tau$. On the source-free stationary family, $dE_b = T_b\,dS_h$ at fixed Φ_b, with $T_b = R/(2\pi LR_b)$.

Proof. Writing the boundary as $v = V(u)$, its dilaton equation and the induced metric give $1 = 4L^2R_b^2\dot{u}^2/P^2 - 4L^2\dot{\Phi}_b\dot{u}/P$. Its positive root is the first equation. Use $\mathcal{C}' = \kappa f P$ for the next two. The gradient identity gives $Ln^\mu\partial_\mu\Phi = S_b$. Along the unit tangent, the JT equation reads $\ddot{\Phi}_b = -\Phi_b/L^2 - \kappa q_\tau + KS_b/L$. This gives K; differentiating E_b proves the work balance. The stationary first law follows by differentiating $E_b = (\Phi_b - \sqrt{\Phi_b^2 - R^2})/(\kappa L)$ and $S_h = S_0 + 2\pi R/\kappa$. □

For example, retaining a constant gap $R_b = r > 0$ under positive flux requires

$$y = \frac{r}{\sqrt{1 + 2\kappa L^2 q_\tau/\Phi_b}}, \qquad \dot{\Phi}_b = \frac{\kappa L y q_\tau}{\Phi_b}.$$

These equations give a nonempty family on regular finite patches. Increasing the cutoff carries the work term in Eq. (96); it is not an uncharged extension of the fixed-cutoff budget. The boundary datum is still externally prescribed. A physical support must produce it and carry its own stress.

The same calculation fixes the conversion to the running horizon clock:

$$z := \frac{d\tau}{d\eta} = \frac{Ly}{R}, \qquad q_\eta = z^2 q_\tau. \tag{97}$$

For a fixed cutoff $y = R_b$, and the stationary temperature is $T_b = (2\pi z)^{-1}$. During a driven contact this is not a global KMS identification. If a local thermal-current regime has separately been justified, entropy flux transforms as $s_\eta = z s_\tau$ and its actual input temperature must satisfy $zT_\tau = 1/(2\pi)$ to use Proposition H.2. Changing coordinates does not prepare that input.

G.8 A thermal detector can be reused without refreshing its field

A finite Gibbs swap provides a physical replacement channel for the detector. Its limitation is equally useful: reset accuracy, retained correlations and heat cannot be specified independently.

Proposition G.10 (Reference-uniform thermal replacement)**.** *During sensing the detector Hamiltonian is degenerate. For reset raise its excited level to $\Delta > 0$ and supply an independent auxiliary bit in $\tau_\beta = \mathrm{diag}(1-\epsilon, \epsilon)$, with $\epsilon = (1 + e^{\beta\Delta})^{-1}$. A resonant SWAP, followed by lowering the detector gap, gives for every joint detector–reference state*

$$\rho_{SA} \longmapsto \tau_\beta \otimes \rho_A.$$

Its half-diamond distance from pure-blank replacement is exactly ϵ. If $p = \Pr(S = 1)$ just before reset, the auxiliary heat and net gap-modulation work are both

$$Q_{\mathrm{reset}} = W_{\mathrm{gap}} = \Delta(p - \epsilon). \tag{98}$$

A finite resonant swap can use $H_{\mathrm{sw}}(\tau) = \chi(\tau)(I - \mathrm{SWAP})$ with $\int \chi\, d\tau = \pi/2$; it commutes with the equal-gap free Hamiltonian. Its net switch work is zero when χ vanishes at both endpoints.

Proof. The swap transfers the whole previous detector state and its reference correlations to the auxiliary, not just its populations. The distance between the two replacement states is ϵ, for every reference, proving the channel statement and attainability. Raising the gap costs Δp, the resonant swap conserves total bare energy, and lowering it returns $\Delta\epsilon$. The outgoing auxiliary has excitation probability p. The expectation of $I-\mathrm{SWAP}$ is constant during the specified resonant contact, so its switching work integrates to zero. □

This replacement process is the standard swap benchmark for Landauer accounting [13, Example 1 and Sec. 3.3]. The auxiliary must be prepared and transported. The old state remains correlated with retained matter; it has not disappeared. If a copy of the outcome is kept, that copy is part of the maintained state in Eq. (28). At finite temperature this finite device never supplies an exact pure blank.

Proposition G.11 (A bounded-heat Gibbs ladder)**.** *Fix $\beta > 0$, a target $0 < \epsilon < 1/2$, and m finite thermal stages. First swap the degenerate detector with an independent zero-gap Gibbs bit $I/2$. Put $x_* = \ln[(1-\epsilon)/\epsilon]$, $x_j = jx_*/m$ and $p_j = (1+e^{x_j})^{-1}$. At stage j raise the detector gap from x_{j-1}/β to x_j/β and resonantly swap with an independent Gibbs bit of that gap. Finally lower its gap to zero. The complete replacement is exactly $\tau_\beta \otimes \rho_A$, with target excitation ϵ, and*

$$\beta Q_{\mathrm{ladder}} = \sum_{j=1}^{m} x_j(p_{j-1} - p_j) \le \ln 2 - h(\epsilon) + \frac{x_*}{m}(\tfrac{1}{2} - \epsilon). \tag{99}$$

The net gap-modulation work equals this heat. Choosing $m \ge x_*/(2\eta)$ gives $\beta Q_{\mathrm{ladder}} \le \ln 2 + \eta$ for every initial detector state and every retained reference.

Proof. The first swap transfers all old correlations to its output. Each later swap leaves an independent Gibbs detector. The heat formula follows from the thermal populations, and summation by parts gives the same net work, including final gap lowering. The sum is a right-endpoint approximation to $\int_0^{x_*} x[-p'(x)]dx = \ln 2 - h(\epsilon)$; its excess is at most $(x_*/m)(p_0 - p_m)$. □

This is an explicit finite swap construction of the type used in Ref. [13, Sec. 6], with a distribution-independent starting step and a bound on its stages. It consumes $m+1$ prepared Gibbs bits, including the zero-gap port, and raises the maximum gap to x_*/β. The finite reset does not hide a pure auxiliary. The first port can leave with less entropy than it had initially; it must be included along with all later outputs and their correlations. For a fixed supplied sequence of operations, N replacements differ from the pure-blank retained-system process by at most the sum of their ϵ_i, including arbitrary intervening instruments on the field. Thus high reset accuracy need not cause divergent heat, but the number of stages and the maximum gap grow. If the replacement changes the realised geometry or timing of those operations, their source and clock errors require the separate joint certificate of Section G.6. The timing, Gibbs preparation, transport and capacity required to execute these stages are additional resources.

Here is an exact two-read law on one field. Let g_1, g_2 have ordered, disjoint compact supports, put $k_i = \langle Q_i^2 \rangle$ and $c_{12} = \langle Q_1 Q_2 \rangle$ in the initial vacuum, and reuse the detector after the above swap. Both contacts start with its state τ_β. Read Z_s and use it for the subsequent command before each reset. A nondemolition dephasing of the pointer can be

implemented by a controlled Z_s using a separate degenerate Gibbs bit $I/2$. Those ports are supplied resources, not pure memories. Since $[Q_1, Q_2] = 0$, the joint outcomes obey

$$\begin{aligned}\Pr(R_1 = r, R_2 = t) = \frac{1}{4}\big[1 + \alpha\{(-1)^r e^{-2k_1} + (-1)^t e^{-2k_2}\} \\ + \alpha^2 (-1)^{r+t} e^{-2(k_1+k_2)} \cosh(4c_{12})\big].\end{aligned} \tag{100}$$

Here $\alpha = 1 - 2\epsilon$. Indeed each conditional pointer probability at a fixed Q_i is $[1 + \alpha(-1)^r \cos(2Q_i)]/2$. The commuting Gaussian characteristic function then gives the displayed law. The same field remains conditioned by the first read: thermal replacement of the detector does not restore a vacuum field. On a stationary test background the contrast of two invariant-dependent single-contact laws is multiplied by α, so it stays nonzero for $\epsilon < 1/2$. The weak-source proof of Proposition G.6 also survives this factor.

A retained body makes the two commands physically consequential. Prepare the body once in $(I + 0.8Z)/2$. After read r, couple it through $P_0 = (I + Z)/2$ or $P_1 = (I + X)/2$ to an independent vacuum control current. The first source population is $\rho_1 = 0.9 - 0.4 \Pr(R_1 = 1)$. If that contact's realised variance is k_c, the first X projection attenuates the body's Z component by $e^{-k_c/2}$. With p_{rt} denoting Eq. (100), its second source population is therefore

$$\rho_2 = \tfrac{1}{2} + 0.4\{p_{00} + e^{-k_c/2} p_{10}\}. \tag{101}$$

This uses the body left by the first command, not a newly prepared copy.

For a finite boundary run take $L = 1$, $\Phi_b = 3$, $\kappa = .01$, initial $(u, a, b, \gamma) = (1, 1, 0, 0)$, $\epsilon = .01$ and $\beta = 1$ in the proper clock. Use the profile $A[1 - (2(\tau - \tau_0)/d - 1)^2]^3$ on an interval of duration d. The two sensor contacts have amplitude .5 on $[0, 2]$ and $[4, 6]$; the two selected control contacts have amplitude .3 on $[2.5, 3.5]$ and $[6.5, 7.5]$. The gaps leave finite windows for readout and reset. Integrating the same mean boundary equations gives

$$k_1 = .068080013, \quad k_2 = .068082993, \quad c_{12} = -.000926643, \quad k_c = .024462001,$$

$$\rho_1 = .871049781, \quad \rho_2 = .870722307, \quad \mathcal{C}_{\text{final}} = 1.030570773.$$

The proper-clock coupling-source charge is .540937293. The two reset auxiliaries separately absorb .573257937, using $\Delta = \ln 99$; this heat has not been inserted as a chiral bulk source. The two outgoing reset bits have joint entropy .519488204, against .112003069 initially, and relative entropy .165772802 to the product Gibbs reference. Thus their heat obeys exactly $\beta Q = \Delta S(B_1 B_2) + D(\rho_{B_1 B_2} \| \tau_\beta^{\otimes 2})$. Other output ports and retained matter belong to the full entropy account. The source charge, reset heat and apparatus work are distinct entries.

Starting instead at $a = 1.4$ with the other initial coefficients unchanged changes both recorded laws and returned source populations on the same proper schedule. The detector and body are reused in both runs. The completion transmits the sensor current and two independent control currents into the bulk. The reset ports, clock and support still require a conserved transport completion; their cost is not removed by the displayed bulk balance. A long sequence of simple Gibbs swaps is also expensive: choosing $\epsilon_N = \varepsilon/N$ to certify total blank-replacement error at most ε requires $\beta\Delta_N = \ln(N/\varepsilon - 1)$. If detector excitation probabilities stay above a fixed positive constant, its heat per reset grows as $\log N$. This particular construction cannot attain the fixed-rate sharp-exhaust conjecture. The ladder of Proposition G.11 avoids this heat divergence, but fitting its stages and transported thermal units into a fixed capacity and operating rate remains necessary.

H Running horizons, source states and clocks

H.1 Source sorting and tracking-clock support

The finite-carrier sorting statement remains Proposition 11.2, with its source and old-reference permissions unchanged.

Proof of Proposition 11.2. The source enters Q-RAIF C's proof in three places: concentration of input eigenlabel strings, the source's fixed neutral charge, and the final source–carrier entropy identity. Its detailed depolarizing eigenvalues are not otherwise used. Each ρ_p has rank two in the same neutral quartet sector and finite information variance at fixed q, ϵ. Independent eigenlabel strings have information variance $O(L)$, including a nonidentical product over the two-symbol alphabet. The sorter's subextensive typical-label margin, thermal-sector state count and physical neutral-sector unitary compiler therefore apply with entropy $\sum_j h_{p_j}$. The source has zero total charge, leaving the carrier's sector weights and sublinear sector correction unchanged. Completing the eigenlabel injection to a sector unitary keeps all bad labels and acts as the identity on external references. The source–carrier entropy identity and controlled full-carrier energy tails give the deficit and slack limits through the Gibbs identity. Trace error alone would not suffice. Uniform local reset error alone does not establish the supplied product marginal for an adaptive history. □

Equation (56) gives the actual clock, not a static-wall substitution. For a tracking datum $R_b = \rho R$, $\Phi_b = \alpha R$, $\alpha = \sqrt{1+\rho^2}$ and $\delta = R^{-1} dR/d\eta$, it gives

$$z = L\sqrt{\rho^2 - 2\alpha\delta}, \qquad \rho^2 > 2\alpha\delta. \tag{102}$$

H.2 The running horizon and its entropy

Long-time entropy limits need an operating geometry, not just a source integral. Positive steady flux can exhaust a fixed u chart. At the boundary $v = -1/u$ the numerator of the retarded dilaton is

$$N_b(u) = 2H_0 - \frac{\kappa}{u}\int^u (u-z)^2 F(z)\, dz.$$

For a constant flux $F_0 > 0$ switched on at $u = 0$, this becomes $2H_0 - \kappa F_0 u^2/3$ and vanishes at $u_* = \sqrt{6H_0/(\kappa F_0)}$. A device at a fixed exterior position cannot therefore be assumed to run at fixed u frequency indefinitely. Finite observation windows and families with larger operating domains are different from an infinite-time protocol in one fixed background.

There is an exact horizon balance inside the retarded solution. In this subsection we use $\hbar = k_B = 1$ and the classical Wald entropy, not an event-horizon generalized entropy. Write the running coefficients as

$$a(u) = H_0 + \kappa\int^u zF(z)\, dz, \qquad b(u) = -\kappa\int^u F(z)\, dz, \qquad \gamma(u) = \kappa\int^u z^2 F(z)\, dz,$$
$$\Phi = \frac{a + bu + \gamma v - auv}{1+uv}, \qquad \mathcal{C}(u) = a^2 + b\gamma, \qquad R(u) = \sqrt{\mathcal{C}(u)}. \tag{103}$$

The lower limits and any source-free initial coefficients are fixed by the matching data. Put $P = bu^2 + 2au - \gamma$ and $\xi = P/(2R)$.

Proposition H.1 (Exact marginal-horizon balance)**.** *On a patch with $\mathcal{C} > 0$, $a + R \neq 0$, $P > 0$ and $1 + uv_h \neq 0$, the curve*

$$v_h(u) = \frac{b}{a + R}$$

is a future outer marginal horizon: $\partial_u \Phi = 0$, $\partial_v \Phi < 0$ and $\partial_v \partial_u \Phi < 0$ there. Its dilaton and Wald entropy obey

$$\Phi_h = R, \qquad S_h = S_0 + \frac{2\pi R}{\kappa}, \qquad \frac{dS_h}{du} = 2\pi F \xi, \qquad \frac{dS_h}{d\eta} = 2\pi F \xi^2, \quad d\eta = \frac{du}{\xi}. \tag{104}$$

For $F \geq 0$ this horizon is spacelike or null and its entropy is nondecreasing.

Proof. The coefficient derivatives $a' = \kappa u F$, $b' = -\kappa F$ and $\gamma' = \kappa u^2 F$ cancel from the first dilaton derivative. With $D = 1 + uv$,

$$\Phi_u = \frac{b - 2av - \gamma v^2}{D^2}, \qquad \Phi_v = -\frac{P}{D^2}.$$

Substitution of v_h gives $\Phi_h = R$ and $\Phi_{uv}|_h = -2R/(1 + uv_h)^2$. Also $\mathcal{C}' = \kappa F P$ and $v_h' = -\kappa F (1 + uv_h)^2/(2R)$. The metric then makes the tangent spacelike for $F > 0$ and null for $F = 0$. Finally $R' = \kappa F \xi$ and $\kappa = 8\pi G$ give Eq. (104). □

The identification of this classical marginal horizon and its area law is a standard JT result [53, Sec. 2.3.1]. Here it assigns an actual horizon observable to the running invariant used in the source account. Negative quantum excess stress need not be monotone, and replacing S_h by S_{gen} would require the matter entanglement contribution and a separate quantum argument.

Proposition H.2 (Conditional entropy-export cost at the horizon)**.** *Suppose the charged heat in the rapidity clock is exactly $\dot{Q}_\eta = F\xi^2$, and the exhaust satisfies Proposition 10.2 in that same clock at $T_H = 1/(2\pi)$. On a window of rapidity duration τ, let X_τ be the net exported maintenance entropy. In the extensive rate regime,*

$$\frac{\Delta S_h}{\tau} \geq \frac{X_\tau}{\tau} + \frac{6}{c_{\mathrm{ex}}} \left(\frac{X_\tau}{\tau} \right)^2 + o(1).$$

Writing bars for the corresponding mean rates gives

$$\overline{\dot{S}_h} \geq \dot{s} + \frac{6}{c_{\mathrm{ex}}} \dot{s}^2. \tag{105}$$

The same pointwise formula requires a justified local-current limit; it does not follow for arbitrary fluctuating packets. A mean horizon-entropy budget $\overline{\dot{S}_h} \leq B$ permits at most

$$\dot{s} \leq \frac{c_{\mathrm{ex}}}{12} \left(\sqrt{1 + \frac{24B}{c_{\mathrm{ex}}}} - 1 \right).$$

Proof. Multiply Eq. (29) at $T = T_H$ by 2π, use the integrated Eq. (104) and the long-window Gibbs account, then invert the increasing quadratic. □

This transfers the entropy removed from the full maintained state to a specified gravitational entropy, including the finite-rate excess. The heat-routing condition is substantive: incoming nonequilibrium energy, other stress sources, charge transfer and the clock's own entropy cannot be omitted. Nor does a change of clock alone prepare a thermal reservoir.

Proposition H.3 (Schwarzian of the running clock)**.** *On a smooth running patch put* $q = F\xi^2$ *and* $\delta = R^{-1}dR/d\eta = \kappa q/R$*. Then*

$$\{u;\eta\} = -\frac{1}{2} - \frac{d\delta}{d\eta} - \frac{\delta^2}{2}. \tag{106}$$

For constant $q > 0$*,* $R(\eta) = R_0 + \kappa q\eta$ *and*

$$\{u;\eta\} = -\frac{1}{2} + \frac{(\kappa q)^2}{2R(\eta)^2}. \tag{107}$$

Proof. Write $j = (bu + a)/R$, and let a dot denote $d/d\eta$. The coefficient equations and $h^2 - 2bR\xi = R^2$, with $h = bu + a$, give

$$\frac{\dot{\xi}}{\xi} = j - \delta, \qquad \dot{j} = \frac{j^2 - 1}{2} - j\delta.$$

Substitute these into $\{u;\eta\} = (\dot{\xi}/\xi)^{\cdot} - (\dot{\xi}/\xi)^2/2$. For constant q, $\dot{\delta} = -\delta^2$, which recovers Eq. (107). □

Constant positive heat current changes the vacuum clock correlation, not merely the horizon size. Positive growth itself, however, does not rule out an exact thermal reference. The following two histories solve the clock condition.

Proposition H.4 (Growing horizons with exact geometric KMS clocks)**.** *Use natural units and the prescribed-mean-source JT equations. An independent chiral conformal reference sector is in its vacuum; its state is not the state of the driven matter supplying* F*. On every finite smooth operating window, the following positive-flux profiles admit the initial matching* $(a, b, \gamma) = (R_0, 0, 0)$ *at* $\eta = 0$*,* $u = u_0 > 0$*:*

1. *Constant fractional growth* $\delta_0 > 0$*:* $R = R_0 e^{\delta_0 \eta}$*,* $q = \delta_0 R/\kappa$*. The reference pulled back to the* η *clock is geometric KMS at* $T = \sqrt{1 + \delta_0^2}/(2\pi)$*.*

2. *Fixed rapidity temperature: for* $\eta_0 > 0$*,*

$$R = R_0(1 + \eta/\eta_0)^2, \qquad q = \frac{2R_0(\eta + \eta_0)}{\kappa\eta_0^2}. \tag{108}$$

   The same reference is exactly geometric KMS at $T_H = 1/(2\pi)$ *despite the growing horizon.*

Here KMS refers to the pulled-back reference algebra and its complete η*-translation dynamics. A finite experiment samples its restriction.*

Proof. Both profiles make Eq. (106) constant, $-s^2/2$, with $s = \sqrt{1+\delta_0^2}$ in the first case and $s = 1$ in the second. Let $\delta(0) = \delta_0$ also in the second case, where $\delta_0 = 2/\eta_0$. An explicit solution with the required initial coordinate and derivatives is

$$u(\eta) = u_0 \left[1 + \frac{e^{s\eta} - 1}{s + m(e^{s\eta} - 1)}\right], \qquad m = \frac{s - 1 + \delta_0}{2}. \tag{109}$$

It has $\xi = \dot{u} > 0$, $\xi(0) = u_0$ and $\dot{\xi}(0)/\xi(0) = 1 - \delta_0$. Set $j = \dot{\xi}/\xi + \delta$ and

$$b = \frac{R(j^2 - 1)}{2\xi}, \qquad a = Rj - bu, \qquad \gamma = bu^2 + 2au - 2R\xi.$$

The constant-Schwarzian equation gives $\dot{j} = (j^2 - 1)/2 - j\delta$. Differentiation then gives $\dot{b} = -\kappa q/\xi$, $\dot{a} = \kappa u q/\xi$ and $\dot{\gamma} = \kappa u^2 q/\xi$. Thus these are the retarded coefficient equations with $F = q/\xi^2$, $a^2 + b\gamma = R^2$ and $P = 2R\xi$. At zero they give $(R_0, 0, 0)$; $a + R$ and the horizon denominator remain nonzero on a sufficiently short initial operating patch, which can be continued while those patch conditions hold.

Equation (109) is a Möbius transform of $e^{s\eta}$. Möbius invariance of the reference vacuum and the geometric KMS construction therefore give inverse temperature $2\pi/s$ [9, Sec. 2.8, Proposition 2.4 and Theorem 2.5]. For a free current the complete Gaussian correlation test is explicit: if $\langle j(u)j(u')\rangle = -C_j/(u - u' - i0)^2$, then

$$\langle \xi(\eta) j(u(\eta))\, \xi(\eta') j(u(\eta'))\rangle = -\frac{C_j s^2}{4\sinh^2[s(\eta - \eta' - i0)/2]}.$$

This is stationary and obeys the KMS boundary condition; the conformal-net argument establishes the state on bounded observables, not just on one stress expectation. □

These are actual nonstationary solutions of the mean-source equations, rather than frozen-background substitutions. The reference sector's thermality is a kinematic input that can now be made compatible with growth. It is not a thermal exhaust realization: the driving sector can be coherent, its state is not KMS merely because the independent reference is, and preparation, heat transport and the physical clock still need their own account. In the second profile q grows linearly, so it does not realize the fixed mean export in Conjecture C.1. As $\eta \to \infty$, u approaches a finite endpoint and $F = q/\xi^2$ diverges. Each finite smooth window has finite source energy; these formulas assert no finite-resource infinite run.

H.3 The actual thermal source changes its clock reference

A thermal spectator field does not make the exhaust thermal. Here the state condition is imposed on the source field itself. Work with the vacuum-subtracted chiral flux of the preceding JT construction, with the state-independent trace background absorbed consistently into the renormalized JT parameters. A complementary chirality may remain in its reference state. This is a semiclassical conformal-matter completion, not a microscopic construction of a thermal emitter.

For the inner patch let $u = f(v_-)$ be its Poincaré null coordinate. The coordinate change $\rho = \Phi/f' - L^2 f''/(f')^2$ gives

$$\{u; v_-\} = -\frac{\mathcal{C}_-}{2L^4}.$$

Specify the source state as the pullback of a geometric KMS state at temperature T_o in η, with $T_H = 1/(2\pi)$ as the comparison temperature. Define the charged excess current and entropy current by

$$q = \frac{\pi c}{12}(T_o^2 - T_H^2) > 0, \qquad s = \frac{\pi c}{6}(T_o - T_H), \qquad 2\pi q = s + \frac{6s^2}{c}. \tag{110}$$

This state condition is imposed on the finite operating patch. Section H.4 gives a local preparation route from supplied thermal reservoirs and also treats nonthermal outputs; the reservoir and contact's complete material realization remain inputs.

The physical stress is a tensor. The Schwarzian enters when the state is expressed relative to the Poincaré vacuum, through the chiral stress transformation law [7, Eq. (2.14)]:

$$\begin{aligned} F_{vv} &= \left(\frac{d\eta}{dv_-}\right)^2 \left(\frac{c}{48\pi} + q\right) - \frac{c}{24\pi}\{\eta; v_-\} + \frac{c}{24\pi}\{u; v_-\} \\ &= \frac{R^2 q}{L^4} - \frac{c}{24\pi}\{\eta; v_-\}. \end{aligned} \tag{111}$$

The last term in the first line is essential. On a static patch, the T_H thermal state is the zero-flux vacuum reference; retaining only its positive thermal energy would falsely insert a source. Conversely, on a growing patch that same running-clock reference need not have zero physical stress.

Proposition H.5 (State–clock–source consistency)**.** *Put $a = \kappa c/(24\pi)$ and $\delta = R'/R$, where primes in this proposition denote η derivatives. The actual source state and the JT equation are jointly satisfied by*

$$R' = R\delta, \qquad \delta' = \frac{\kappa q - R\delta}{a} - \frac{\delta^2}{2}. \tag{112}$$

The physical flux in this clock and the horizon entropy account are

$$\begin{aligned} F_\eta &= q + q_{\mathrm{ref}} = R'/\kappa, & q_{\mathrm{ref}} &= -\frac{c}{24\pi}(\delta' + \delta^2/2), \\ \Delta S_{h,-} &= 2\pi q \Delta\eta - \frac{c}{12}\Delta\delta - \frac{c}{24}\int \delta^2 d\eta. \end{aligned} \tag{113}$$

For $R_0^2 > 2a\kappa q$ and $\kappa q/(2R_0) \le \delta_0 \le 2\kappa q/R_0$, the source–clock ODE has a global positive solution in η with

$$\frac{\kappa q}{2R} \le \delta \le \frac{2\kappa q}{R}, \quad R = R_0 + \kappa q \eta + O(1), \quad \int_0^\infty \delta^2 d\eta \le \frac{8\kappa q}{R_0}. \tag{114}$$

Thus $\Delta S_{h,-}/\Delta\eta \to 2\pi q = s + 6s^2/c$. This global statement concerns the source and clock, not the lifetime of a finite material energy reserve.

Proof. Since $dv_-/d\eta = L^2/R$, its Schwarzian is $\{v_-; \eta\} = -\delta' - \delta^2/2$. The chain rule in Eq. (111) gives $F_\eta = q - c(\delta' + \delta^2/2)/(24\pi)$. Together with $\partial_v \mathcal{C}_- = 2\kappa L^2 F_{vv}$ this is Eq. (112); integration gives Eq. (113). At the upper boundary $\delta = 2\kappa q/R$, the derivative of $\delta - 2\kappa q/R$ is $-\kappa q/a + 2(\kappa q)^2/R^2 < 0$. At the lower boundary the derivative of $\delta - \kappa q/(2R)$ is $\kappa q/(2a) + (\kappa q)^2/(8R^2) > 0$. The strip is invariant and $R' \ge \kappa q/2$. Its upper bound gives $\int \delta^2 \le \int_0^\infty 4(\kappa q)^2 (R_0 + \kappa q\eta/2)^{-2} d\eta$. Finally $R' + a\delta' + a\delta^2/2 = \kappa q$ gives the bounded correction to linear growth. All variables stay finite on finite η intervals. □

The charged heat q and physical source F_η differ by an explicitly accounted reference stress. Proposition H.2 remains conditional on their equality; it cannot be applied pointwise to q after discarding q_{ref}. Here the reference correction is nonextensive, so the sharp long-time rate is recovered by the stronger state–clock calculation. A finite patch does not establish a globally finite-energy state across a limiting Poincaré coordinate endpoint.

For any fixed finite proper-time window, nonempty operating solutions also follow without relying on the numerical table. At $q = 0$, start from the positive static support, $e > 0$, $R_0 > 0$ and $\delta = 0$. This is a regular equilibrium of the combined equations. Continuous dependence on q, with $\delta_0 = \kappa q/R_0$, preserves positive mass, energy reserve and exterior domain on that window for sufficiently small $q > 0$. This is not a fixed-rate infinite-lifetime statement.

Combining Eq. (112) with Eq. (133) uses $P_{\mathrm{out}} = R'/[\kappa z^2]$ and $P_{\mathrm{in}} = 0$. For the support above, initial $e = 2.34$ and $\delta_0 = \kappa q/R_0$, all three examples below run for proper duration 80 with positive reserves and $\Phi > R$:

q	η_{end}	Φ_{end}	e_{end}
.005	85.53862	3.003858	1.882696
.010	85.60190	3.007607	1.424039
.020	85.73040	3.015345	.502569

The mass-junction residual is below 1.2×10^{-12} on the checked grid. These solve the moving mean support and the state condition together. The internal reserve's decline is part of the solution. It has not been derived from the seven-qubit core's local release Hamiltonian.

H.4 A local thermal contact supplies the actual source

The source condition can be replaced by a scattering calculation. Use two complex chiral fermion ports, each with $c = 1$, at a local conformal contact. This is the doubled free-fermion defect of Bernard, Doyon and Viti [5, Sections 3 and 5]. The zero-charge input streams are jointly independent and thermal at T_H and T_o in the same supplied η frame. Their fields mix by $S_\theta = \cos\theta I - i \sin\theta \sigma_x$ at the contact. Put $\chi = \sin^2\theta$. For positive frequency, the source occupation is

$$n_\chi(\omega) = (1-\chi) n_{T_H}(\omega) + \chi n_{T_o}(\omega), \qquad n_T(\omega) = \frac{1}{1 + e^{\omega/T}}. \tag{115}$$

The actual output is hot KMS for $\chi = 1$. For $0 < \chi < 1$ it is stationary but nonthermal: its high-frequency tail has coefficient χ rather than one. Its return-port cross covariance is nonzero. These are retained field correlations, not fresh independent thermal maps. Q-RAIF C derives the complete contact law and its energy account [14, Section on a local thermal contact].

Proposition H.6 (Stationary transfer without a KMS output)**.** *For a constant contact angle set*

$$q_\chi = \frac{\pi\chi}{12}(T_o^2 - T_H^2), \qquad s_\chi = \frac{1}{\pi}\int_0^\infty [h(n_\chi) - h(n_{T_H})]\, d\omega. \tag{116}$$

Here $h(x) = -x \ln x - (1-x)\ln(1-x)$. *The state–clock equations, positive growing family and finite reference correction in Proposition H.5 hold with* $c = 1$ *and* $q = q_\chi$.

Their asymptotic horizon rate is

$$\frac{\Delta S_{h,-}}{\Delta\eta} \longrightarrow 2\pi q_\chi \geq s_\chi + 6s_\chi^2. \tag{117}$$

The inequality is strict for $0 < \chi < 1$ *and* $T_o > T_H$.

Proof. The stationary output's stress in η is $1/(48\pi) + q_\chi$. Its coordinate/state transformation has the same central charge and projective-vacuum subtraction as Eq. (111). That derivation uses only this stress expectation, not the KMS property. The source–clock ODE and its invariant strip therefore follow unchanged. At fixed energy the Gibbs state maximizes the fermion entropy current. Its temperature is $T_{\rm eff} = \sqrt{T_H^2 + 12q_\chi/\pi}$ and its entropy excess is $\pi(T_{\rm eff} - T_H)/6$. Since the partially transmitted state is not that Gibbs state, the entropy inequality is strict. Eliminate $T_{\rm eff}$ and use $T_H = 1/(2\pi)$. □

For a hot-input excess .02 and $\chi = .5$, the output has $q_\chi = .01$, $s_\chi = .04818703$ and thermal-floor excess $q_\chi - T_H s_\chi - 3s_\chi^2/\pi = .000113460$. A KMS output with the same heat has $s = .04863795$. Their mean stress, geometry and clock can be identical despite their different states and entropy currents. A geometric response alone does not establish thermalization or identify the source's complete state.

A program that changes the contact also changes its stress. For a smooth prescribed angle, point splitting the full unequal-time covariance gives

$$q_{\rm eff}(\eta) = \sin^2\theta(\eta) q_{\rm hot} + \frac{\theta'(\eta)^2}{4\pi}, \qquad F_\eta = q_{\rm eff} + q_{\rm ref}. \tag{118}$$

Each output port receives the same pump term. The total field work is $\int \theta'^2 d\eta/(2\pi)$ and is at least $(\Delta\theta)^2/(2\pi\Delta\eta)$ over a switching interval. Thus switching the selected interface has a source cost in the same interaction, rather than triggering an unrelated emitter. The running clock equations use $q_{\rm eff}(\eta)$, retaining their reference stress. They also permit a stronger statement for changing programs.

Proposition H.7 (Positive growth under bounded changing sources)**.** *Suppose* $q_{\rm eff}$ *is smooth,* $0 < q_{\min} \leq q_{\rm eff}(\eta) \leq q_{\max}$ *and* $R_0^2 > 2a\kappa q_{\max}$*, with* $\kappa q_{\min}/(2R_0) \leq \delta_0 \leq 2\kappa q_{\max}/R_0$*. Then the source–clock ODE with* $q_{\rm eff}$ *has a global positive solution satisfying*

$$\begin{aligned} \frac{\kappa q_{\min}}{2R} \leq \delta \leq \frac{2\kappa q_{\max}}{R}, \qquad & R' \geq \frac{\kappa q_{\min}}{2}, \\ \int_0^\infty \delta^2 d\eta \leq \frac{8\kappa q_{\max}^2}{q_{\min} R_0}, \qquad & R - R_0 = \kappa \int q_{\rm eff} d\eta - a\Delta\delta - \frac{a}{2}\int \delta^2 d\eta. \end{aligned} \tag{119}$$

The reference correction stays bounded. If the actual source has a long-time mean, the mean horizon entropy rate is 2π *times that mean. This concerns the source–clock sector, not an unlimited body energy reserve.*

Proof. At the upper barrier, the relative derivative is at most $-\kappa q_{\max}/a + 2(\kappa q_{\max})^2/R^2 < 0$. At the lower barrier it is at least $\kappa q_{\min}/(2a) + (\kappa q_{\min})^2/(8R^2) > 0$. Thus the strip is invariant. Integrate its upper bound using $R \geq R_0 + \kappa q_{\min}\eta/2$ to obtain the integral estimate. Integration of $R' + a\delta' + a\delta^2/2 = \kappa q_{\rm eff}$ gives the last identity. The strip also prevents finite-time blowup. □

A bounded clock-reference correction does not remove the program's work. If N disjoint contact changes each have angle increment at least $\Delta_{\min}$ and duration at most ν^{-1}, their total pump work is at least $N\nu\Delta_{\min}^2/(2\pi)$. At $N/\Delta\eta = \alpha\nu$ the mean work current is at least $\alpha\nu^2\Delta_{\min}^2/(2\pi)$, half in each output. This is an additional switching-charge requirement of this contact architecture, not a universal heat bound to be added again when that same output energy has already been counted in the entropy account.

The source is prepared by the contact, but the input reservoir is still prepared in a specified clock frame. The contact's local scattering law does not supply a microscopic moving material body or its controller. Its other port, reservoir preparation, switching work, clock conversion and recoil must be included before it can replace the complete source in the mass-10 table. Nor may a stationary entropy formula be applied instantaneously during the switch. For a stationary frequency cutoff Λ with the common vacuum retained, a separate current bound is

$$|\Delta q| \le \frac{1}{\pi} \sum_{T=T_H,T_o} T(\Lambda + T)e^{-\Lambda/T}.$$

This controls omitted thermal source energy, not the field's global trace distance or the stress of turning the contact on.

I Finite preparation and quantum material source certificates

I.1 Finite preparation on a predicted running clock

The finite interval need not be supplied by an infinite-energy KMS state. Q-RAIF C constructs a compact diffeomorphism of the vacuum that agrees with $\exp[2\pi T\eta(v_-)]$ on the regular operating interval [14, Section on thermal correlations from a finite-energy preparation]. Its full local covariance equals the thermal η covariance, while the global state is pure and has finite energy and its second moment. Two independent copies give the local contact's required inputs, retaining each copy's outside purification. Their stress is exactly Eq. (111) after the same Poincaré subtraction. A local bounded regularization and positive history clock approximate this preparation and its smooth stress to any fixed finite tolerance. The field-energy gain is paid by the clock-interaction energy; this is not a free reservoir renewal.

The prepared states and inertial contact now have a common fermion carrier [14, Proposition on a positive contact on the full CAR carrier]. A compact-gauge compensation proves the single-contact Hamiltonian positive and self-adjoint even when its static dressing is not implemented. The flavor rotation retains the original two fields' correlations instead of replacing them by independent current baths. On this carrier the actual reaction and the exterior smooth-source certificate apply together. Matching that quantum contact to the moving material action remains a separate requirement.

A record selected by earlier local operations can also choose the next thermal preparation on this same field [14, Proposition on record-selected local thermal preparation]. If its past field protocol lies in an interval of length a, the next preparation support has length b, and their gap is d, then the two complex chiral fields give $I_0 = \frac{1}{3}\log[1+ab/(d(d+a+b))]$. The probability-weighted conditional thermal error is at most $\sqrt{I_0/2}$. A smooth source test with actual and comparison second-moment norms M_a, M_i has error at most

$\sqrt{I_0}(M_a + M_i)$, in addition to the preparation-gate and clock errors. The outside purification and prior records remain in the device. The same relative-entropy bound includes a controller and its initially retained references obtained by the local past protocol; it passes through later joint channels on that controller and window. Additional incoming fields must be included rather than treated as independent. Separation, physical record transport and a new preparation are charged; the comparison state is not a physical reset of a correlated field. This permits a conditional finite update, while causal transport in the moving material and renewal of its clock, blanks and reference remain to be realized together.

The field contact can also acquire a finite-mass quantum position [14, Section on the contact position]. The same full-CAR Hamiltonian then conserves body-plus-field momentum; a finite anchor contributes its own recoil. Its moving-frame term is $(p - H_f)^2/(2M)$, so the fixed contact is a fixed-momentum large-mass limit, not a free prescribed trajectory. Position tails require a generator-weighted locality error before using the record-controlled thermal certificate. This support is nonrelativistic and does not yet provide the covariant material stress or the physical clock required below. Prescribing a constant velocity is a different limit: the harmonic support has a stiffness-dependent stability boundary, while the full finite-mass recoil Hamiltonian remains positive.

Energy conservation in the unfolded carrier does not identify its translation generator with bulk momentum. Folding its two flavors into right- and left-moving fields makes the finite-width mixing a mirror coupling. A local bidirectional current instead needs two flavors per direction, with $P = H_R - H_L$ and recoil $(p - P)^2/(2M)$ [14, Proposition on bidirectional current and physical recoil]. Its harmonic prescribed-velocity threshold is $\sqrt{a_0/(a_0 + d_R + d_L)}$, while the full finite-mass energy remains positive. The added preparations and purifications enter the source account, and each exhaust capacity counts its actually routed fields. For the same interval lengths a, b and gap $g > 0$, their product free vacuum gives $I_0 = \frac{2}{3}\log[1+ab/(g(g+a+b))]$, twice the two-field renewal value above; the conditional errors use that value. This supplies a physical momentum and local field density; deriving the propagating material's renormalized quantum stress is still needed before inserting that density as the complete matter in a JT apparatus.

For a flat-coordinate interval of length ℓ, the compact vacuum-orbit preparations at temperature T have infimum energy

$$E_{\min} = \frac{\pi c T^2 \ell}{12} - \frac{cT}{6}\tanh\frac{\pi T \ell}{2}.$$

On a changing clock the actual map, including its compact completion, sets the physical preparation cost. Its waveform cannot be redefined after a read changes the future trajectory.

Proposition I.1 (Source error from a predicted thermal clock)**.** *Prepare with $\widehat{\eta}$ while the actual clock is η. On a fixed regular source interval suppose both first derivatives are at least $m > 0$, both jth derivatives have absolute value at most M_j, and $\|\eta^{(j)} - \widehat{\eta}^{(j)}\|_\infty \le \epsilon_j$ for $j = 1, 2, 3$. Put $J_T = \pi c T^2/12$. The prepared one-point stress differs from the exact η-thermal stress by at most*

$$B_{\text{clock}} = 2J_T M_1 \epsilon_1 + \frac{c}{24\pi}\left(\frac{\epsilon_3}{m} + \frac{M_3\epsilon_1}{m^2} + \frac{3M_2\epsilon_2}{m^2} + \frac{3M_2^2\epsilon_1}{m^3}\right). \tag{120}$$

A constant two-port contact combines its two input bounds with weights $1-\chi, \chi$. A smooth source test f has additional clock error at most $\|f\|_1 B_{\text{clock}}$, separate from the autonomous execution's state and stress error.

Proof. Apply $|a^2 - b^2| \le 2M_1|a-b|$ to the squared Jacobian. The ratio of third to first derivative changes by at most $\epsilon_3/m + M_3\epsilon_1/m^2$. Factoring the difference of the squared second-to-first ratios gives $2M_2\epsilon_2/m^2 + 2M_2^2\epsilon_1/m^3$. The Schwarzian's factor $3/2$ yields Eq. (120). The same coordinate-vacuum subtraction cancels between the two preparations, not between different geometric backgrounds. □

This tolerance makes a record-selected prediction testable without an exactly known future geometry. It controls mean source, not complete state thermality or entropy. Learning a predictor that maintains the stated derivative tube, and embedding the local gate, clock and two-port stress in the moving material action, remain physical tasks. The thermal restriction has an exterior purification; it cannot be inserted unchanged into a finite-window entropy-reset argument.

I.2 A matched quantum material source

Positivity of an excitation Hamiltonian leaves its vacuum contribution to the source undetermined. A local Gaussian material makes the required comparison explicit. On a finite interval with Dirichlet boundaries, let $a \in C_c^\infty$ vanish near the endpoints and $m > 0$. The spatial form for an active field ϕ and a normalized angle Θ is

$$q_a = \int \left[|\phi' + a\Theta|^2 + |\Theta'|^2 + m^2|\Theta|^2\right] dx.$$

It is closed and coercive. Its operator, with the same boundary conditions for $a = 0$, is

$$K_a = \begin{pmatrix} -\partial_x^2 & -\partial_x a \\ a\partial_x & -\partial_x^2 + a^2 + m^2 \end{pmatrix}.$$

Here $-\partial_x a$ denotes $-\partial_x \circ a$, including the derivative of the field. A constant positive angle–storage mass matrix and free spectator fields leave the following coefficient unchanged.

Proposition I.2 (Matched material vacuum coefficient)**.** *For the proper-time cutoff comparison*

$$\Delta E_\Lambda = -\frac{1}{4\sqrt{\pi}} \int_{\Lambda^{-2}}^\infty t^{-3/2} \operatorname{Tr}(e^{-tK_a} - e^{-tK_0})\, dt,$$

and a fixed reference scale $\mu > 0$,

$$\Delta E_\Lambda = \frac{\log(\Lambda/\mu)}{8\pi} \int a^2 dx + C_\mu[a] + o(1). \tag{121}$$

The finite part requires a common material normalization condition.

Proof. Write $K_a = -\partial_x^2 + B\partial_x + C$, where $B = \left(\begin{smallmatrix} 0 & -a \\ a & 0 \end{smallmatrix}\right)$ and $C = \left(\begin{smallmatrix} 0 & -a' \\ 0 & a^2+m^2 \end{smallmatrix}\right)$. The antisymmetric connection $A = -B/2$ gives $K_a = -(\partial_x + A)^2 + P$, with

$$P = C - B'/2 + B^2/4 = \begin{pmatrix} -a^2/4 & -a'/2 \\ -a'/2 & m^2 + 3a^2/4 \end{pmatrix}, \qquad \operatorname{tr}(P - P_0) = a^2/2.$$

The heat coefficient for $-(\nabla^2 + E)$ uses $E = -P$ [3, Eq. (4.27)]. Matched boundary jets cancel, so the trace difference is $-\sqrt{t}\int a^2/(4\sqrt{\pi}) + O(t^{3/2})$. Its proper-time integral gives Eq. (121). Coercivity controls the large-t end of that integral. □

The coefficient can be assigned to a covariant material action. For the positive target metric

$$ds^2_{\rm mat} = (d\phi + \lambda\theta f(Z)dZ)^2 + m_Z^2 dZ^2 + I d\theta^2,$$

linearization about a static texture Z_0, with $\theta = 0$, gives $a = \lambda f(Z_0)Z_0'/\sqrt{I}$. Direct curvature calculation yields $\mathrm{Ric}_{ZZ} = -\lambda^2 f(Z)^2/(2I)$ there. The usual one-loop kinetic counterterm in the Euclidean action [4, Section 7.2] is, in the present normalization,

$$S_{\mathrm{ct},E} = \frac{\log(\Lambda/\mu)}{4\pi} \int \sqrt{g}\, \mathrm{Ric}_{AB}\, \partial_\mu X^A \partial^\mu X^B.$$

On the texture it cancels the logarithm in Eq. (121). Other potential, boundary and metric counterterms must be matched as well. The finite Gaussian model is not the full charged-fermion contact or an all-order continuum construction.

The charged carrier can nevertheless be retained at finite material resolution. Q-RAIF C's positive material construction [14] uses the full fixed-spin charge/current basis, including the unused oscillator modes, and keeps the material momentum squares. Its boson–fermion correspondence is an operator equivalence, not a neutral-sector replacement. This supplies a positive nonlinear regulated Hamiltonian. The local continuum, its thermal states and its renormalized source remain to be matched; the correspondence does not determine a new vacuum subtraction for each contact. Smoothing a permitted bounded gate under this interacting Hamiltonian also supplies the energy regularity needed for autonomous record-controlled operation. Its positive regulated energy tests have a source certificate, while their covariant continuum stress still requires a separate limit. A local positive binding potential also produces finite normalizable angle and storage modes with the finite contact's mechanical coefficients [14]. Their omitted material modes give an explicit self-energy and a filtered causal-response error. This avoids an infinite uniform angle, but an exact local light front and quantum thermal state are not supplied by the collective-mode projection. Local record-selected drives can instead be compared on the same complete Gaussian Hamiltonian and initial state [14], Proposition on local actuation with the initial state retained. The actual field's characteristic function controls the joint state error with references retained, and a centred Gaussian source changes by the classical full retarded response. Its force work is charged; an old mean contributes a cross term. This supplies a quantum actuation/source bridge in the fixed-texture sector, not a new vacuum for the projected body or a proof of the moving clock's stress and thermal renewal.

A source can be derived from this driven material rather than specified as a separate emitter. In the corresponding flat Gaussian sector, ξ_Θ is the complete record-driven angle response, including its coupling to the carrier. Outside the compact contact, the actual right and left profiles satisfy

$$\begin{aligned} \xi_R(u) &= -\frac{1}{2}\int a(y)\xi_\Theta(y, u+y)dy, & \xi_L(v) &= \frac{1}{2}\int a(y)\xi_\Theta(y, v-y)dy, \\ \Delta T_{uu} &= (\partial_u \xi_R)^2, & \Delta T_{vv} &= (\partial_v \xi_L)^2. \end{aligned} \tag{122}$$

Here $u = t - x$, $v = t + x$, the past displacement is zero and the initial Gaussian mean is zero. The carrier equation is $(\partial_t^2 - \partial_x^2)\xi_\phi = \partial_x(a\xi_\Theta)$. Its retarded Green function, integrated once in y, proves the two profiles. Unchanged covariance and common vacuum subtraction prove the stress identities. The integrated two-side carrier radiation is at most the actual force work; the rest is retained material energy. Different finite drive records

can therefore change the source through actual material motion. This is not injective: opposite drives have equal quadratic stress on centred input, and $a = 0$ emits no carrier source while material energy can remain. Putting these flat-background data into JT requires a common quantum-state and coordinate map plus the driver, moving support and clock's stress; the formulas alone do not supply that map.

Finite autonomous execution also needs a source certificate for this material, rather than the free-carrier estimate. On a fixed Dirichlet interval, Q-RAIF C derives a relative form bound for each compact-time quadratic energy or exterior null-stress test O:

$$\|(H+\Lambda)^{-1/2}O(H+\Lambda)^{-1/2}\| \le C_O, \qquad H = d\Gamma(\sqrt{K}), \quad \Lambda > 0.$$

The complete Gaussian mode tower and its common vacuum subtraction are retained. If actual and target output purifications have $e = \|\psi - \varphi\|$, $g_H = \|H(\psi - \varphi)\|$, then

$$|\langle O\rangle_\psi - \langle O\rangle_\varphi| \le C_O\sqrt{\Lambda e^2 + e g_H} \times \left[\sqrt{\Lambda + \|H\psi\|} + \sqrt{\Lambda + \|H\varphi\|}\right]. \tag{123}$$

Energy-regular return under this same H therefore certifies a common output source reading, with old references retained. Uniform physical rms bounds M_a, M_i give $g_H \le M_a + M_i$, so the certificate converges with its state error despite the growing history clock. The bound does not use ordinary trace distance on an unbounded stress. It is an output comparison under the agreed subsequent evolution, not an account of the driver and clock stress during their interaction. Its constants depend on the finite interval. Matched source tests can then enter the response estimate of this paper only after the state, coordinate and total-source identification is supplied; neither a uniform real-line limit nor the full moving JT apparatus follows from this Gaussian certificate. The unbounded-time weighted moments in the JT source hypothesis also need their own tail control; finite smooth tests alone do not certify them.

A physical change of the coupling or texture changes this fluctuation operator. It must use the same measured material parameters and vacuum normalization as the original configuration. Independently setting each contact's vacuum energy to zero would also discard its response to those changes. Conversely, the bare logarithm supplies no finite universal gravitational price. A covariantly renormalized material functional Γ, with $T_{\mu\nu} = -2\delta\Gamma/(\sqrt{-g}\,\delta g^{\mu\nu})$ and $E_i = \delta\Gamma/(\sqrt{-g}\,\delta X^i)$ for its scalar arguments, obeys

$$\nabla_\mu T^\mu{}_\nu = E_A\partial_\nu X^A + E_\Phi\partial_\nu\Phi,$$

where E_A, E_Φ are its variational equations. On the material equations this retains the compensating dilaton force. The same functional must be varied for the material and gravity; a static Euclidean vacuum comparison cannot replace the causal, initial-state response of a controller.

J Supported apparatus and an acquired-record source return

J.1 A material support with its own conserved core

A boundary action gives its conjugate energy but not the matter that holds the boundary. A metric-only local apparatus is also blind to the dilaton on a fixed AdS_2 patch. We now

change the matter model explicitly, without applying the preceding boundary readout laws to this new apparatus. In natural units add a timelike worldline Γ to the JT action:

$$I = \frac{1}{2\kappa}\int \sqrt{-g}\,\Phi(R+2/L^2)\,d^2x + I_{\rm outer} + I_{\rm fields} - \int_\Gamma M(\Phi)\,d\tau,$$
$$M = m_{\rm rest} + U(\Phi) + \langle H_{\rm core}\rangle. \tag{124}$$

The usual outer boundary terms are included. Take $m_{\rm rest} > 0$ and $U \geq 0$. The core has worldline quantum action $\int d\tau\,[i\psi^\dagger D_\tau\psi - \psi^\dagger(m_{\rm rest}+U+H_{\rm core})\psi]$; its expectation provides a semiclassical worldline source; the apparatus's internal spatial geometry is not resolved in this approximation. Q-RAIF C constructs a finite time-independent, nonnegative $H_{\rm core}$ containing the battery, clock-dependent interactions, detector, retained body and every thermal output cell [14]. Its mean energy is conserved. Output cells remain inside the apparatus until a release operation is supplied.

Let δ_Γ be the covariant line delta and $M_\Phi = \partial M/\partial\Phi$. Variation gives

$$R + 2/L^2 = 2\kappa M_\Phi \delta_\Gamma, \tag{125}$$
$$E_{\mu\nu} := g_{\mu\nu}\Box\Phi - \nabla_\mu\nabla_\nu\Phi - g_{\mu\nu}\Phi/L^2 = \kappa T^{\rm total}_{\mu\nu}. \tag{126}$$

The material stress is not separately divergence-free:

$$\nabla_\mu T^{{\rm total}\,\mu\nu} = -M_\Phi\delta_\Gamma\nabla^\nu\Phi, \qquad \nabla_\mu E^{\mu\nu} = -\tfrac{1}{2}(R+2/L^2)\nabla^\nu\Phi. \tag{127}$$

The identities agree by Eq. (125), including energy exchange with the dilaton. Before radiation forces are added, $Ma^\mu = -(g^{\mu\nu}+u^\mu u^\nu)\nabla_\nu M$. Direct dilaton coupling therefore changes the curvature equation at the wall.

Proposition J.1 (Positive static support)**.** *In a symmetric thin-wall limit choose a common normal toward increasing dilaton. At a static wall let $\Phi = \phi > 0$, with derivatives $d_\pm = \partial_n\Phi|_\pm$ and extrinsic curvatures $K_\pm$. The vacuum junction and constraint equations are*

$$[d] = -\kappa M, \quad [K] = -\kappa M_\Phi, \quad K_\pm d_\pm = \phi/L^2, \quad M\overline{K} + M_\Phi\overline{d} = 0. \tag{128}$$

For $M > 0$ and $D > \kappa M/2$, they have the solution

$$d_\pm = D \mp \kappa M/2, \qquad K_\pm = \frac{\phi}{L^2 d_\pm}, \qquad M_\Phi = -\frac{\phi M}{L^2 d_+ d_-}.$$

A positive quadratic material potential realizes these data whenever its stiffness a obeys $M - M_\Phi^2/(2a) > \langle H_{\rm core}\rangle$. If a positive inner invariant is required, also choose $\phi > Ld_-$. For every finite conserved core mean energy, a choice of M, D and sufficiently large a gives positive rest mass and linear radial stability.

Proof. In Gaussian normal coordinates $ds^2 = dr^2 - N(r)^2dt^2$, $R = -2N''/N$ gives the curvature jump. The singular tt metric equation gives $[d] = -\kappa M$, and the regular rr equation gives $Kd = \phi/L^2$. Its jump gives the averaged force equation. The displayed data satisfy all four conditions. Choose

$$\phi_* = \phi - M_\Phi/a, \quad U(\Phi) = \tfrac{a}{2}(\Phi-\phi_*)^2, \quad m_{\rm rest} = M - \frac{M_\Phi^2}{2a} - \langle H_{\rm core}\rangle > 0.$$

The regular patches near the wall are $N_\pm = \cosh(r/L) + LK_\pm\sinh(r/L)$ and $\Phi_\pm = \phi\cosh(r/L) + Ld_\pm\sinh(r/L)$. They have positive lapse there and solve the vacuum equations. □

For $L = 1$, $\kappa = .01$, $M = 10$, $\phi = 3$, $D = 2$ and $a = 100$,

$$d_+ = 1.95, \quad d_- = 2.05, \quad \mathcal{C}_+ = 5.1975, \quad \mathcal{C}_- = 4.7975, \qquad \mathcal{C}_\pm = \phi^2 - L^2 d_\pm^2.$$

Here $M_\Phi = -7.50469043$, $\phi_* = 3.07504690$, and $m_{\rm rest} + \langle H_{\rm core}\rangle = 9.71839811$. The two-cycle core with 39 battery levels has mean 2.34, including its clock-dependent interaction, so $m_{\rm rest} = 7.37839811 > 0$. The 135-level core has mean 7.14 and still fits with positive rest mass. Support and a running finite internal circuit thus have jointly satisfiable semiclassical source data, with clock and battery included in the mass.

The reference support is locally stable in radial vacuum dynamics. On the positive-normal branch let $\beta_\pm = \sqrt{\dot\Phi^2 + (\Phi^2 - \mathcal{C}_\pm)/L^2}$. The moving-wall junction $\beta_- - \beta_+ = \kappa M(\Phi)$ gives

$$\dot\Phi^2 + V(\Phi) = 0, \qquad V = \frac{\Phi^2 - \mathcal{C}_+}{L^2} - \left[\frac{\mathcal{C}_+ - \mathcal{C}_-}{2\kappa L^2 M(\Phi)} - \frac{\kappa M(\Phi)}{2}\right]^2. \tag{129}$$

At the reference, $V = V' = 0$, $V'' = 388805871/5681780 > 0$, and the proper-time linear frequency is 5.84937147. Increasing $\mathcal{C}_+$ by 10^{-5} gives a small oscillation between approximately 2.99755 and 3.00244. This proves linear radial stability at one vacuum support, not stability through an arbitrarily long export history.

The stability construction extends beyond the numerical example. At the supported point, differentiation of Eq. (129) with fixed $\mathcal{C}_\pm$ gives

$$V''(\phi) = \frac{2}{L^2} - \frac{2\phi^2}{L^4 d_+^2} - \frac{4D\phi^2}{L^4 d_+ d_-^2} + \frac{2d_+ d_-}{M}\, a. \tag{130}$$

Choose $M > \langle H_{\rm core}\rangle$ and $D > \kappa M/2$ first. The coefficient of a is positive, so finite stiffness can satisfy both $m_{\rm rest} > 0$ and $V'' > 0$. This supports any fixed finite compiled core; increasing accuracy or operating length may require a larger material device.

J.2 An output pulse must recoil against its source

A thermal cell becomes radiation only through a release mechanism. Even an idealized event must conserve momentum as well as energy. Suppose the initially static wall emits an inward null pulse of energy q in its initial local rest frame. Take an isolated instantaneous event with $0 < q < M/2$, with no external holding impulse. Local four-momentum conservation fixes

$$M_f = \sqrt{M^2 - 2Mq}, \quad v_f = \frac{q}{M-q}, \quad \dot\Phi_f = \frac{d_+ q}{M_f}, \quad \beta_{+,f} = \frac{d_+(M-q)}{M_f}. \tag{131}$$

Proposition J.2 (The source–support charge balance)**.** *Keeping $\mathcal{C}_+$ fixed and imposing the moving-wall junction after the event gives*

$$\mathcal{C}_{-,f} - \mathcal{C}_- = 2\kappa L^2 d_- q. \tag{132}$$

The wall's Killing charge $E_\Gamma = (\mathcal{C}_+ - \mathcal{C}_-)/(2\kappa L^2)$ loses exactly $d_- q$, the pulse's initial interior Killing energy. For an internal mass drop $\Delta = M - M_f$, $q = \Delta - \Delta^2/(2M)$ and recoil energy is $\Delta^2/(2M)$.

Proof. Initially $E_\Gamma = Md_+ + \kappa M^2/2$; after emission it is $M_f\beta_{+,f} + \kappa M_f^2/2$. Equation (131) gives their difference $(d_+ + \kappa M)q = d_- q$. Solving the mass shell for q gives the last claim. □

At $q = .06$ in this support, $M_f = 9.93981891$ and $\Delta\mathcal{C}_- = .00246$. Reducing mass while pinning the wall would introduce the spurious term $-\kappa^2 L^2 q^2$. The inner horizon entropy has infinitesimal response

$$\left.\frac{dS_{h,-}}{dq}\right|_{q=0} = \frac{2\pi L^2 d_-}{\sqrt{\mathcal{C}_-}}, \qquad S_{h,-} = \frac{2\pi}{\kappa}\sqrt{\mathcal{C}_-}.$$

The number $q = .06$ is an emission example, not the finite core's net heat automatically sent into the bulk. Its cells contain gross energy and an initial thermal contribution. The reference and all correlated ports must be transported before an excess source can be identified with charged heat. The finite energy-sector gates do not establish a local field release Hamiltonian or prepare a KMS exhaust on a running horizon clock. This branch supplies a positive support and its charge balance while leaving those transport operations explicit.

J.3 Continuous radiation moves the supported apparatus

The mass, recoil and radiation can be evolved together. On each side of the material wall use an ingoing Eddington–Finkelstein coordinate:

$$ds_\pm^2 = -f_\pm dv_\pm^2 + 2dv_\pm d\Phi, \qquad f_\pm = \frac{\Phi^2 - \mathcal{C}_\pm(v_\pm)}{L^2}.$$

Keep the normal toward increasing Φ and write $w = d\Phi/d\tau$, $\beta_\pm = \sqrt{w^2 + f_\pm}$. Let $P_{\rm in}$ be the proper power of inward radiation absorbed from the outer side, and $P_{\rm out}$ the inward power sent into the inner side. These are the actual renormalized source powers; an excess over a thermal reference is specified separately below. Put $M = m_{\rm rest} + U(\Phi) + e$ and $A = U'(\Phi)$, where e is usable internal mean energy. The mean worldline approximation is that of Eq. (124); junctions with dilaton-coupled surface matter are standard gravitational constructions [6, Sec. V].

Proposition J.3 (Radiating material junction)**.** *For smooth supplied powers and initial data with $M > 0$, $\beta_- - \beta_+ = \kappa M$ and $\beta_\pm - w > 0$, the coupled equations are*

$$\begin{aligned}
\dot e &= P_{\rm in} - P_{\rm out}, & \dot M &= Aw + P_{\rm in} - P_{\rm out},\\
\dot w &= \frac{\beta_+ P_{\rm out} - \beta_- P_{\rm in} - A\beta_+\beta_-}{M} - \frac{\Phi}{L^2}, & &\\
\dot{\mathcal{C}}_+ &= 2\kappa L^2 P_{\rm in}(\beta_+ - w), & \dot{\mathcal{C}}_- &= 2\kappa L^2 P_{\rm out}(\beta_- - w),\\
\dot v_\pm &= (\beta_\pm - w)^{-1}. &&
\end{aligned} \tag{133}$$

Dots denote proper time. These equations preserve the mass junction and obey $[K] = -\kappa A$. They have a local solution while the stated domain and energy reserves hold.

Proof. The regular JT equation gives $T_{v_\pm v_\pm} = \partial_{v_\pm}\mathcal{C}_\pm/(2\kappa L^2)$. Its contraction with the wall velocity gives the two invariant equations. The tangent and normal projections of

the material force balance give $\dot{M} - Aw = P_{\rm in} - P_{\rm out}$ and $M\overline{K} + A\overline{\beta} = P_{\rm out} - P_{\rm in}$. Direct evaluation of the extrinsic curvature gives

$$K_+ = \frac{\dot{w} + \Phi/L^2 + \kappa P_{\rm in}}{\beta_+}, \qquad K_- = \frac{\dot{w} + \Phi/L^2 + \kappa P_{\rm out}}{\beta_-}.$$

Their jump and the mass junction give Eq. (133). Differentiating $\beta_- - \beta_+ - \kappa M$ using these equations gives zero. The right-hand sides are smooth inside the domain, giving local existence and uniqueness. Positivity of quantum excess stress is not used in this algebraic consistency statement. □

For $P_{\rm in} = 0$, $R = \sqrt{\mathcal{C}_-} > 0$ and $d\eta/dv_- = R/L^2$, the actual clock conversion is

$$z := \frac{d\tau}{d\eta} = \frac{L^2(\beta_- - w)}{R}, \qquad \frac{dS_{h,-}}{d\eta} = 2\pi z^2 P_{\rm out}. \tag{134}$$

Recoil enters both the source and the clock; keeping the wall fixed while its mass decreases does not solve these equations.

J.4 Stored mechanical energy, radiation and the moving support

The angle need not be specified externally. Q-RAIF C gives a positive inertial current Hamiltonian with a storage coordinate, its retarded reaction and force spectrum [14, Section on an inertial contact]. A displacement selected by a record changes the actual excess current stress, not merely a reconstructed parameter. That local rest-frame model still needs its infrared representation, moving ports and complete quantum stress to be matched to the material action. In particular, $\gamma\langle\dot{\theta}^2\rangle$ alone is not its net heat: the noise–velocity correlation contributes to the work.

A distinct classical branch makes a common moving action explicit. Put a real massless scalar on each side of the wall, with actions $-\frac{1}{2}\int_{\mathcal{M}_\pm}\sqrt{-g}(\nabla\varphi_\pm)^2$. Append the mechanical worldline action

$$S_\Gamma = \int d\tau \left[-m_{\rm rest} - U(\Phi) + \frac{I}{2}\dot{\theta}^2 + \frac{J}{2}\dot{y}^2 - \frac{k}{2}(\theta - y)^2 - \frac{K}{2}y^2\right], \qquad I, J, k, K > 0, \tag{135}$$

and impose $\varphi_\pm|_\Gamma = g_\pm\theta$ with worldline Lagrange multipliers. The wall, storage and angle are all varied. These are full nonchiral scalar fields, not two auxiliary right movers reinterpreted as an anomaly-free bulk theory. With no classical incoming waves, their outgoing solutions are $\varphi_\pm(\xi_\pm) = g_\pm\theta(\tau_{\rm ret})$. Their proper powers and the mechanical equations are

$$\begin{aligned} P_\pm &= g_\pm^2\dot{\theta}^2, & \gamma &= g_+^2 + g_-^2, \\ I\ddot{\theta} + k(\theta - y) &= -\gamma\dot{\theta}, & J\ddot{y} + k(y - \theta) + Ky &= 0, \\ e &= \tfrac{I}{2}\dot{\theta}^2 + \tfrac{J}{2}\dot{y}^2 + \tfrac{k}{2}(\theta - y)^2 + \tfrac{K}{2}y^2, & \dot{e} &= -P_+ - P_-. \end{aligned} \tag{136}$$

Boundary variation gives the mechanical force; characteristics give the powers. There is no independent supplied radiation function. Incoming waves would add their force and work and are excluded in this branch. Quantizing a moving boundary also requires its vacuum stress and fluctuations; the classical formulas do not supply them.

The outer radiation now moves outward, unlike the outer input of Eq. (133). Use outgoing EF coordinate u_+ outside and ingoing v_- inside. Set $M = m_{\rm rest} + U + e$, adding any fixed probe offset, and keep $\beta_- - \beta_+ = \kappa M$. The same action gives

$$\begin{aligned} \dot w &= \frac{\beta_+ P_- - \beta_- P_+ - U'\beta_+\beta_-}{M} - \frac{\Phi}{L^2}, \\ \dot{\mathcal{C}}_+ &= -2\kappa L^2 P_+(\beta_+ + w), & \dot{\mathcal{C}}_- &= 2\kappa L^2 P_-(\beta_- - w), \\ \dot u_+ &= (\beta_+ + w)^{-1}, & \dot v_- &= (\beta_- - w)^{-1}. \end{aligned} \tag{137}$$

The opposite invariant signs follow from $T_{u_+u_+} = -\partial_{u_+}\mathcal{C}_+/(2\kappa L^2)$ and $T_{v_-v_-} = \partial_{v_-}\mathcal{C}_-/(2\kappa L^2)$. The tangent Ward equation is $\dot M = U'w - P_+ - P_-$ and the normal equation is $M\overline{K} + U'\overline{\beta} = P_- - P_+$. Here $K_\pm = (\dot w + \Phi/L^2 + \kappa P_\pm)/\beta_\pm$. These identities give both $[K] = -\kappa U'$ and conservation of the mass junction. They do not follow by putting a negative outer input into Eq. (133); the null direction also changes.

Proposition J.4 (A finite mechanical-source window)**.** *Equations* (136) *and* (137) *have a unique local solution in the domain* $M > 0$, $\beta_\pm > 0$, $\beta_+ + w > 0$, $\beta_- - w > 0$, $\mathcal{C}_- > 0$. *They preserve the mass junction,* $e \ge 0$, *and*

$$\int_0^\tau (P_+ + P_-)\, ds = e(0) - e(\tau) \le e(0). \tag{138}$$

For every fixed finite proper-time window, positive static-support data admit a sufficiently small positive coupling with nonzero radiation throughout a nontrivial subinterval and a solution inside that domain.

Proof. The right-hand side is smooth in the stated domain. Direct differentiation of the junction gives zero, and the positive quadratic mechanical energy obeys the displayed budget. At $g_\pm = 0$ the supported geometry is static and the internal oscillator energy is conserved. Continuous dependence on the couplings preserves its strict domain margins on any specified compact time interval for sufficiently small positive $g_\pm^2$. Nonstationary mechanical data give nonzero angle velocity. □

Balanced outgoing power also gives a necessary operating-lifetime bound for this same apparatus. If its fixed operating domain satisfies $0 < \Phi \le \Phi_{\max}$, $\Phi^2 > \mathcal{C}_-$ and $\beta_- - \dot\Phi \ge b_{\min} > 0$, then

$$\begin{aligned} \mathcal{C}_-(\tau) - \mathcal{C}_-(0) &\ge \kappa L^2 b_{\min} Q_{\rm out}(\tau), \\ Q_{\rm out}(\tau) &< \frac{\Phi_{\max}^2 - \mathcal{C}_-(0)}{\kappa L^2 b_{\min}}. \end{aligned}$$

Equation (137) integrates this inequality for $P_- = P_+ = P/2$. If each complete episode exports at least $q_{\min} > 0$, its count is below the latter bound divided by $q_{\min}$, as well as limited by the installed program and prepared bank. The two acquisitions in Foundation III are one episode, not two renewals. Enlarging the domain or changing its null-clock margin changes this necessary bound; it is not a general impossibility theorem.

The common energy and geometry can be held fixed across records. Use $I = 10, J = 20, k = 1, K = .1$, $g_+^2 = g_-^2 = 1/(4\pi)$ and $e(0) = .95$. Install $\theta(0) = \dot\theta(0) = 0$, $y(0) = A_R$ and $\dot y(0) = \sqrt{2[.95 - .55A_R^2]/J}$ for $A_R = .5, 1.3$. Both records then use the same mass-10

support, potential and initial invariants of Section J.1. With the positive probe offset .02 below, their common m_{rest} is 8.74839811. At proper time 80 the two solutions give

A_R	.5	1.3
$\int (P_+ + P_-)d\tau$	.428637558	.522051307
$e(80)$	.521362442	.427948693
$\mathcal{C}_-(80) - \mathcal{C}_-(0)$	.008793748	.010712134
$\Phi(80)$	3.003737133	3.004545378

(139)

Both remain outside the inner horizon with positive reserves. Integrating both invariants independently preserves the junction to 8.3×10^{-15}; another integrator agrees to 3.7×10^{-12}. The phase probe below, at $\lambda = .02$, yields Y-outcome probabilities .503428742 and .504187486. The difference is created during evolution, not by a different initial mass or an externally selected source. The numerical 80-window check is not an interval-arithmetic domain proof.

The inner rapidity clock is determined by the same solution: $\dot{\eta} = \sqrt{\mathcal{C}_-}/[L^2(\beta_- - w)]$ and $dS_{h,-}/d\eta = 2\pi(d\tau/d\eta)^2 P_-$. This source is coherent classical radiation, not an η-KMS heat output, so its horizon response cannot be substituted into the thermal entropy bound. Installing the mechanical record has a preparation cost. The finite return in Section 9.1 instead lets an acquired quantum record exert force on the same storage coordinate. Preparing these initial mechanical data and renewing the whole apparatus are distinct from that return.

Proposition J.5 (A rest-frame mean-stress bridge)**.** *Use Q-RAIF C's quadratic current–mechanical contact on a fixed inertial line, with the same I, J, k, K and a common centered Gaussian state for mechanics and incoming currents. Displace only the mean mechanical data; take the mean initial angle to be zero. Keep all centered correlations and old references. For independent physical outgoing directions $s = \pm$, let the contact coefficients obey $\kappa_s = \sum_i q_{si}^2$ and $\gamma_s = \kappa_s/(4\pi)$, where q_{si} are the effective angle-coupling coefficients, not a change of the carrier's physical charge sectors. Assume the renormalized current products exist on the common domain. In the zero-width limit of the retarded mean solution, away from initial/contact singularities, the outgoing stress difference and mean mechanical equations are*

$$\Delta\langle T_{\mathrm{out},s}\rangle = \gamma_s v(\tau_{\mathrm{ret}})^2, \qquad I\ddot{\theta} + k(\theta - y) = -(\gamma_+ + \gamma_-)\dot{\theta}, \quad v = \dot{\theta}. \tag{140}$$

Thus $g_s^2 = \gamma_s$ identifies this excess mean stress and its mechanical energy loss with Eq. (136) *in that fixed local frame. The balanced scalar example requires $\kappa_+ = \kappa_- = 1$. Q-RAIF C's unscaled two-flavor bidirectional contact has $\kappa_+ = \kappa_- = 2$ and twice that total damping; scaling its angle coupling by $1/\sqrt{2}$ gives the balanced normalization.*

Proof. Take $f_a(x) = a^{-1}f(x/a)$ with nonnegative $f \in C_c^\infty(\mathbb{R})$ and $\int f = 1$. At finite width the displacement adds the classical linear solution without changing the common covariance. In direction s its current mean shift is $q_{si}h_a/(2\pi)$, where h_a is the retarded convolution of v with f_a along that characteristic. The Sugawara stress $\pi \sum_i : j_{si}^2 :$ has zero cross term with the centered baseline. Its difference is therefore $\gamma_s h_a^2$. Eliminating the mean currents gives the retarded kernel $\Gamma_{a,s}(t) = \kappa_s \int f_a(x) f_a(x - t)dx/(2\pi)$, whose one-sided integral is γ_s. For smooth mean histories its convolution error for a smoothly extended velocity is bounded by $\|v'\|_\infty \int_0^\infty t\Gamma_{a,s}(t)dt = O(a)$. For the initial-value solution, the initial kernel tail has integrated size $O(a)$ as well. The common kernel mass bounds

the linear Volterra solution uniformly on fixed windows; Gronwall then gives convergence to the displayed damped equation. Beyond the contact, h_a tends to the retarded velocity, proving the stress statement. The resulting positive quadratic displacement energy obeys $\dot{e} = -(\gamma_+ + \gamma_-)v^2$, the scalar-source balance. Incoming noise–velocity work and centered quantum stress remain in the common baseline, rather than being set to zero. □

This identifies an excess mean radiation channel. Absolute quantum stress, a quantum zero-width channel limit, dilaton-dependent mass, moving-frame work, physical support and recoil, acquired-record spring forces and stress fluctuations require further matching. In particular a statistical mixture of mechanical records has branch covariance and cannot be replaced by one Gaussian displacement in this proposition.

J.5 A quantum phase reads the moving dilaton

The material model also permits a direct finite sensor. For $\lambda > 0$, add a probe with

$$H_P(\Phi) = \epsilon_P I + \lambda(\Phi - \phi_0)Z_P, \qquad \epsilon_P > \lambda \sup_\Gamma |\Phi - \phi_0|, \tag{141}$$

initially in $|+\rangle$. This is the same kind of explicit dilaton coupling as the support, now in an internal quantum degree of freedom. It does not follow from metric-only matter. During the reading, $\langle Z_P \rangle = 0$ and Z_P is conserved. In the mean worldline approximation its energy is ϵ_P and its mean additional dilaton force is zero. Count ϵ_P in the mass, reducing m_{rest} by that amount in the displayed examples. The probe's phase gives

$$\Pr(Y_P = +1) = \frac{1 + \sin[2\lambda \int (\Phi - \phi_0) d\tau]}{2}. \tag{142}$$

With $\lambda = \epsilon_P = .02$, the $q = .01$ and $.02$ histories give .50617417 and .51230214 at the same proper duration. Their actual bit laws have total variation distance .00612797. No numerical dilaton value is supplied to the controller.

The terminal read itself can be implemented. Q-RAIF C gives a positive autonomous clock whose last two gates write the actual Y result into a record, then apply a true SWAP with a paid coherent blank [14]. The probe is renewed; all old correlations move into the blank and record. For a supplied finite $\Phi(\tau)$ trajectory, its joint error relative to the actual phase at $\tau_0 < T$ is bounded by

$$d_{\text{joint}} \le 2A\tau_0\sqrt{P_0} + 2\sqrt{P_0} + \lambda D(T - \tau_0), \quad D = \sup |\Phi - \phi_0|, \tag{143}$$

where P_0 is the clock's terminal binomial tail and A is the sum of its last two edge strengths. Finite resources suffice for each specified error; the clock mean energy is Jm, and must enter the supported mass. The sufficient $m = 10^5$, $T = 80$, $\tau_0 = 78$ example uses clock mean energy 1963.4954, not the mass-10 apparatus above. During writing and renewal the mean additional force need not vanish: the material equation uses $A_{\text{mat}} = U' + \lambda\langle Z_P \rangle$, with $|A_{\text{mat}} - U'| \le \lambda$. The probe, clock, blank and record belong in the same matter action. If e now includes their Φ-dependent internal energy, its equation is $\dot{e} = \lambda\dot{\Phi}\langle Z_P \rangle + P_{\text{in}} - P_{\text{out}}$; then $\dot{M} = A_{\text{mat}}\dot{\Phi} + P_{\text{in}} - P_{\text{out}}$ as required by the material Ward balance.

A smaller clock can be included directly in this mean material action, instead of using the large worst-case certificate. Take $m = 128$, $J = \pi/160$, $\epsilon_P = \lambda = .02$, and retain the complete 129-state clock, probe, record and blank throughout the proper-duration

80 evolution. Its initial internal mean energy is 2.5532741. Choose $m_{\mathrm{rest}} = 4.8251240$ so that the initial mass 10 already includes it and the declared source reserve 2.34. Evolve the joint Schrödinger equation for

$$H_{\mathrm{int}}(\Phi) = H_{\mathrm{hist}} + 2\epsilon_P I + \lambda(\Phi - \phi_0)Z_P$$

with the radiating junction and its actual η clock. The two contact outputs $\chi = .5$ and 1, at hot-input excess .02, give

q_χ	actual $\Pr(R = 1)$	final source reserve	probe reset distance
.010	.50529766	1.42403868	.00075428
.020	.51055894	.50256908	.00150041

The last column is the distance to the closest pure product of a renewed probe and a normalized retained internal reference, not an error certificate against the ideal meter's entire reference state. The actual record laws differ by .00526128. Norm, mass-junction and internal dilaton-work residuals are below 4×10^{-11} in this integration; a second integrator agrees on the displayed trajectory and record values within 1.3×10^{-9}. These are numerical checks, not interval-certified error bounds. For these gates and initial data, the vanishing mean force has an exact reason.

Proposition J.6 (A force-neutral read and replacement)**.** *Let K denote conjugation in the clock and computational bases and set*

$$\mathsf{C} = (-1)^{N_{\mathrm{clock}}} X_P X_B K, \qquad H_0 = H_{\mathrm{int}} - (Jm + 2\epsilon_P)I.$$

For the identity-edge clock with terminal C_Y and true SWAP, $\mathsf{C}H_0(\Phi)\mathsf{C}^{-1} = -H_0(\Phi)$ and $\mathsf{C}Z_P\mathsf{C}^{-1} = -Z_P$. With initial clock state 0 and data $|+\rangle_P|0\rangle_R|+\rangle_B$, every real dilaton trajectory has $\langle Z_P\rangle = 0$ and $\langle H_{\mathrm{int}}\rangle = Jm + 2\epsilon_P$ throughout the read.

Proof. Conjugation exchanges the two Y projections, and X_P exchanges them again. Thus $X_P X_B C_Y^* X_P X_B = C_Y$; the real SWAP commutes with $X_P X_B$. Clock parity reverses every off-diagonal edge, while X_P reverses Z_P. This proves both operator identities. Remove the constant Hamiltonian phase. Since C is antiunitary, it reverses both i and H_0 in the Schrödinger equation, and so maps its solution to another solution with the same initial state. Uniqueness makes that solution C-invariant. Expectations of the two odd Hermitian operators must vanish. □

The clock energy is therefore included from the start without changing the previously supported mean trajectory. This does not set the quantum force operator or its fluctuations to zero, and different blanks or gates need not have the symmetry. The general coupled force equation, not a zero-force constraint, was used in the numerical calculation. The thermal reservoir's quantum degrees, both-port material stress and its local preparation in η are not thereby included in that body.

The same finite meter also reads the mechanical-source branch of Section J.4. Its complete mean energy 2.5532741, rather than just the probe offset, is included from the start by using the common positive rest mass 6.21512398 for both equal-energy records. Proposition J.6 leaves their mean mechanical and geometric trajectories unchanged. Their actual acquired record probabilities are .502991099 and .503655046, a total variation distance .000663947. Keeping clock, blank and record gives distances .000376618 and

.000458340 to the closest renewed-probe pure product with its normalized internal reference. These are not errors against an ideal reference state. The full vector evolution preserves norm to 1.6×10^{-11} and internal mean energy to 2.6×10^{-12}; a second integration agrees on the record probability to 3.1×10^{-14}. This finite episode obtains the record through physical gates, rather than reading the numerical solution into a controller. In this force-neutral example the record does not act on the mechanics. The next section adds that interaction and includes its reaction during acquisition. The scalar source has no thermal reservoir whose regeneration could be inferred from either construction.

For input streams jointly independent of the record and prepared in a common clock frame, the record can select either of two contact angles in the next stationary window. If their transmissions are $\chi_0 \neq \chi_1$, its source-current mean is

$$q_{\mathrm{next}} = q_{\mathrm{hot}}[\chi_0 + (\chi_1 - \chi_0)\Pr(R = 1)].$$

It depends on the actual phase bit, not a supplied numerical dilaton. The angle change's pump work in Eq. (118) is also charged. This gives the conditional response-to-source map; it does not yet autonomously prepare the reservoirs or renew the entire controller. Quantum fluctuations of the probe force also lie outside the mean-source approximation. The phase sensor and the earlier fixed-boundary current sensor are different realizations, not two freely composable laws for one apparatus.

J.6 Checks of the acquired-record source return

A complete integration of Section 9.1 uses the supported mechanical parameters I, J, k, K above, $\gamma = 1/(2\pi)$, $e(0) = .95$, $y_0 = .5$ and $d = -.5$, with $\lambda = .2$, $\chi_R = .05$ and $E_{\mathrm{off}} = .04$. The 33-state clock has edge scale $\pi/160$, $r = 3$, mean energy .628318531 and spectral maximum 1.256637061. The initial probe has $\upsilon = .8$, its two-dimensional old purification is retained, and B starts in $|+\rangle$. The quadratic support has stiffness 100 and $U'(3) = -30/(1.95 \cdot 2.05)$, as in the static mass-10 support. Both geometries have initial $\Phi = 3$, $w = 0$, $M = 10$ and $\mathcal{C}_- = 4.7975 \pm .1$, with the outer invariant fixed by the junction. The initial core energy .668318531 is already in the mass; the common rest mass is 8.100079577. At proper time 100:

$\mathcal{C}_-(0) - 4.7975$	$\Pr(R = 1)$	Q_{out}	$\mathcal{C}_-(100) - \mathcal{C}_-(0)$
$-.1$	.506528009	.518808111	.010786169
$+.1$	.496292018	.517763845	.010480933

The largest proper-power difference on this window is 1.0350613×10^{-4}; with either interaction removed it is below 3.2×10^{-16}. With feedback removed the two record laws still differ, while their scalar sources agree. Thus acquisition and actuation are separately tested. The nonzero-source result concerns the emitted power, not the already different initial geometries.

The full core–reference vector is evolved along with both invariants. Norm, junction and the three work balances, including U's dilaton work, have residuals below 3.7×10^{-14} in the primary integration. A separate tensor-neighbour implementation with another integrator agrees on the record and source to 6.0×10^{-13} and 2.0×10^{-15}, respectively, and on the mass to 1.2×10^{-8}. The largest probe force is about .0082 and is retained. Clock energy changes during acquisition; it is not an uncharged timer. Only the C_Y edge changes P_R, giving

$$|\dot{p}| \leq 2a_r \|\Psi_{r-1}\| \|\Psi_r\|.$$

This bound is checked on the interacting clock, not a free-clock tail. It supplies finite record retention when that edge is sparsely occupied, not permanent halt. The construction closes an actual finite geometry–record–force–source return in the mean material branch. Full quantum recoil, covariant thermal transport and renewal of the clock, blank and reference remain required for Conjecture C.1.

J.7 A spatial return with a physical clock

The geometry and support below are semiclassical mean fields. The internal oscillators and both retained spins are quantum. The regulator is part of the physical action; the statement does not remove its bandwidth cutoff.

Take JT gravity with radius R and coupling $\kappa > 0$, a full material label a, and a full synchronization scalar θ, with

$$S_b = -\int \sqrt{-g}\, M \left[mn + \frac{K}{2}(n-n_0)^2 \right], \quad n^2 = (\nabla a)^2, \qquad S_\theta = -\frac{J}{2}\int \sqrt{-g}(\nabla\theta)^2. \tag{144}$$

Require $m > Kn_0 > 0$, n near n_0, and a time function $\theta_t > 0$ with timelike gradient. Use the unrestricted inverse embedding $\theta(X(\tau,\xi)) = \tau$, $a(X(\tau,\xi)) = \xi$ and $h = X^* g$. Only the internal expansions are finite: $\sqrt{I}\chi = \sum q_{c\alpha} e_\alpha(\xi)$ and $u = \sum q_{u\alpha} e_\alpha(\xi)$, for smooth supported real e_α. Substitute them into the two scalar kinetic actions and the potential

$$V = \frac{I\mu^2\chi^2}{2} + \frac{1}{2}\left[\omega u + \frac{A\chi}{\omega}\right]^2 + E_o\rho_o - D_o, \quad A = \lambda + gwY_P + \gamma w_2 Y_S, \tag{145}$$

before Legendre transform and metric/embedding variation. Here

$$D_o = \frac{C_\chi}{\omega^2}(\lambda g w Y_P + \lambda\gamma w_2 Y_S + g\gamma w w_2 Y_P Y_S), \qquad C_\chi(\xi) = e^T Q_c e / I.$$

Q_c is the fixed prepared clock covariance. Choose the paid scalar offset to dominate $|D_o|$. Quantize the resulting Hamiltonian with Weyl ordering and a joint-state Berry term on the actual θ leaves. Its kinetic matrices are

$$C = -\int \sqrt{-h}\, h^{\tau\tau} e e^T, \quad D = -\int \sqrt{-h}\, h^{\tau\xi} e (e')^T, \quad K_{\mathrm{grad}} = \int \sqrt{-h}\, h^{\xi\xi} e' (e')^T,$$

giving $(p - Dq)^T C^{-1}(p - Dq)/2 + q^T K q/2$ plus the paid offset/calibration terms. A proper-number sensor contributes

$$H_S = \int n_S(\xi) N [E_S + h_s \Phi(X) Z_S]\, \mathrm{d}\xi, \quad N = \sqrt{-h_{\tau\tau}}, \quad \int n_S \mathrm{d}\xi = 1. \tag{146}$$

n_S is smooth, symmetric, and has nonzero width. Include both spins, their Berry terms, all profile reactions, and any old reference in the same retained state. Restrict $|h_s| \sup|\Phi| < E_S/2$. No profile is an external time-dependent signal.

Proposition J.7 (Finite-bandwidth spatial return)**.** *There are finite positive I, M, J, R, smooth finite-width preparations, and nonzero finite h_s, γ for which this one mean-action system has a positive local solution and two actual histories with the same initial geometric constraints, differing only in their retained P record, whose next-program probabilities*

for $(1+Y_S)/2$ differ at the same central proper time. The retained Y_S enters the subsequent actual material Hamiltonian, with a nonzero mixed sensing/return response of the material mean acceleration. All clock, support, interaction and probe sources are metric variations of this action, and satisfy the full Ward identity. This is an existence result, not a uniform gain, state-independent CPTP map, continuum-controller or thermal-renewal statement.

Proof. Prepare $P = |\pm Y_P\rangle$, $S = |+X_S\rangle$, a common free product Gaussian field state with $\langle q\rangle = 0$, zero symmetrized qp covariance, and clock momentum $\sqrt{I}\Omega F$. References are retained, not reset. At $\gamma = h_s = 0$, calibration cancels the initial expected record-dependent potential as a functional of X. Thus the complete initial metric stresses coincide, not just their integrals. Take common static AdS data, $\theta = t$ on a finite prepared plateau, and hydrostatic bar density $n = n_0 + (m/K)[\cos(x/R) - 1]$ on a small causal patch. The common initial JT constraint is a linear spatial equation for Φ; adding a sufficiently large finite positive homogeneous solution $B\sec(x/R)$ makes $\Phi > 0$. This changes initial geometric data, not the charged positive source.

We first check the physical time equation. At rest, varying $k = X^t_\tau$ gives $C = \int x' e e^T/k$, $K_{\rm grad} = \int k e'(e')^T/x'$ and $K_{\rm mass} = \int e^{2\rho} k x' e e^T \mathcal{M}_\pm$. Hence the local time-momentum current $\delta H/\delta X^t_\tau$ contains only q^2, p^2 and commuting spin constants. Its commutator with the initial Hamiltonian is Weyl qp; its expectation, and the mixed time/spatial current expectation, vanish. The corresponding initial mixed principal block vanishes too. The full time EL equation therefore gives $\theta_{tt}(0) = 0$, or $\theta = t + O(t^3)$. It changes no potential trace coefficient of order t^2. Nevertheless, if $\theta = t + \theta_3 t^3/6 + \cdots$, the actual $T_{\theta,tt} = T_{\theta,xx} = J/2 + J\theta_3 t^2/2 + \cdots$ is retained; only its trace is identically zero.

Here is a nonempty analytic trace certificate. Choose one normalized even smooth mode e, constant and positive in an open central neighborhood, $w = 1$ on its support, and $F = 1$. Put $k = \int (e')^2$, $\beta = \int \sec^2(x/R) e^2$, $Q_c = [2\sqrt{k + \mu^2\beta}]^{-1}$, $Q_u = [2\sqrt{k + \omega^2\beta}]^{-1}$. The exact static central trace-contrast coefficient is

$$d_2 = -D_0 + B_1/I + B_2/I^2, \quad D_0 = 4\lambda g \Omega^2 e(0)^2/\omega^2,$$
$$B_1/D_0 = \beta[\mu^2 Q_c + \omega^2(Q_c + Q_u)]/\Omega^2, \tag{147}$$
$$B_2/D_0 = 2\beta(\lambda^2 + g^2) Q_c/(\omega^2\Omega^2). \tag{148}$$

These follow by $[t^2]Q = -\{K_\pm - K_0, Q_0\}/2$; the off-diagonal species covariance is included. The initial spatial momentum current is locally proportional to $ee'\,\mathrm{sym}(qp)$, so its force contrast and spatial derivatives vanish on the central plateau. No bar recoil trace term of order $1/I$ survives there. For $(\mu, \omega, \lambda, g, \Omega) = (.3, 1.4, .6, .2, .8)$, $B_1/D_0 \le 6.432292\beta$ and $B_2/D_0 \le 1.062926\beta$. Take finite R with $\beta < 1.01$, and $I = 100$. Then $d_2 < -0.9349 D_0$. Scale $M = \alpha L$, $J = \beta_b L$ together, with fixed positive ratios. Actual common-acceleration corrections to (148) are $O_I(L^{-1})$; hydrostatic preparation cancels the order-M force, not the quantum reaction. A sufficiently large finite L preserves $d_{2,\rm total} < -D_0/2$.

For completeness the actual coupled equations have a local solution. Let $\mathcal{A} = 1 + \sum(q_j^2 + p_j^2)$. A degree-two Hamiltonian gives $\|\mathcal{A}^s U_z \psi\| \le e^{C_s T}\|\mathcal{A}^s\psi\|$ and

$$\|\mathcal{A}^2(H_z - H_{\hat z})\psi\| \le C\|z - \hat z\|_{H^{r-1}}\|\mathcal{A}^3\psi\|, \qquad r \ge 4. \tag{149}$$

Here z consists of first-order classical variables. Quadratic principal/source moments and quartic differentiated-current forces are Lipschitz in $D(\mathcal{A}^2)$ on this graph ball. The latter degree, which arises from noncommuting spin coefficients, is not suppressed. The full

bar and synchronization principal forms have size L and strict positive kinetic/spatial margins; their finite quantum corrections, including smooth finite-rank kernels, have bounded norm. Large finite L preserves those margins.

To express geometry on actual leaves, use ambient-frame derivatives $P = (\rho_t, \rho_x)$ and $Q = (\Phi_t, \Phi_x)$ pulled back to X. Both first-order wave systems have the symmetric matrices

$$A_\tau = \begin{pmatrix} \theta_t & -\theta_x \\ -\theta_x & \theta_t \end{pmatrix} > 0, \qquad A_\xi = \begin{pmatrix} a_t & -a_x \\ -a_x & a_t \end{pmatrix}.$$

Their sources are respectively $(\kappa e^{2\rho} J_\Phi - e^{2\rho}/R^2, 0)$ and $(-2e^{2\rho}\Phi/R^2 - \kappa e^{2\rho} T, 0)$. No second embedding derivative enters this geometric principal block. The first-order classical energy in an H^r ball contracts in H^{r-1}; (149) contracts the state in $D(\mathcal{A}^2)$ using only a uniform $D(\mathcal{A}^3)$ bound. Both difference estimates have a factor T. Time derivatives of principal quadratic moments are at most quartic, so there is no higher-graph difference cycle. Picard iteration, followed by closedness and finite smooth-data bootstrap, gives the full unprojected EL solution. Smooth preparations with the additional finite set of graph moments give six time derivatives and the two coupling derivatives used below. Unrestricted X, including its time equation, gives $\nabla_\mu T^{\mu\nu} = -J_\Phi \nabla^\nu \Phi$; consequently the JT constraints propagate. The sources include the variation of the regulator and proper-number probe. At nonzero γ, retain $J_\Phi = h_s n_S \langle Z_S \rangle / \ell_\perp$ and the resulting $R(g) + 2/R^2 = 2\kappa J_\Phi$.

At $\gamma = 0$, Z_S is conserved with mean zero; the source evolution is independent of h_s, and $\langle Y_S \rangle = \sin(2h_s\Theta)$, where $\Theta = \int n_S N \Phi \, \mathrm{d}\tau \mathrm{d}\xi$. At the parity-fixed center, $X = (T(\tau), 0)$ and $N = T'$. Composing with each history's endpoint $\tau_\pm(t) = \theta_\pm(t, 0)$ therefore gives exactly $\Theta_\pm(t) = \int_0^t \Phi_\pm(v, 0) \mathrm{d}v$, with t central proper time. The wave equation with common initial data yields

$$\Delta\Phi(t, 0) = -\kappa d_{2,\mathrm{total}} t^4/12 + O(t^5), \quad [t^5]\partial_{h_s} \Delta \langle Y_S \rangle \big|_0 = -\kappa d_{2,\mathrm{total}}/30 \neq 0. \tag{150}$$

Finite-width proper-time phases converge in this fifth jet to (150). Scaling M, J together keeps background-dilaton times recoil jets bounded; the center endpoint contribution cancels exactly. Choose a small but nonzero symmetric width preserving the coefficient, then finite L preserving all margins. Lower-order recoil jets are allowed. Thus some finite short $t > 0$ and small nonzero h_s give nonzero next-program contrast. Smooth parameter dependence preserves it for sufficiently small finite $\gamma \neq 0$, when the retained program enters the actual material square (145).

One can also check a complete mean-force response, without attributing every term to geometry. Put $w_2 = 1$ on $\operatorname{supp} e$ and write $\bar{\Phi}_0 = \int n_S N \Phi(0) > 0$, $b = \beta$, $c = \langle q_c \rangle / \sqrt{I}$, $y = \langle q_u \rangle$, $a_\pm = \lambda \pm g$. At the static-support large-L limit the unperturbed means satisfy $\ddot{c} = -\nu_c^2 c - a_\pm^2 bc/(I\omega^2) - a_\pm by/I$, $\ddot{y} = -\nu_u^2 y - a_\pm bc$. For $\delta = \partial_\gamma \partial_{h_s}|_0$, independence at $\gamma = 0$ and $\partial_{h_s}\langle Y_S \rangle = 2\bar{\Phi}_0 t + O(t^3)$ give

$$\begin{aligned} \delta\ddot{c} = & -(\nu_c^2 + a_\pm^2 b/(I\omega^2))\delta c - a_\pm b \delta y/I - 2\bar{\Phi}_0 t(by/I + 2a_\pm bc/(I\omega^2)), \\ \delta\ddot{y} = & -\nu_u^2 \delta y - a_\pm b \delta c - 2b\bar{\Phi}_0 tc. \end{aligned}$$

The common $O(t^3)$ sensor term cancels in the branch fourth-jet. With $\dot{c}(0) = \Omega$, all other initial means zero, these equations include the clock/state variation and give

$$[t^4]\delta(\ddot{y}_+ - \ddot{y}_-) = \bar{\Phi}_0 \left[\frac{8\lambda g b^2 \Omega}{3I\omega^2} + O_I(L^{-1}) \right] \neq 0. \tag{151}$$

The finite-L correction includes proper-time pullback: mixed synchronization displacement can start at t^4, spatial displacement at t^5, while mixed curvature starts at t^5. Their fourth-jet mean-force contributions, divided by $\bar{\Phi}_0$, are $O_I(L^{-1})$ by the scaled positive EL inverse. Initial mixed source differences vanish. Ordinary phase terms are present: (151) is not a purely geometry-induced coefficient. Geometry's nonzero path was separately certified by (150). Choose finite L also preserving the strict sign in (151), then small finite h_s, γ and a short nonzero event. All parameters and the positive operating window are finite; no external read/reinstall is performed. □

J.8 Covariant bulk evolution of a retained quartet

The full-mode carrier requires a covariant source, not just a conserved finite-mode Hamiltonian. Use a physical metric $g = e^{2\rho}(-dt^2 + dx^2)$ and an independent unit material director $U = e^{-\rho}(\cosh\eta, \sinh\eta)$, with orthogonal unit V. The work field χ and director have actions

$$S_\chi = \frac{I_\chi}{2}\int \sqrt{-g}[(U\nabla\chi)^2 - C_\chi(V\nabla\chi)^2], \qquad S_\eta = \frac{I_\eta}{2}\int \sqrt{-g}[(U\Omega)^2 - C_\eta(V\Omega)^2], \quad (152)$$

where $\Omega = d\eta$ plus the physical spin connection and $0 < C_i < 1$. A positive elastic bar with coordinate a anchors this director through $-\Lambda\int\sqrt{-g}(U\nabla a)^2/2$. Prepare a strict bar domain $B - \Lambda > K_a > 0$ and finite alignment overcoming the work-clock orientation curvature $I_\chi(C_\chi - 1)$ and the quantum pressure debit. These are paid material parameters. Deriving U from ∇a instead would put second bar derivatives into the optical coefficients and lose the same-order estimate below.

The quartet comprises two nonchiral Majorana sectors of central charge $c_s = 1/2$, and all three spectator bosons of $c_s = 1$. Each sector has both chiralities and $\gamma_s = c_s/(24\pi)$; the total central charge is four. Its actual velocity $0 < v_s(\chi) < 1$ defines $\bar{g}_s = -v_s^2 U^\flat U^\flat + V^\flat V^\flat$. Supply the original global representation, density operator and every old reference. Choose optical-metric covariant point splitting with no additional finite two-derivative medium counterterm. The anomaly normalization is the conventional one [2, Section 2]; the actual causal state, rather than a vacuum inverse-curvature functional, determines its homogeneous stress.

Proposition J.8 (Actual-state covariant quartet bulk)**.** *Supply a consistent global full-mode representation and a smooth actual initial state with the required energy moments and old references. Prepare smooth local clock/bar/director data with strict paid margins, the causal Hessian conditions* (162) *and the JT constraints* (165)*. The paid fields, retained quartet state and renormalized constrained JT equations then have a smooth common local-time bulk solution. Its mean solution and source are independent of admissible exterior data continuations inside the physical domain of dependence. Equality of its quantum restrictions additionally requires equality of the initial local joint-state restrictions, not merely stress means. This result does not prescribe arbitrary local quantum means or a physical moving boundary.*

Proof. For one sector, set

$$d = \cosh^2\eta - v^2\sinh^2\eta, \quad N = v/d, \quad \beta = (v^2-1)\sinh\eta\cosh\eta/d, \quad \omega = \rho + \tfrac{1}{2}\log d. \quad (153)$$

Thus $\bar{g} = e^{2\omega}\hat{g}$ with $\hat{g} = -N^2dt^2 + (dx + \beta dt)^2$. Its canonical chiral coefficients are $h_\sigma = N - \sigma\beta > 0$. Write $D_t = \partial_t - \beta\partial_x$, $k = \beta_x/N$, $E = T_+ + T_-$ and $P = T_+ - T_-$. The actual Virasoro evolution gives

$$D_tE + NP_x + 2N_xP - 2\beta_xE = -2\gamma\beta_{xxx}, \tag{154}$$
$$D_tP + NE_x + 2N_xE - 2\beta_xP = 2\gamma N_{xxx}. \tag{155}$$

The exact conversion to the orthonormal $\hat{g}$ frame is

$$\hat{E} = E - \gamma k^2, \quad \hat{P} = P + 2\gamma k_x, \quad \hat{S} = \hat{E} + \gamma\hat{R}, \qquad \hat{R} = -2(N_{xx} + D_tk - Nk^2)/N. \tag{156}$$

Substitution gives both covariant conservation equations with zero residual and trace $\gamma\hat{R}$. The lapse and shift variations of $-\gamma\int \beta_x^2/N$ give precisely this projective conversion. The Weyl cocycle $\gamma\int\sqrt{-\hat{g}}[(\hat{\nabla}\omega)^2 + \omega\hat{R}]$ then yields

$$\bar{T}_{ab} = \hat{T}_{ab} + 2\gamma\left[\hat{\nabla}_a\hat{\nabla}_b\omega - \omega_a\omega_b - \hat{g}_{ab}\hat{\Box}\omega + \tfrac{1}{2}\hat{g}_{ab}(\hat{\nabla}\omega)^2\right]. \tag{157}$$

Its trace is $\gamma\bar{R}$ and its $\bar{g}$ divergence vanishes. In the bulk, the combined conversion is the first-jet density

$$\mathcal{L}_{\mathrm{loc},s} = \gamma_s\left[-\frac{(D_t\omega_s - \beta_{s,x})^2}{N_s} + N_s\omega_{s,x}^2 + 2\omega_{s,x}N_{s,x}\right]. \tag{158}$$

The integrated-by-parts difference is $-2\gamma_s[\partial_x(\omega_sN_{s,x}) + \partial_t(\omega_sk_s) - \partial_x(\omega_s\beta_sk_s)]$. It cannot be dropped at a physical wall.

Define the canonical closed-time-path functional by $Z_{\mathrm{can}}[z_+, z_-] = \mathrm{tr}\big(U[z_+]\varrho_0U[z_-]^\dagger\big)$. Add $\Gamma_{\mathrm{loc}}[z_+] - \Gamma_{\mathrm{loc}}[z_-]$, with $\Gamma_{\mathrm{loc}} = \sum_s\int\mathcal{L}_{\mathrm{loc},s}$. Variation at coincident branches gives the actual canonical expectation and the local forces in (158). No new homogeneous Polyakov field, independent purifier or in-out state is introduced. Positive h_σ makes each canonical Hamiltonian lower bounded, not nonnegative: diffeomorphism conjugation includes the debit $-\gamma_s\int h_{\sigma,x}^2/h_\sigma\,/2$ and the chosen Casimir term. On the prepared smooth coefficient ball these have a finite uniform bound. One common paid scalar offset covers it; its stress is included in both histories.

An exact local reduction supplies the same-order estimate. For each $u_j = \sigma_jh_j$, solve

$$\begin{aligned} w_{j,t} + u_jw_{j,x} + u_{j,x}w_j &= -u_{j,xx}, & w_j(0,x) &= 0, \\ A_{j,t} + u_jA_{j,x} + 2u_{j,x}A_j &= 0, & A_j(0,x) &= T_j(0,x). \end{aligned} \tag{159}$$

Differentiation gives $(\partial_t + u_j\partial_x + 2u_{j,x})(w_{j,x} - w_j^2/2) = -u_{j,xxx}$. Hence, with $Z_j = \gamma_jw_j$,

$$T_j = A_j - Z_{j,x} + Z_j^2/(2\gamma_j). \tag{160}$$

This is the exact Ward solution. If $f_{j,t} + u_jf_{j,x} = 0$, $f_j(0,x) = x$, then $w_j = \partial_x\log f_{j,x}$, $A_j = f_{j,x}^2T_j(0,f_j)$ and $T_j = f_{j,x}^2T_j(0,f_j) - \gamma_j\mathcal{S}(f_j)$. These are the actual characteristic connection and transported initial stress, not additional quantum fields or a new source action.

Put $q = (\chi,\eta)$, $P_q = q_t$, $Q_q = q_x$. Let K_{tt}, K_{tx}, K_{xx} be the q Hessians of the complete first-jet classical action plus (158), with $K_{tx} = \mathcal{L}_{P_qQ_q} + \mathcal{L}_{Q_qP_q}$, and let G have columns $g_j = \nabla_qh_j$. The matter principal matrix and its symmetrizer are

$$A_m = \begin{pmatrix} K_{tt}^{-1}K_{tx} & K_{tt}^{-1}K_{xx} & -K_{tt}^{-1}G \\ -1 & 0 & 0 \\ 0 & \mathrm{diag}(\gamma_j\sigma_j)G^T & \mathrm{diag}(\sigma_jh_j) \end{pmatrix}, \quad H_m = \begin{pmatrix} K_{tt} & 0 & 0 \\ 0 & -K_{xx} & G \\ 0 & G^T & \mathrm{diag}(h_j/\gamma_j) \end{pmatrix}. \tag{161}$$

Multiplication gives $H_m A_m = (H_m A_m)^T$. The Schur conditions for H_m and $H_m \pm H_m A_m$ are exactly

$$K_{tt} > 0, \qquad -K_{xx} - \sum_j \gamma_j g_j g_j^T / h_j > 0,$$
$$K_{tt} + K_{xx} \pm K_{tx} \pm \sum_j \frac{\gamma_j \sigma_j g_j g_j^T}{1 \pm \sigma_j h_j} > 0. \tag{162}$$

At rest the director stiffness is corrected by $\Delta D = \sum_s 2\gamma_s (1 - v_s^2)^2 / v_s$. Finite large inertias with strict $0 < C_i < 1$ give a nonempty domain; (162) uses its actual nonlinear coefficients, including nonzero director boost. These inequalities imply a uniform matter cone strictly inside the physical light cone.

The metric and dilaton each have light speeds $\lambda = \pm 1$. There is no resonant trace coupling between them. For the full (χ, η, ρ) Hessians of a quantum sector, direct substitution of (153) gives the exact identity

$$(K^3_{tt,s} - \lambda K^3_{tx,s} + K^3_{xx,s}) \begin{pmatrix} 0 \\ \lambda \\ 1 \end{pmatrix} = \sum_{\sigma = \pm 1} \frac{\gamma_s \sigma \lambda h_{\sigma,s,\eta}}{\lambda - \sigma h_{\sigma,s}} \nabla_{(\chi,\eta,\rho)} h_{\sigma,s}. \tag{163}$$

The right side has zero ρ component. The director connection also vanishes on this light mode; its clock component is zero. The corresponding stress is $Z_{\sigma,s} = \gamma_s \sigma h_{\sigma,s,\eta} \lambda p / (\lambda - \sigma h_{\sigma,s})$, with $P_\eta = -p$, $Q_\eta = \lambda p$ and $P_\chi = Q_\chi = 0$. Hence the complete trace residue is zero. The strict matter cone solves the matter-to-light Sylvester equations; the zero resonant projection solves the remaining metric-to-dilaton equations. Pulling back the block positive form produces a smooth full symmetrizer. The ten A_j variables append the diagonal principal block $\mathrm{diag}(u_j)$ and positive form 1_{10}; they introduce no new light resonance.

For $m > 5/2$, use $q \in H^{m+1}$ and $P_q, Q_q, Z, A \in H^m$; (160) lies in H^{m-1}. The exact local split is

$$Z_{j,t} + u_j Z_{j,x} + \gamma_j \sigma_j g_j^T Q_{q,x} = -\sigma_j (g_j^T Q_q) Z_j - \gamma_j \sigma_j h_{j,qq}[Q_q, Q_q],$$
$$A_{j,t} + u_j A_{j,x} = -2\sigma_j (g_j^T Q_q) A_j. \tag{164}$$

The force is $\sum_j g_j Z_{j,x} - \sum_j g_j [A_j + Z_j^2/(2\gamma_j)]$. All remainders are smooth local products in the same H^m algebra. The conversion contributes first-jet Hessian principal terms and quadratic first-jet remainders. No spatial inverse, omitted mean or higher derivative of the quantum state enters the iteration. Symmetric-hyperbolic iteration gives $d\mathcal{E}_m/dt \le C_R(1 + \mathcal{E}_m)$ on the declared operating ball. Tame higher-order estimates keep smooth data smooth on that same local time. The actual full-mode propagator therefore exists for the solved coefficients; uniqueness of the distributional Ward transport identifies its stress with the solved means on the interior patch, with the same initial state and all references retained. A norm-controlled exterior extension of the reduced initial variables can be chosen for the local iteration. It does not assign artificial means to that quantum state outside the initial interval, and no common-action solution is inferred there. Its bound fixes the local existence time; arbitrary exterior continuations are compared only on their common interval of existence.

This local reduction also proves physical bulk causality. Let two admissible continuations agree initially on (a, b), including scalar values and actual stress means. Their

local difference energy on $(a+t, b-t)$ has endpoint fluxes $-V^T(H-HA)V$ on the left and $-V^T(H+HA)V$ on the right. They are nonpositive; the interior is bounded by its energy. Gronwall therefore gives identical mean solutions until the interval closes or the operating domain is left. The interval boundaries are the null curves of g. For quantum restrictions require the same initial joint local state on the causal interval and the compared old references. The actual sector diffeomorphisms pull local observables into that interval, giving the corresponding restriction equality. Stress means alone do not give quantum-state equality, nor is a Type-III local density matrix assumed.

Finally write the minimally coupled JT action as $(2\kappa)^{-1}\int \sqrt{-g}\Phi(R-2K_{\mathrm{geo}})$. Its metric curvature equation is $\rho_{tt}-\rho_{xx}=K_{\mathrm{geo}}e^{2\rho}$. Set $w=\Phi_x$, $\pi=\Phi_t$. The initial constraints are

$$\Phi_x = w, \qquad w_x = \rho_x w + \rho_t \pi - K_{\mathrm{geo}} e^{2\rho}\Phi - \kappa T_{00}, \qquad \pi_x = \rho_x \pi + \rho_t w - \kappa T_{0x}. \tag{165}$$

Here T is the complete source just constructed, including initial conversion jets; needed accelerations come from its invertible time matrix. This spatial ODE has a local solution for three integration constants, with positive Φ on a sufficiently small interval. Conservation propagates its two traceless constraints locally along physical light directions. Arbitrary periodic positive-energy JT data are not assumed. A consistent global representation and state realize the quantum solution; the causal interior restriction no longer depends on an arbitrary exterior coefficient continuation. □

This closes the quartet's nonlinear bulk source–clock–director–JT interface. It supplies neither moving-wall boundary terms nor the other gate windows' errors. The original mixed target and its purifier remain in the actual state, but clock and geometry changes alter gate areas. Their same-history finite-horizon comparison and the physical exhaust still have to be incorporated before the complete thermal apparatus is obtained.

J.9 A physical wall retains its thermal friction

A finite rest mass cannot repair a perfect mirror by ignoring its high-frequency reaction. Partial transmission has a causal passive linear-response formulation [1]. For the four actual Dirac flavours, the corresponding force can be derived from a smooth material mass layer rather than imported from a scalar mirror. In a static material frame let $\mathsf{h}_z = -iv\sigma_z\partial_x + M(x-z)\sigma_x$, with $M \in C_c^\infty$, $M \geq 0$, and positive bare mechanical mass m_b. The same mass action gives the translation force $F = \int M'(x-z)\psi^\dagger\sigma_x\psi\,dx + F_{\mathrm{ct}}$, distributed bar reaction and metric mass stress. Use one fixed local counterterm action, an actual static interacting Hadamard ground or KMS preparation, and smooth flat-start displacement tests.

Proposition J.9 (Passive Dirac-layer response)**.** *For the static ground-state linear response, the full-continuum induced vacuum inertia is finite. With positive bare mass and total positive elastic stiffness, the unreduced causal mechanical impedance is positive-real. In the static zero-charge KMS continuum the layer instead has positive ohmic friction. Its vacuum-style low-frequency inertia integral diverges, while the unreduced thermal impedance remains convergent and positive-real. These are linear statements about the specified passive preparation, not arbitrary driven quartet states.*

Proof. Let $u_{E,a}$, $a = L, R$, be energy-normalized scattering modes with incoming amplitude $(2\pi v)^{-1/2}$. Define $A = \int M/v$, $Q = \|M'\|_1$ and $B = e^A/\sqrt{\pi v}$. The exact transfer

equation $u_x = (iE\sigma_z + M\sigma_y)u/v$ bounds $\|u\| \le B$. Integration by parts gives

$$f_{ab}(E,E') := \int M' u^\dagger_{E,a}\sigma_x u_{E',b} = -(E+E')\int (M/v) u^\dagger_{E,a}\sigma_y u_{E',b},$$
$$|f_{ab}(E,E')| \le B^2 \min\{Q, A|E+E'|\}. \tag{166}$$

The mass anticommutator vanishes and M removes the endpoint term. For $N = 4$ flavours the filled-negative-state force spectrum obeys

$$\chi_I(\nu) = \pi N \sum_{a,b}\int_0^\nu |f_{ab}(E, E-\nu)|^2\, dE \le 4\pi N B^4 \min\{Q^2\nu, A^2\nu^3/3\},$$
$$\mu = \frac{2}{\pi}\int_0^\infty \frac{\chi_I(\nu)}{\nu^3}\, d\nu \le \frac{16 N B^4 A Q}{\sqrt{3}}. \tag{167}$$

At zero energy the exact transfer gives reflection $\tanh^2 A$, and hence $\chi_I(\nu) = N\tanh^2 A\,\nu^3/(6\pi v^2) + O(\nu^4)$. Retain the complete causal memory. For $\operatorname{Re} s > 0$ its impedance is

$$Z(s) = m_b s + k/s + \frac{2}{\pi} s\int_0^\infty \frac{\chi_I(\nu)}{\nu(\nu^2+s^2)}\, d\nu. \tag{168}$$

The measure is positive and $s/(s^2+\nu^2) = \frac{1}{2}[(s-i\nu)^{-1} + (s+i\nu)^{-1}]$ has positive real part. Thus $\operatorname{Re} Z(s) > 0$; the response has no right-half-plane pole. The observed vacuum inertial mass is $m_{\text{phys}} = m_b + \mu$, not the action's bare m_b. For two separated layers the same supportwise identity gives a positive matrix spectral measure. Adding their actual finite static polarization/Casimir Hessian to the paid elastic stiffness extends this argument when that total stiffness and the bare mass matrix are positive.

In the KMS continuum, with Fermi function $n_T(E)$,

$$\chi_{I,T}(\nu) = \pi N \sum_{a,b}\int_{\mathbb{R}} [n_T(E) - n_T(E+\nu)]|f_{ab}(E+\nu, E)|^2\, dE. \tag{169}$$

Its weights are positive. On shell, $\sum_{a,b}|f_{ab}(E,E)|^2 = 2E^2R(E)/(\pi^2v^2)$, where $R(E)$ is the actual reflection probability. Therefore

$$\chi_{I,T}(\nu) = \gamma_T\nu + o(\nu), \qquad \gamma_T = \frac{2N}{\pi v^2}\int E^2 R(E)[-n_T'(E)]\, dE > 0. \tag{170}$$

The vacuum μ integral now diverges in the infrared. Equation (168) itself stays finite: its infrared integrand uses this ohmic law, while $\int[n_T(E) - n_T(E+\nu)]dE = \nu$ and (166) give an ultraviolet linear bound. Its positive-real proof is unchanged. A finite prepared exterior has the corresponding discrete positive Kubo measure; this continuum friction limit is not an uncharged infinite thermal reservoir. □

All transmitted exterior channels, old purifiers, static polarization and motion work remain in this construction. Smooth spatial profiles alone do not justify applying an instantaneous force to the old free sea as a finite-energy vector; the valid response tests are also smooth in time. More importantly, the actual driven body need not be passive. Its opposite-chirality mass mixing and non-Gaussian quartet require a joint local-source and task-error estimate. The common-velocity Ward closure inside a massless gate does not close that boundary hierarchy. The passive wall result establishes a physical linear margin, without substituting that margin for the retained mixed-target reset proof.

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